

Chapter 22

Vibration and Noise

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Vibration is one of the most vexing problems with pumping machinery, and it is the cause of considerable altercation and litigation. Noise can become a significant, annoying problem. Excessive vibration from primary equipment can be transmitted directly to the building structure, which causes uncomfortable (and sometimes dangerous) structural vibration levels. Excessive vibration of equipment and piping can destroy portions of the equipment (such as drive shafts, bearings, and seals), loosen or break pipe anchors, and even cause pipes to burst under certain conditions.

To make this chapter easier to use, the presentation is divided into four parts:

- Avoiding vibration problems. Sections 22-1 and 22-2.
- Troubleshooting excessive vibration. Section 22-3.
- Vibration analysis. Sections 22-4 through 22-12.
- Noise analysis. Sections 22-13 and 22-14.

The first two parts require no background knowledge of the subject and are easy to read. The third part is written for those who wish to delve more deeply into

the subject of vibration. The fourth part is, of necessity, a simplified mathematical presentation of noise but with enough worked examples to make it easy to follow.

References to a standard or code are given in abbreviated form such as ANSI/HI/9.8-1998. Specific sections may be referenced as, for example, ANSI/HI 9.8, 2.2.3-1998.

22-1. Problems of Vibration and Noise

For the reader's convenience, the problems of vibration and noise in pumps and drivers are treated separately.

Pumps

Sources or causes of vibration include:

- The reaction of the impeller vane as it passes the casing cutwater.

- Pumps operating off the best efficiency point (BEP) and thereby creating eddies within the pump.
- Eddies created by water flowing through bends, valves, and other obstructions.
- In every rotating (e.g., motor) or reciprocating (e.g., engine) machine, some imbalance and shaft misalignment always exists.
- Resonance. If the natural frequency of the machine or system is nearly equal to the frequency of excitation, however small, the resulting vibration can become destructive.

Much of this chapter is devoted to the problems of lateral or translational vibrations, where the vibrating element actually moves back and forth. Rotating shafts may vibrate in both translational and/or torsional modes. In the torsional mode, the centerline of the shaft may not physically move in space, but the shaft does rotate at slightly different instantaneous rotational speeds with different phases at various points along the length of the shaft, which causes an oscillating torque. This torsional (twisting) vibration of the shaft can develop stresses high enough to cause the shaft to fail without any audible or visible warning signs prior to the failure. The subtle character of torsional vibration makes it all the worse.

The impact of a new pumping station on environmental noise levels in the community should be evaluated prior to final site selection, because noise adversely affects residential property values, causes sleep interference and general annoyance, and may lead to legal action against the owner of the plant. Building codes often include noise level limits at property lines. Noise problems are most likely to occur when facilities are located within 300 m (1000 ft) of a residential neighborhood. To protect the owner from legal challenges should a subsequent complaint develop into a court case, documentation of environmental noise levels by a qualified acoustical consultant prior to the start of construction in these instances is vital.

All too often, nothing whatever is done to reduce the noise level within the pumping station. Excessively high noise levels here can impair speech intelligibility and cause temporary or even permanent hearing loss to operating personnel exposed over extended periods of time. It is usually impractical to enclose a noisy machine in a sound-absorbing housing, partly because of the need to service the machine and partly because such measures are relatively ineffective. Common practice, therefore, is to enclose all the noisy machinery within thick concrete walls and ceilings. Requiring workers to wear ear protection and designing for reduced maintenance time and

worker occupancy is helpful. The noise level in the reverberation field can be reduced by acoustical treatment (see “Indoor Sound Propagation” in Section 22-13). Although the noise level cannot be reduced by much more than 6 to 8 decibels, this reduction is often enough to permit conversation and substantially increase the level of comfort. At 2005 prices, the cost of acoustical treatment varies from \$40–\$130/m² (\$4–\$12/ft²).

Drivers

Sometimes electric motors are the cause of excessive noise or vibration. Vibration tends to have unusually high spikes at one and/or two times the line frequency (60 and/or 120 Hz). If there is a broken rotor bar or bad stator winding, a high-pitched noise (and accompanying vibration acceleration) is often evident at the “slot pass” frequency (about 100 times running speed). In motors driven by AF controllers (particularly the older models), strong vibration pulses are evident even for healthy motors at 6×, 12×, 18×, and sometimes 24× the frequency with which the controller feeds the motor at any given speed. These frequencies seldom cause dynamic or structural resonances of any significance. However, these harmonics can sometimes excite linked-rotor system torsional critical speeds that are difficult to track down with standard vibration equipment that cannot directly sense torsional oscillations, even if shaft stresses are becoming excessive. If shaft failures are occurring at the shaft shoulder near the pump impeller or coupling hub, this situation should be investigated.

Engines

If the driver is an internal combustion engine, the primary driving frequencies (besides one times and two times running speed and the pump vane passing frequency) are the cylinder stroke frequency (i.e., the number of cylinders times the engine speed) and its first 10 harmonics. For most engines, the second harmonic (i.e., two times engine speed times number of cylinders) is the strongest, and no natural frequency of the engine-support floor structure or the engine-gear-pump train should be allowed to exist anywhere in the typical operating speed range. Resonances of the other first 10 harmonics cannot, however, be ruled out, and they should be investigated if there is excessive noise or vibration in such installations. Offset or right-angle gear boxes can also be a torsional excitation source at the gear mesh

frequency (number of gear teeth times running speed). Gear mesh excitations can also cause gear tooth damage and/or excessive noise due to axial resonances of the drive shafts, particularly if the drive shafts are hollow. Filling a hollow drive shaft with grease may eliminate the problem by energy-absorbing damping.

For every engine-driven pump—whatever its size—designers should specify that the pump manufacturer shall make (or have a qualified analyst make) at least a torsional vibration analysis of the pump-driver system. A translational vibration analysis should also be made for the engine-pump-support structure. Many low-cost pump bidders are not qualified for such work, so prequalify bidders to protect both the client and your reputation as a designer. Alternatively, prequalify third-party specialists to perform the analyses and require the pump manufacturer or the contractor to retain only a firm from the list. This approach avoids the potential of a conflict of interest on the part of the individuals performing the analyses. See Appendix C Paragraph 11050–1.05B.3 for details.

Active Noise Control

The use of electronically generated sound waves to cancel or reduce unwanted sound (noise) is called “active noise control.” This technique has been known for decades, but it has not been widely implemented because of practical limitations in its application. Recent advances in signal processing hardware and software have improved the situation considerably, but active noise control is still generally restricted to low-frequency applications (typically below 200 Hz) and to noise contained inside a duct or a pipe. Once the noise has escaped into the listening environment, this technique is no longer a viable means of achieving a significant noise reduction over an extended region of space.

22-2. Avoiding Vibration Problems

By following a few simple rules, a nonspecialist can—fortunately—avoid about 90% of all vibration problems. The presentation herein is focused on minimizing vibration in pumping station machinery and its related systems. Necessary concepts are explained in simple terms, and their practical use is described. The information is relevant to small- or medium-sized centrifugal pumps of all types, reciprocating pumps, ventilation fans, and electric motor and diesel drivers.

For stations with pumps larger than 75 kW (100 hp), or with engine or variable-frequency drives, review the more detailed procedures presented in later sections of this chapter or have a formal vibration analysis made.

Pump Selection

Select a pump that will operate close to its best efficient point (BEP). Plot the maximum and minimum headloss envelope, and choose a pump suitable for the entire range. Then, as conditions change (flow rate increases, pipe becomes rougher), it is necessary only to change the impeller to meet the new conditions. Beware of applying large factors of safety to the flow rate, because this blunder could result in an oversized and therefore unreliable pump.

Balancing

Adequately balance all rotating components of diameter equal to at least one-third of the pump impeller diameter. Balance all components (in two planes) that have a length of at least two-thirds their diameter. The entire rotor system should be check-balanced in the assembled condition. A balance standard shown to be conservative for pumping station equipment is $e = kW/N$, where k is a constant equal to 0.423×10^{-3} for SI units, e is the imbalance in $k \cdot m$, W is the mass of the balanced component in kg, and N is the peak operational speed in Hz. In U.S. customary units, the equation is $e = 16 W/N$, where e is in oz · in., W is weight in lb, and N is speed in rev/min. Reports to demonstrate compliance with the balancing standards should be submitted as product data. Do not accept any loose rotating component fits where the mass of the component multiplied by the radial clearance approaches the above criterion. Check the shafts to ensure that, if shaft bow is detectable, the amount of bow times the supported component weight also does not exceed the above criterion.

Accurate alignment is necessary to make the centerline of the driving and driven equipment coincide as closely as possible both in concentricity and parallelism. To specify adequate coupled shaft alignment for a rigid coupling or one with one axial location of flexibility (“single engagement”), a good rule is to maintain an error of no more than 37 to 50 μ g (1.5 to 2 mils) in the concentricity between coupling hub centerlines. This concentricity criterion may be loosened by an additional 1 to 2 mm per m (1 to 2 mils

per in.) of length between coupling hubs for double engagement or “spacer” flexible couplings, depending on the coupling type. The maximum permissible angle between shafts at the point of coupling engagement is 5 to 10 min, which is automatically maintained if the above offset specifications are maintained. Coupling manufacturers often quote larger amounts of offset and parallel misalignment than the above values, but these quotes account only for coupling survivability, not equipment bearing survivability, and they also do not account for the increased equipment vibration that misalignment might cause.

Pumps with stuffing boxes (packing glands) are less sensitive to misalignment than those with mechanical seals, and the maximum criteria above are satisfactory where stuffing boxes are used. Use the minimum criteria for pumps with mechanical seals.

Alignment should be performed only by qualified technicians or millwrights. Alignment is best determined by the reverse dial indicator method or its modern optical equivalent (although the latter is best performed only by an expert) as described in detail on pages 2.413ff in the *Pump Handbook* [1]. Keep in mind that the above alignment criteria can be applied with the equipment cold, but steps must be taken to ensure that they also apply to the “hot” alignment (i.e., with the pump or fan running, after warm-up). Also, be sure to account for significant machining inaccuracies in whichever coupling faces or rims are used for measurement (reverse dial indicator largely accounts for that). Furthermore, compensate for gauge support arm sag, because the sag can be larger than the alignment specification level. Once alignment is achieved, the pump or fan feet should be doweled to the baseplate to avoid loss of alignment from slip at bolted joints. Grout baseplates to the foundation pad with nonshrinking epoxy grout (per manufacturer’s recommendations), and take care to leave no air pockets of significance between the grout and baseplate by the liberal use of vent holes—some located by tapping the baseplate top and listening for hollow sounds to find voids. See Section 12.10, Items 27–31 and Section 25–8.

Pump Support

The suction and discharge piping end flanges and the opposing pump nozzles must be properly supported. Avoid the use of unrestrained pressure-bearing “expansion” or “flexible” joints at pump nozzles for pipes equal to or larger than 150 mm (6 in.) in diameter unless the contained pressure times the nozzle

cross-sectional area is within the manufacturer’s nozzle load limits, and the pump and piping natural frequencies are well removed from the range of vane pass frequencies. Although such joints relieve any piping thermal expansion or Bourdon-tube effects from “loading” the equipment nozzles, they do not allow the piping to absorb the cross-sectional nozzle hydraulic load (i.e., the pressure times the open area). This nozzle hydraulic load can produce a large thrust perpendicular to the nozzle opening and severely load the pump casing. Failure to account for such loads has caused serious operating alignment problems, casing rubs, and system damage in many installations.

Natural Frequencies of Pump and Piping Supports

Design piping and pump supporting structures (including floors) to have natural frequencies that are at least 25% less or 25% more than the operating range of the key excitation frequencies in the pump. Typically, these excitation frequencies equal (1) the running speed, and (2) the number of vanes times running speed. Loaded-floor natural frequencies can be significantly different from those of the bare floor. In addition, the floor stiffness can influence the natural frequency of the pump because the manufacturer calculates the natural frequency on the basis of an infinitely stiff floor. For horizontal and vertical non-clog pumps, include the pump mass (plus the water contained) and pedestal stiffness when calculating loaded-floor natural frequencies. For vertical turbine pumps, include the motor and pump bowl assembly masses, and the discharge head and column piping stiffness when calculating loaded-floor natural frequencies. The component masses and stiffnesses are available from the manufacturers upon request.

These calculations can be made by the finite element method, but it is possible that the same models used for the structural design can be used to calculate the building floor and pipe structural natural frequencies after bulk elements are included for the pump and motor, as described above. Natural frequencies for the pump itself with added mass and stiffness of the floor and piping added are best found by the finite element method.

Pipe Support Spacing

Use a liberal number of unequally spaced pipe supports. This rule includes discharge piping for submersible pumps. To avoid vibration problems, piping must

be well supported in three perpendicular directions (axes x, y, and z)—not just in the vertical direction. To avoid or suppress vibration, both the weight of the piping and adequate stiffness must be considered in designing proper supports (see Section 22-10).

Drive Shaft Offsets

Do not offset drive shafts at an angle. In the past, drive shaft manufacturers have often recommended at least a 3-degree offset to help lubricate needle bearings, but it has been found that an offset is not needed for bearing lubrication. Furthermore, an offset excites vibration at a frequency of two and four times the running speed.

Baseplates and Foundation Pads

Foundation pads should extend beyond the machinery baseplate by at least 75 mm (3 in.) or one-half of the floor-plus-pad thickness, whichever is larger. The effective thickness from subfloor beams and gussets need not be taken into account in this calculation. The mass of machinery foundations should be 5 to 10 times that of the machine itself. The foundation should have an adequate aspect ratio. That is, the “footing” lines from the centerline of the pump inclined 30 degrees to the vertical should pass through the bottom of the foundation mass—not through its sides. The centerline of the pump is considered to be at the center of the impeller hub except for vertical turbines.

Pump Suction Piping

If possible, use straight suction piping up to the pump suction nozzle, and keep any required bends in a single plane. As a minimum, avoid piping reducers or bends within 5 to 10 suction pipe diameters of the pump inlet, unless the pump has been specially designed to accommodate such fittings closer to the pump (as in some vertical nonclog pumps). If in doubt, seek the pump manufacturer’s advice.

Single and Double Volute

The vane passing force is generally two to three times higher for single volute pumps than for “twin” or “double” volute pumps. Where practical, choose double volute pumps (preferably with an odd number

of impeller vanes), because they minimize the vane passing force and maximize both BEP and, particularly, off-design efficiency. Single volutes are often necessary because of cost or solids-passing capability. The hydraulic imbalance present in single volutes is minimized, at some decrease of efficiency, by increasing the clearance between impeller and volute tongue.

Pump Suction Conditions

Ensure adequate pump suction conditions with sufficient NPSHA including all significant effects (see Figure 10-13) and a sump design in which surface and subsurface vortices are suppressed.

Recommended Detailed Design Practices

The potential for vibration problems is often given insufficient attention at the beginning of the pumping station design process. By following appropriate design rules, and with a bit of luck, it is likely that the vibrational behavior of the installed equipment will be acceptable. However, this is a risk not worth taking for pump drivers of more than 56 kW (75 hp), and the cost to all parties in those situations where vibration problems do occur quickly overshadows the “savings” of not performing analyses and test procedures such as those recommended below.

Controlling Excitation Forces

The first step in controlling excitation forces is to limit mechanical operational forces that can act to drive machinery vibration. The primary forces that must be controlled are caused either by imbalance of large diameter rotating components such as coupling hubs or pump impellers, or by misalignment of coupled shafts. These issues are discussed above.

Although mechanical forces may be the most common cause of excessive vibration, another common problem is the use of a pipe roughness coefficient that is too conservative and a resulting specified total discharge head (TDH) that is higher than the one encountered. A pump selected on such a basis operates well to the right of its BEP, discharges at a higher flow rate than it should, and increases the hydraulic excitation forces. In many instances, the pump will be operating in a cavitation region. The motor draws more current and may be overtaxed. Eliminate this problem by ensuring the proper range of *C* values is used for selecting both pump and impeller.

The primary frequency of hydraulic excitations is the vane-passing frequency. The forces and the resulting vibration depend upon several factors, controllable at the design and specification stage. The most important method of minimizing vane-passing vibration is to design proper suction piping, to use double volute or centered volute pump designs wherever possible, and—most important—to operate the pump at or near the BEP.

For all types of volutes, it is possible to lower vane passing forces at some expense of efficiency by opening up the clearance between the impeller vanes and the volute tongue or diffuser vanes. This opening is called the “B-Gap.” Designers should, in general, be wary of excessive vane-passing vibrations from any pump with a diametral B-Gap of less than 4% of the impeller diameter, and a B-Gap of 6 to 10% or more is preferable, although excessive B-Gap can encourage discharge recirculation, as discussed below, when operating below the BEP. A sound approach is to follow manufacturer guidance on the setting of the B-Gap, while enforcing appropriate vibration specifications (discussed below) and performance guarantees throughout the operating range.

Reduced Flow

For all types of centrifugal pumps, the vibrations, shaft deflections, and bearing loads usually increase as the pump is run at lower flows. They do not decrease as is commonly assumed. In fact, if centrifugal pumps are operated well below BEP (say, at 50% of the design flow at any given speed), suction and/or discharge recirculation will probably occur, accompanied by possibly large excitation forces at vane pass and at frequencies below running speed. The problem in both suction and discharge recirculation is that stalling is induced on the vanes or tongue similar to the stalling that occurs around an airplane wing at a bad angle of attack. This condition causes strong eddies, which resemble and act like miniature tornadoes, to be spawned and “kicked upstream” of the stalled passage. These “stall cells” often subsequently rotate near the impeller at a speed somewhat less than running speed (usually about 10 to 40% in the diffuser or volute or about 60 to 90% at the inlet of the impeller). The high local velocity and low local pressure associated with these cells (keep in mind the analogy of a tornado) interact with the vanes of the impeller and cause new and usually unexpected excitation force frequencies. In summary, *running a centrifugal machine at flows (loads) below its rated capacity (i.e., well below its BEP) typically causes*

more, not less, stress and vibration, and decreases machine life and reliability. (Of course, vibration due to unbalance or misalignment decreases at lower speeds.) These consequences should be considered in the lifetime cost and factored into design and purchasing decisions. Pumps can be selected for both present and future (larger) capacities by choosing a model that can meet both requirements merely by changing the impeller. The motor must be capable of operation at both flow rates as well. In many high-specific-speed pumps, power requirements at reduced (less than BEP) flow rates can be higher than at BEP.

Selection of Machines

For reliable operation and long life, select centrifugal machines that will operate as close to their BEP as possible throughout their operating range, and keep the flow rate within the manufacturer’s maximum and (especially) minimum limits. Designers should pay attention to system changes with time because (1) flow resistance can increase with time due to increasing pipe roughness (particularly in long force mains), (2) the pump capacity can decrease significantly as clearance around the wear rings increases, and (3) erosion further decreases discharge capacity. The effect of system changes on suction line losses, and therefore on NPSHA, must also be analyzed and compared to the manufacturer’s requirements. In reciprocating pumps, the effect of “acceleration head” due to the unsteady pulsing flow in long suction lines must be considered (see ANSI/HI 6.1–6.5-2000).

Acoustic “organ pipe” resonances must be avoided (as discussed later in this chapter), or they may become exciting forces. Concern with acoustic resonance is particularly apt in reciprocating machines. Their strong flow and pressure pulsations at piston or plunger frequency are able to excite certain lengths of manifold or piping, and they are often run across a broad speed range. Accumulators are often needed near the inlet or discharge manifolds to detune and/or dampen potential acoustic resonances.

Net Positive Suction Head

To minimize hydraulic forces for all types of pumps, it is important to operate the pump with sufficient NPSHA (see Section 10-4). In addition to causing cavitation, inadequate NPSHA can excite unexpectedly high vibrations at various natural frequencies. From Bernoulli’s equation, any increase in local velocity decreases static pressure and thus increases the

cavitation potential, so it is important to keep the pump inlet flow velocities low (preferably below about 2 m/s or 6 ft/s) and evenly distributed. Hence, a large suction pipe (at least equal to the pump suction flange ID) should be used. Pipe reducers should preferably be at least five pipe diameters upstream from the pump flange. Maintaining this distance from more aggressive suction disturbances such as valves and elbows is even more important. Besides avoiding cavitation, this practice also ensures well-distributed flow velocity and pressure at the inlet and throughout the pump, and thus minimizes the hydraulic excitations in general. High velocity in the discharge is not usually important because the pump increases the static pressure well above the cavitation point.

Wet Wells

Minimize the possibility of vortex formation and discourage whirling flows around column piping by using care in designing the wet well. Follow the advice in Chapter 12 (preferably) or in ANSI/HI 9.8-1998.

Avoidance of Resonance

Resonance may cause excessive vibration even when mechanical and hydraulic forces are properly controlled. To avoid this problem, distribute the installed system (pump, piping, pedestal, and floor) mass and stiffness in such a manner that no natural frequencies are close (within at least 25%) to excitation frequencies, such as one or two times the running speed or one times the vane-passing frequency. This analysis is done in the design stage by successive trials, using appropriate computer models, such as finite element analysis. In such a model, pump and motor component drawings of sufficient detail must be obtained from the manufacturers—at least overall external dimensions, weight, and center of gravity (relative to the mounting flange)—and pedestal or base (including gussets), floor (including subfloor beam and column dimensions and locations), piping (if rigidly connected, including joint directional rigidity), and water mass must be included. Simple manual calculations are not likely to be sufficiently accurate to avoid resonances because of the complexity of these interlinked components. So much effort may be overkill for small stations, but it should be included in the design of large stations.

A common misconception is that if only the pump is designed stiffly enough, then natural frequency resonances will be avoided. Because the lower natural frequencies of assembled pumping and compres-

sion systems usually depend strongly on the base and floor design as well, as discussed by Reichert et al. [2], any reliable analysis must include the pump, base, and floor together. Similarly, any test for the lower natural frequencies of installed machines and possible resonances is unlikely to be valid in the pump manufacturer's shop (although hydraulic/aerodynamic performance tests and rotordynamic testing should be valid). Only when all equipment is installed on its final foundation, filled with water, and with all bolts tight, will representative natural frequencies be present in the lower frequency ranges and detectable by a shaker or impact test.

Avoiding all matching between significant forcing frequencies and all natural frequencies may not be feasible with some pumps, particularly those operating at variable speed with a turn-down ratio of 30% or more. For example, if the first natural frequency occurs at 70% of the running speed in a vertical pump with a variable frequency 2:1 turn-down range, it is unlikely that mechanical design changes can be made that would move such a natural frequency up enough or down enough to be out of the speed range added to the resonance avoidance margin of 25%. Although vibration isolation pads can be used to decrease the reed frequency as much as necessary, the resulting flexibility of the revised system is likely to cause alignment problems, and also is likely to decrease the second natural frequency into the running speed range, thereby trading the original resonance for a new one. If resonance occurs in such instances, there are several alternative solutions:

- Tighten the balance specifications, and tighten the maintenance schedule and procedures, for example, by field trim balancing in which mass is added to the coupling hubs without disassembly.
- Reduce the natural frequency as much as possible by, for example, installing vibration isolation pads under foundation bolts until alignment approaches a potential problem due to static or low-frequency hydraulic forces. Unfortunately, reducing the natural frequency is accomplished by successive trials until the second natural frequency is within 25% of the running speed range.
- Use “lockout speed range” settings on the speed controller to avoid operation within $\pm 25\%$ of the problem natural frequency. The use of lockout settings may cause a larger “hole” in the operating speed range than is tolerable from an operational standpoint. If the amplification factor at the edges of the compromised range results in acceptable vibrations at maximum imbalance and misalignment and at minimum flow, the lockout range can be

relaxed to $\pm 15\%$ and even perhaps to $\pm 10\%$. The natural frequency sometimes shifts with ambient temperature or with water level in the sump.

Rotordynamic Analysis

Rotordynamic analysis of the pump, compressor, or fan rotating system should be performed by the manufacturer or a third-party consultant. The authors prefer that the work be the product of third-party specialists who have a demonstrated record in the field and are registered professional engineers. This analysis requires specialists, and details are not presented here. Some of the items that should be requested, however, are discussed below.

Usually the rotordynamic analysis performed by the manufacturer is for lateral vibrations only, not for torsional or axial vibrations. All three analyses should be performed, particularly for variable-speed systems. It should be made clear in the request for quotes (or in the equipment specifications) that a complete analysis is required so that all manufacturers are bidding on an equal basis and will include the cost of the analysis in their bid. A paper by Corbo and Malanoski [3] is an excellent reference for writing a purchase order or project specification. It describes step-by-step procedures and the results required to demonstrate compliance with requirements as outlined in Appendix C. The cost of the rotordynamic analyses and the torsional analyses, sometimes called the dynamic analyses of the system, should also include the pump and motor frames, the supports for the motor or engine, and other appurtenances such as intermediate bearings and gear reducers. A complete analysis, including finite element stress analyses for fatigue strength, usually costs in the neighborhood of \$20,000 (in 2005 dollars) per set of like pumps. One consulting firm has found that 80% of the pumps submitted failed to meet these requirements and needed some kind of remediation. In all instances, the changes resulted in improved performance and exceptionally smooth-operating equipment. See Section 1.05.B.3 in Appendix C for more details.

Regardless of whether a structural vibration and rotordynamic analysis is performed on a pumping station machine and drive train, acceptance should be based upon the concurrence with the specifications of the results of a vibration test performed on-site in the final, fully assembled and running condition. If the vibration is found to be excessive, the manufacturer should be required either to fix the problem or prove that the problem is not related to the equipment. If the latter is true, a knowledgeable consultant

should be retained to ensure that the manufacturer is correct and to help find a fix to the problem.

Lateral Rotordynamic Analysis

The minimum rotordynamic analysis should consist of the determination of rotor critical speeds (i.e., non-critically damped rotor natural frequencies) and mode shapes from a frequency of zero up to the maximum excitation frequency. If the bearing stiffness has a large tolerance or might shift significantly with time, such as from the Lomakin effect (see subsection “Description of Basic Vibration Terms, Concepts, and Equipment” later in this chapter) in centrifugal pumps (as the clearances increase due to wear in normal service), a critical speed map should be constructed in which the value of these frequencies versus the range of possible bearing stiffness values is plotted. The predicted critical speeds need to be compared with the excitation frequencies to determine whether a resonance is possible. If so, the offending natural frequency must be moved or sufficiently damped by, for example, using a different style of bearing or coupling.

If any of the rotor natural frequencies might (with reasonable allowance for system degradation) match the frequency of an excitation force within the intended speed range, a “forced response” analysis should be performed. Sometimes this analysis is done only for one time running speed excitation. It is then called an “unbalance response” analysis. In such an analysis, it is assumed that the rotor is driven by a worst-case excitation and the amount of vibration that can occur at the various critical locations along the rotor (e.g., at bearings, wear rings, and mechanical seals) is calculated. These results can be compared with available clearances to determine whether rubbing is likely and whether the bearings might become overloaded.

Vertical turbine pumps exhibit nonlinear shaft dynamics because of the large shaft excursions that occur in the lightly loaded long length/diameter ratio bearings. Given the flexibility of the lineshaft and the weak support provided by the pump casing column piping, and given the relatively large assembly tolerances and misalignments in the multiple lineshaft bearings of these machines, the contribution of each bearing to the net rotordynamic stiffness is nearly random and is constantly changing, as explained by Marscher [4]. As a result of the changing stiffness, there is no single value for each of the various theoretically predicted natural frequencies. Instead, the natural frequencies of the lineshafting and shaft in the bowl assembly must be considered on a time-averaged and location-averaged

basis and they constantly vary between two limiting states. A useful simplified method of predicting lineshaft reliability with a worst-case model is known as the “jump rope” model, and is documented by Marscher [4]. Another important vibration problem in certain vertical pumps is the vibration of the shaft enclosure tubes that provide a jacket of contaminant-free lubricating fluid to the lineshaft bearings. The natural frequencies of these tubes (supported by “spider” assemblies) can be analyzed for various end supports using finite element analysis or by using the multiply-supported beam models of Blevins [5]. Vertical pump vibration analysis of the stationary structure, the lineshafting, the shaft enclosure tubes, and the pump and motor rotors should be done simultaneously by means of finite element analysis.

The possibility of rotordynamic instability due to lateral motion needs to be analyzed in centrifugal and axial flow machinery, particularly in large fans and certain motors wherein cross-coupling forces (those induced by rotor motion perpendicular to the motion, as discussed in the subsection “Basic Vibration Terms, Concepts, and Equipment” later in this chapter) are often large enough to overwhelm the damping. Rotordynamic instability refers to phenomena whereby the rotor and its system of reactive support forces are able to get “out-of-synch” with each other and become self-excited. Potentially catastrophic vibration levels ensue even if the original excitation forces are quite low. The characteristic of rotordynamic instability, sometimes called “shaft whip,” is that it is whirling at about one-half running speed and begins when running speed exceeds twice the first bending natural frequency of the shaft. Very few pumps have natural frequencies in the running speed range with damping low enough that this process can occur, but motors and fans sometimes do.

If an unstable machine is encountered, a typical design modification to reduce the tendency for rotordynamic instability involves bearing changes. The type of bearing most likely to participate in instability problems is the plain journal bearing, because it has very high cross-coupling (although it also has high beneficial damping). Bearings that discourage whirling lubricant flow tend to decrease cross-coupling—dramatically so in terms of the axially grooved and tilt-pad-style bearings described in the *Pump Handbook* [1].

Torsional Vibration Analysis

Unlike lateral rotordynamic analysis, which usually can be performed independently by various manufac-

turers, torsional analysis is valid only when performed for the entire linked-up drive train. Torsional analysis of only the driver or the pump or compressor alone is without value. Also, the flexibility, backlash, and gear ratio of couplings and gear assemblies, if any, must be estimated and included in any accurate model. Methods of manually calculating the first several torsional natural frequencies for simple rotor systems are given in Blevins [5], but a finite element analysis is generally needed for complicated pumping station systems. A detailed description of how such analyses should be performed (and an excellent basis for a specification on the conduct of such analyses) is given by Corbo and Malanoski [6].

Forced response torsional analysis is necessary if any of the torsional natural frequencies coincide with a strong excitation frequency. To determine the frequencies at which large values of vibratory excitation torque are expected and the value of the torque occurring at each of these frequencies, the pump torque at any given speed and capacity can be multiplied by a “per unit factor.” The per unit factor at important frequencies can be obtained from pump, compressor, motor, and control manufacturers for a specific system, and it is typically in the range of 0.01 to 0.1 zero-to-peak for important excitations. The most important torsional excitation frequency from an electric motor is the motor rotating speed times the number of motor poles. Unsteady hydraulic torque from the pump is also present at a frequency equal to the running speed times the number of impeller vanes, and the maximum intensity equals the delivery torque divided by the number of impeller vanes, although typically the per unit factor is 0.03 to 0.1. The gear mesh frequency (number of teeth times the rotational speed for a given gear) is usually a strong torsional excitation frequency with a typical per unit factor of about 0.02.

The worst-case torsional vibrations in pumping station rotors often occur due to temporary excitations during start-up, trip, and motor control transients. Therefore, it is wise to make a time-integration analysis to determine the transient peak stresses caused by these transients.

Particular care should be taken with systems involving adjustable-frequency drives (AFDs). Besides sweeping the excitation frequencies through a large excursion and therefore increasing the chances of a resonant encounter (Marscher [7]), AFD controllers provide new control pulse excitations at multiples of the motor running speed, commonly at 6 $\times$, 12 $\times$, and 18 $\times$, and often at whole-fraction submultiples as well. The controls manufacturer can predict these frequencies and their associated per unit factors.

Judgment of the acceptability of the assembly's torsional vibration characteristics should be based on whether the forced response shaft stresses are below the fatigue limit by a sufficient factor of safety at all operating conditions. The recommended factor of safety is 3.0 if all stress concentrations (such as keyways) are taken into account. If specific stress concentration factors are not known, assume that they have a value of 3.0 and use a revised factor of safety of 9.0 at any shaft location that has a step, attached part, thread, or keyway.

Vibration Specifications

Specifications concerning acceptable vibration levels, the method and frequency range of measurement, and the method of interpretation of the results should be clear and reasonable. Vibration-minimization responsibilities, such as who will do the pre-design analysis and who will do the post-installation testing, must be delineated and the responsible parties must be identified. There should be a list of items required, such as results from certain types of validated computer programs or testing procedures. Responsibility for each item should be assigned in the bid request so that bids are on an equal basis. The inclination or ability of low-bid suppliers to provide an appropriate level of technology should be specifically required—not assumed—in the bid and contract.

Injury to machines due to excessive rotor vibration consists of wear or fatigue damage to the pump internal components, such as bearings, annular seals, mechanical seals, and shafts. Most of this damage depends on the total displacement associated with the vibration. However, as machines are made faster, they become smaller, and hence the amount of vibration displacement they can tolerate decreases proportionally to the machine speed. Therefore, the allowable running speed vibration velocity (displacement times running speed) is roughly constant regardless of the running speed of the machine. Historically, machine survival versus failure data support the use of constant vibration velocity versus speed as an acceptance criterion in assessing vibration severity (see Rathbone [8], Blake [9], Baxter [10], and Hancock [11]). However, the raw information in these references was based on measurement equipment that could not distinguish between various frequency components. Therefore, vibration severity could be plotted only as unfiltered (total vibration including all frequencies) displacement readings versus machine running speed and not filtered (individual values at specific frequencies) velocity values

versus frequency. Unfortunately, in many specifications (with the notable exception of ANSI/HI 3.1–3.5-2000 and ANSI/HI 6.1–6.5-2000), it is assumed that the original data can be interpreted as velocity versus frequency, and the specifications are written on that basis. Be very cautious in using such specifications because instability and hydraulic problems that cause rubbing at low frequencies tend to be overlooked, and the specifications may require unnecessarily small vibration displacements at high frequencies. Such specifications cause the rejection of good equipment at the expense of all involved, as discussed by Marscher [12].

Typical specifications for pumping stations to establish vibration test types, measurement locations, and acceptance criteria are contained in ANSI/HI 9.6.4-2000, API 610, Mil-STD-167-1 (Ships), and various ISO machinery vibration specifications, such as ISO 2372. These specifications require the monitoring of bearing housing vibration by using an accelerometer or velocity probe in three perpendicular directions at the location of each of the bearings in the drive train. The acceleration readings are typically integrated to obtain vibration displacement and/or velocity values—the typical terms for the criteria. Occasionally, the specifications contain procedures for the installation and evaluation of shaft proximity probes mounted in some bearing housings to measure shaft versus housing displacement.

Acceptability criteria vary among the various specifications. A conservative approach, consistent with all of the quoted specifications, is to require that vibration does not exceed any of these three criteria at any frequency: 0.05 mm (2.0 mils) displacement peak-to-peak, 6 mm/s (0.25 in./s) zero-to-peak (or just peak) velocity, and 1.0 *g* peak acceleration.

In essence, vibration velocity can be thought of as displacement times the frequency at which it occurs times a scaling constant. Likewise, acceleration can be thought of as displacement times frequency squared, times a scaling constant. The value of scaling constants is not a key to the discussion here, although knowledge that the above relationships exist brings out the important point that vibration displacement, velocity, and acceleration are all measurements of the same quantity, but with different emphasis given to vibration that occurs in different frequency ranges. At low frequencies, for example, displacement is the most stringent criterion, while at high frequencies acceleration is the most stringent. Depending on the type of equipment, displacement may be more likely to reject unreliable machines and pass reliable machines than acceleration would be. On the other hand, for machines prone to problems

that show up at high frequencies (usually not true of pumping station equipment), the opposite would be true. Vibration velocity is used in many specifications as a “middle-of-the-road” single acceptance criterion, but this practice is not recommended because it tends to be blind to potential rums at low frequencies and is too stringent as a general criterion for most machinery at high frequencies. In practice, the 1.0 g peak acceleration specification given above is less stringent than the displacement and velocity specifications except for high frequencies. Generally speaking, high acceleration measurement is useful only as a flag that there is something unusual (possibly harmless) in the system, and by itself does not necessarily indicate that there is a problem with the machine.

For vertical pumps, the maximum allowable displacement may be significantly higher than 0.05 mm (2 mils) due to the geometrical lever principle acting to magnify the motion associated with a tall motor or pump bearing tower (ANSI/HI 9.6.4-2000).

Piping vibration is in a special class. Pipes can be allowed to vibrate up to roughly three times that allowed for the machine to which it is attached (see Wachel and Bates [13]).

22-3. Troubleshooting Excessive Vibration

Vibration problems may sometimes occur in existing stations, in new stations because of corners cut during the design process, or in new stations in spite of following recommended practice. These problems can often be solved by a combination of inspection and common sense. If the problems persist, then vibration experts with appropriate vibration measurement equipment can be called in. Make sure that these experts focus on solving the problem at hand instead of just gathering data. Prior to calling in consultants, the following procedures can serve as a guide, but be prepared to modify them to fit the particulars of your situation.

Basic Troubleshooting Procedures

Obtain a history of the problem: Is it recent or has it always occurred? Has the system or its operation recently changed? If the problem is recent in an existing installation, it is likely to be caused by looseness, clogging, erosion, or wear. Pump disassembly may be required for answers, but even so, delay for the moment. Before dismantling the pump for inspection, follow the procedures given below as far as conditions permit, and answer the following questions:

(1) Does the problem relate to flow, the sump level, or some other circumstance? (2) Is the vibration constant or intermittent? (3) Have components been failing? (4) Which locations shake beyond the specifications, and which do not? (5) Is there a pattern to the locations that shake and/or make noise? A list of checking procedures follows, more or less in the order to be performed.

- A common fault is a pump operating off the BEP. Install a calibrated pressure gauge on the discharge and another on the suction pipe. Read static pressures and total dynamic pressures when the pump(s) is (are) running. It is helpful to obtain the flow rate (see Section 3-9). Compare with the manufacturer's curve for the impeller. Check for a worn impeller.
- Check for loose or broken connections, and faulty or improperly set valves. Check all bearing housing, foundation, and piping bolts, gasketed joints, and isolation pads for proper installation and tightness.
- Check for pump baseplate “soft foot” (bottoms of all feet not coplanar) by following the procedures on page 2.288 in the *Pump Handbook* [1].
- Inspect all baseplates and foundation blocks for significant cracking that might decrease machine support stiffness.
- Have a millwright check driver/pump shaft alignment.
- Check the shafts for straightness (e.g., by rotating the shaft by hand while a dial indicator is mechanically held against it).
- Listen near the pump inlet for the crackling sound associated with cavitation. Crackling noises heard near the pump suction suggest inlet cavitation due to insufficient NPSHA, severe impeller vane stalling, internal recirculation due to swirling or skewed flow at the inlet, operation too far from BEP, or submerged vortices in the sump. If any noise is detected, place a calibrated pressure gauge of the proper range close to the pump suction. Ensure that there is sufficient NPSHA by noting static pressure at a tap at the pump inlet flange, and compare it with the manufacturer's NPSHR versus capacity curve. Is NPSHA adequate? If not, perhaps the suction line is clogged or there is a sump problem. If, in spite of inlet crackling noise, NPSHA is adequate, recirculation or sump problems are suggested. Other clues are: (1) inlet cavitation noise is relatively constant, whereas (2) recirculation noise tends to be throbbing, and (3) sump noise tends to come and go without following any pattern.

- Place a calibrated pressure gauge of the proper range at the discharge, preferably about three to five diameters downstream. Use a certified pump system head/capacity curve or a flowmeter in the discharge line to determine flow rate. For variable-speed systems, use a tachometer or strobe to determine pump speed. At a given speed and flow, compare the discharge head minus the suction head with the value obtained from the last test performed, and to the manufacturer's head/capacity curve. If the TDH is larger than expected and/or flow is lower than expected, some form of discharge blockage is possible. If the TDH is lower than expected, wear of the wear rings or erosion of the impeller vanes or volute tongues is possible. If the TDH indicates that the operating point lies outside the range of 75 to 110% of BEP, especially after years of service, the impeller or the volute tongue may be excessively worn. If the station is new or if the vibration has always occurred, the designer may have used an unrealistic roughness coefficient for the piping.
- Ensure that expansion joints and flexible joints such as Dresser[®] couplings are constrained properly, especially if the pressure is high.
- What are the bearing shell or lubricant temperatures, at least approximately? Bearing lubricant temperatures are typically 60 to 82°C (140 to 180°F), and the shell and housing temperatures are usually 11°C (20°F) lower. The loss of oil viscosity at elevated temperatures can reduce bearing support film thickness and allow scuffing and excessive bearing wear to occur, and thermal growth mismatch between rotating and stationary parts may allow radial or axial rubbing. Excessive housing temperatures indicate severe pump/driver misalignment, need for lubricant addition or change, or (contrary to intuition) excessive grease in a packed bearing cavity.
- Are any wear particles or pumpage contamination visible in samples taken from the lubricant? The most common cause of bearing failure in pumps is lubricant contamination—usually by water passing a leaky seal or packing. Condensation from humid air onto the cool walls of an oil sump or inside a grease-packed bearing with badly worn lip seals that allow the free ingress of air is often overlooked. Sometimes the contamination is varnish, carbon deposits, or metal flakes from bearing surface fatigue. If the lubricant change interval has passed or if the bearing was overloaded or overheated for even a few minutes, the effect snowballs to more degradation.
- Following disassembly, if rubbing was involved, was it on just one side of the rotating or stationary

components, or full circumference on each? Does corrosion appear to be a factor? If fracture occurred, are fatigue striations evident? If so, what does a high-power microscope indicate concerning how many cycles it took from crack initiation to failure? Such information is useful in determining whether the problem was some sort of brief transient event or is inherent in the pump or system. If fatigue occurred in relatively few cycles, there was a severe overload condition such as recirculation caused by running the pump below its recommended minimum continuous flow rate. Millions of cycles indicate gradual degradation, such as gradually worsening rotor imbalance or pump/driver misalignment. Microscopy performed by a qualified metallurgist can often determine the root cause of failure so that it can be avoided in the future.

- Check the balance and fit of the disassembled pump impeller and coupling hub. Poor fit can be remedied by reaming the bore and installing a sleeve of proper fit or, sometimes, by chrome plating or knurling the shaft followed by grinding to the proper dimensions. The impeller/shaft assembly should then be rebalanced.

More Detailed Test Procedures

If the above approach does not lead to a solution of the problem, detailed vibration testing may be required. In the most common type, “signature analysis,” an accelerometer output is sent to a fast Fourier transform (FFT) analyzer to document the amount of vibration at each frequency within a tested range. Typically, this range is from several Hz to beyond the pump vane pass frequency. The frequencies at which most of the vibration is occurring and the locations where the vibration is the greatest are used as clues to determine the cause of the vibration.

Vibration testing often ends here. However, it is recommended that vibration testing includes experimental modal analysis (EMA). EMA involves artificially exciting a machine or structure (e.g., with an impact hammer), preferably while the machinery is running so that all bearing stiffnesses are representative (see Marscher [14]). The purpose of EMA is to determine the natural frequencies of a pump (or other machines), its rotor system, and the attached system components. These frequencies can be compared with excitation frequencies to ensure that resonance will not occur. They can also be used to confirm that all equipment and supporting structures have adequate separation margins between the excitation frequencies and natural frequencies.

Recognizing Vibration Problems

The great majority of pump vibration problems can be solved by: (1) re-balancing the rotor assembly, (2) alignment of the couplings when the system is at its rated conditions—especially if it is hot (see Dodd [15]), and/or (3) running the pump within the bounds of its specified head versus capacity curve. Remaining vibration problems are usually caused by a resonance of a pump internal natural frequency or a systemwide natural frequency. During resonance, the rotor vibrations can exceed internal clearances, and excessive bearing loads can occur, even if loads such as imbalance are within normally acceptable limits.

In performing vibration troubleshooting, generalized charts (such as those in the *Pump Handbook* [1]) for matching symptoms to possible causes can be useful for many typical or simple problems. However, do not rely too heavily on such lists, especially if their initial application does not lead to an immediate resolution of the problem. Persistent pump vibration problems are usually due to an unexpected combination of factors, some of which are specific to the particular pumping system, such as mechanical or acoustical piping resonances, or hot-running misalignment of the pump/driver due to thermal distortions of the piping or baseplate. With this warning, a list of some of the more common pump vibration symptoms and their causes is given in Table 22-1. The following predictive maintenance and troubleshooting list can be used for interpreting the reason for many common vibrations and therefore could be the kernel of any future predictive maintenance software. The list is *not* meant to be all-inclusive and is in the order of the observed frequency—not in the order of likelihood or importance for reliability.

Typical Fixes for Vibration Problems

Although many pump vibration problems are caused by operation off the BEP, most machinery vibration problems are traceable to excessive imbalance or misalignment. Determination of the reason for the imbalance or misalignment (and its subsequent elimination) is the cure in these cases. Most vibration problems that persist after proper balance and alignment, however, are related to some form of resonance.

When resonance occurs, simple trial-and-error field fixes, such as addition of gussets at appropriate locations for stiffening, can often be effective. Such “fixes” may, however, only shift the problem to a slightly different frequency. If “cut-and-try” methods

cannot be made effective within a reasonable time span, it is usually cost effective to bring in a vibration expert from either the manufacturer or from a specialty consulting firm. The main priority in selection should be competence and experience. Fee differential is unlikely to compensate for your own personnel time, the cost of hardware changes, and lost operating time wasted by an inexperienced consultant.

In solving resonance problems, vibration experts may employ one of the following:

- Change the stiffness of the resonant component and/or its support (possibly including the floor). An increase in stiffness raises the natural frequency—possibly enough to drive it out of the resonance range. However, it is possible that the increase leaves the natural frequency still within the operating speed range but now excited by larger forces at the top of the speed range. It is also possible that a lower natural frequency, not previously a problem, can be moved into the operating speed range. For these reasons, it might be better to decrease rather than increase stiffness to drop the natural frequency below the operating speed range. Drawbacks to this latter approach are: (1) a higher natural frequency that was not previously resonant might drop into the operating speed range, and (2) decreasing stiffness may make static forces (such as weight) and alignment more difficult to manage.
- Add weight to locations of the largest vibratory motion, thus decreasing the problem natural frequency. Measurements need to be taken to confirm whether the new frequency is out of the operating speed range. Potential drawbacks are greater sensitivity of the revised structure to seismic loads and the possibility of reducing a previously higher natural frequency into the operating speed range.
- Install vibration isolators between the vibration excitation source and the problem vibration location by adding flexibility between the two. Be careful to ensure that the vibration is sufficiently diminished in all potential problem locations and that the isolation does not hide an excessive excitation force that could cause other problems if untreated. The use of isolators is not often practical.
- Design and attach a dynamic absorber (made of a dead weight and a suspending rod) to a point of maximum motion for the problem natural frequency’s mode shape. The mass and adjusted rod stiffness is chosen so that the mass/rod reed frequency equals the problem natural frequency. If the system is tuned just right, the motions of both natural frequency/mode shapes cancel each other at

Table 22-1. Troubleshooting Tips for Identifying the Source of Vibration Problems

Frequency of vibration (in terms of multiples of running speed)	Other symptoms	Probable causes
0.05× to 0.3× (may be broadband)	Vibration response is unsteady and fluctuates somewhat in frequency.	Diffuser or volute wall stationary passage stall.
Exactly $\frac{1}{2}\times$, $\frac{1}{3}\times$, or $\frac{1}{4}\times$	Vibration may increase dramatically shortly after the above (0.05× to 0.3×) appears, except in vertical pumps with four or more lineshaft bearings, where it is common, and generally not harmful.	Light rub or unexpected looseness combined with shaft or bearing support natural frequency.
0.42× to 0.48×	Near shaft natural frequency, and orbit “pulses,” forming an inside loop. Vibration onset is sudden at a speed roughly twice the excited natural frequency, and locks onto the natural frequency in spite of speed increase.	Rotodynamic instability due to fluid whirl in close clearance, e.g., “shaft whip.”
0.6× to 0.93×	Smaller peaks at $f\times$ and at $\pm(1 - f)\times$ “sidebands” of the first several multiples of running speed where f is the frequency of the amplitude modulation caused by pressure pulsations. Amplitude modulation is manifested as a beat frequency f . Often accompanied by rumbling noise and beating. Occurs at capacities below BEP, but improves at very low capacity. Independently, depends on speed and flow.	Internal flow recirculation in the impeller, probably at its suction, but possibly at the discharge.
Less than 1×	Increased broadband vibration and noise level below running speed as NPSHA decreases, especially at high flows. Often accompanied by decreased vibration and noise above 1×.	Cavitation without recirculation.
1×	(1) Stronger on shaft than on housing. Hydraulic performance and/or suction pressure normal. Axial vibrations within normal limits and vibrations above runout increase roughly with speed squared. (a) Vibration highest on drive and pump IB (inboard) housing. (b) Vibration highest on driver IB housing. (c) Vibration high on machine inboard (IB) or outboard (OB) housing, low on driver. (d) Natural frequency, 1×. (2) Axial vibration is over $\frac{1}{2}$ of radial vibration, or/and vibration increases much slower than the square of the speed. Also, bearing temperature is high. (3) Discharge pressure pulsations are strong at 1× but not at impeller vane pass in a single volute machine. (4) Same as (3), but vane pass also strong, especially at flows far above or below the design point.	Imbalance in rotating assembly. Coupling imbalance. Driver rotor imbalance. Machine imbalance. Resonance. Machine/driver misalignment at the coupling. Clogged or damaged impeller passage. Volute tongue designed too close to impeller OD, or clogged or damaged volute, or excess impeller eccentricity.
2×	(1) Axial vibrations are low. (a) Both shaft and housing vibrations are strong, and discharge pressure pulses are strong, but impeller vane pass vibrations are low in a twin volute pump.	Clogged or damaged impeller passage.

Table 22-1. (Continued)

Frequency of vibration (in terms of multiples of running speed)	Other symptoms	Probable causes
2× (Continued)	(b) Same as (a), but with unusually high vane pass vibration and discharge pulsations. (c) Shaft vibrations stronger than housing vibrations, maybe exceeding clearance. Decrease in shaft first natural frequency. Multiples of running speed may be strong. (d) Vibration of one bearing housing high, shaft may be quiet. (e) Shaft vibrations stronger than housing. (f) Motor driver frequency equals electrical line frequency and highest vibration on motor. (2) Axial vibrations are over $\frac{1}{2}$ of horizontal, 1× vibrations are also high, and bearing oil temperature is high.	Volute vanes designed too close to impeller OD, clogged or damaged volute, or excessive impeller/volute eccentricity. Looseness in bearing retainer, or cracked shaft. Loose or cracked bearing housing. Torsional excitation. Electrical problem with motor. Pump/driver misalignment at the coupling.
Number of impeller vanes × running speed	(1) Discharge pressure pulsations reasonably high and both shaft and housing vibrations high. (2) Same as (1), but housing vibrations much higher than shaft vibrations. (3) Discharge pressure pulsations high at vane pass frequency but suction pressure pulsations reasonably low. (4) Suction pressure pulsations high. (5) Rotor or casing natural frequency close to vane pass.	Volute too close to impeller OD due to design or excessive rotor eccentricity. Piping mechanical resonance at vane pass or loose bearing housing. Acoustic resonance in discharge pipe. Acoustice resonance in suction pipe. Resonance.
Several multiples of running speed, including 1×, 2×, 3×, 4×, and possibly higher	(1) Orbit shows sharp angles or shows evidence of “ringing,” and/or spectrum shows exactly $\frac{1}{2}$ or $\frac{1}{3}$ ×. Grinding noises may occur. (2) Orbit is fuzzy but does not pulse or ring. Seal coolant flow is unexpectedly high or low, and may exhibit high temperature. Spectrum may also exhibit $\frac{1}{2}$ or $\frac{1}{3}$ ×. (3) Orbit pulses, usually in one direction much more than the other and shaft vibrates more than the housing. May also observe at exactly 0.5×, 1.5×, 2.5×, etc. (4) Shaft orbit fairly steady, and housing vibrates more than shaft. Often combined vibration over a broad range of low frequencies.	Internal rub or poorly lubricated gear coupling. Jammed, clogged, or damaged seal. Shaft support looseness, especially in bearing insert or cap retention. Looseness in casing, pedestal, or foundation or bearing housing.
Non-integer multiples of running speed	(1) Vibrations stronger at problem frequency at certain points in the piping or on the foundation than on the pump. (2) Vibrations high in piping but not foundation. Suction or discharge pressure pulses as strong or stronger relative to background of spectrum than the vibration is in the piping or the machine. (3) Vibrations low in piping and foundation at a fluid machine natural frequency.	System structural resonance. Hydraulic piping dynamics. Machine resonance, excited by turbulence.

the running speed and dramatically reduce the vibration level at the absorber attachment point. The disadvantages of this approach are: (1) it can be unsightly, (2) if the absorber mass is not large enough it can be difficult to maintain proper adjustment, and (3) the concept works only for constant-speed applications. The use of dynamic absorbers is tricky and best left to vibration experts.

Vibration problems that are not related to imbalance, misalignment, or resonance are usually caused by excessive hydraulic forces. If the cause is cavitation from insufficient NPSHA, the NPSHA should be increased to the manufacturer's requirement or even more. If vibration is caused by excessive vane pass interaction, then the impeller/stator vane gap may be too small, and the gap should be opened to about 10% of the impeller diameter (if acceptable to the manufacturer and if the required flow rate and head can be met). If a centrifugal pump is being run at relatively low flow, suction or discharge recirculation may be at fault—possibly with accompanying rotating stall. If the flow cannot be increased, then a bypass line can be placed between the pump discharge and a location in the suction plenum or suction line at least five pipe diameters upstream from the pump suction flange. A valve can be placed on the bypass line, and bypass flow can be increased gradually until the vibration problem ceases. Alternatively, a new impeller less sensitive to low flow can be installed. Although efficiency may decrease several percent, the operating cost is usually less than it is with a bypass line.

Description of Basic Vibration Terms, Concepts, and Equipment

Vibration analysis of machinery and its piping and foundations may be divided into two parts:

- *Structural dynamics*—the vibration of stationary components such as housings, piping, and supporting structures. An important factor in considering structural vibrations is that the amount of vibration at the point of interest may be strongly related to other system components far removed.
- *Rotordynamics*—the vibration of the rotor assembly. The rotor vibration is influenced by all factors relevant to structural vibration but, in addition, may be strongly affected by unexpected reaction forces and gyroscopic effects.

Concepts of key importance for both structures and rotordynamics in all types of rotating machinery are defined and explained below:

- *Excitation forces* supply the energy in a vibrating system. They include imbalance, at a frequency of one times running speed and misalignment, usually at both one and two times running speed. In machinery with vanes or lobes, vane-passing pulsations occur at the number of impeller vanes times running speed. In centrifugal pumps, the level of vane-passing pulsations (and the vibrations they cause) are dependent on the ratio of the capacity at the BEP to the actual pump flow rate. As the departure from BEP increases, the flow path directions inside the pump become less well matched with the impeller vane and volute tongue angles. Running a centrifugal machine at flows below its rated capacity at a given speed causes increased stress and vibration, and decreased machine life and reliability as dramatically stated by Agostinelli et al. [16]. In contrast, the pulsations of screw and reciprocating machines usually increase with load. Strong pulsation frequencies in screw pumps occur at one and two times the number of screw lobes, whereas in reciprocating machines, strong pulsations occur up to four or more times the plunger or cylinder stroke frequency (a value equal to the number of cylinders times running speed).
- *Natural frequencies* are those frequencies at which mechanical systems vibrate freely after energy is imparted, such as after striking a tuning fork. Machinery systems have not only one, but an infinite number of natural frequencies. Only natural frequencies close in frequency to strong excitation forces are of practical significance and merit the designer's or user's attention.
- *Mode shapes* are the vibrating movements associated with a given natural frequency. Each natural frequency has a unique mode shape. The mode shape becomes more complicated as the natural frequency becomes higher. At lower frequencies, simple mode shapes are most likely to possess motion consistent with the action of exciting forces, and these are the most troublesome to the plant designers and maintenance engineers. The lowest natural frequency of vertical pumps and motors resembles a reed blowing in the wind and is, therefore, called the "reed frequency" (ANSI/HI 9.6.4-2000). The second natural frequency has an initially similar motion that starts at the base in one direction, but then it turns and finishes with reed motion at the top in the opposite direction. The lowest natural frequency in horizontal pumps is usually a swaying back-and-forth motion of the pump casing allowed through flexure of its support base, and a second natural frequency is a see-saw motion of the pump casing (as viewed from above).

Both sets of these vertical and horizontal pump mode shapes and natural frequencies are often strongly influenced by (1) the stiffness and/or mass of the foundation and building floor, and (2) the added mass of rigidly attached piping. Hence, the floor and piping should be considered together with the pump as a total, connected system. Remember these statements during the design process and during troubleshooting.

- *Damping* is the absorption of vibration energy. The strongest damping typically occurs in components where there are tight fluid-filled clearances such as in journal bearings and impeller wear rings. Handbook charts and readily available computer programs exist to estimate and optimize the damping of bearings and seals.
- *Resonance* occurs when the frequency of an excitation force equals the natural frequency. Resonance stores vibration energy from cycle to cycle. Very large vibrations build up during resonance even if the excitation forces are reasonably low—especially if the force is applied near a point of maximum vibration in the natural frequency’s mode shape and if the damping is small. The affected machine is then observed to be “balance-sensitive” or “alignment-sensitive,” because the resulting high vibration can be temporarily eliminated by very fine balancing or alignment. However, as these vibrations degrade slightly after brief operation, the vibration problem usually returns.
- *Amplification factor* refers to how much more (or less) a pump or system component moves during vibration than it would move if the force causing the vibration were static instead of oscillating. If the excitation frequency is well below the system’s first natural frequency (<75%), the amplification factor is about 1—that is, the structure’s maximum motion in response to the force is essentially the same as if the slowly oscillating force were static. Systems responding as though to a static force are known as “stiffness-dominated” or “rigid structures,” and they are unlikely to have balance-related resonances (by far the most common type of resonance problem). At the other extreme are systems with important excitation frequencies well above (by at least 25%) those natural frequencies that have simple, easily excited mode shapes. These are known as “mass-dominated” or “flexible structures,” and the amplification factor for them is less than 1. Hence, there is even less vibration motion than would be expected if the excitation force were applied statically. Although amplification factors less than unity sounds like an ideal situation, the high degree of flexibility implied in such systems makes them very prone to distortion from static forces and may lead to serious misalignment between the driver and the pump. The distortion may be accompanied by internal rubs and possible large dynamic misalignment in spite of the low amplification factor. Between the rigid and flexible structures are “resonant” structures, in which the excitation force frequency and natural frequency are close to each other—a situation to be avoided because of the high amplification factor and consequent high vibration levels.
- *Parametric resonances* refer to large nonlinear vibrations that can occur in response to looseness or rubbing. Such resonances are typically caused by bearing support looseness or a rub at a bearing, seal (or other close running clearance), or slip in a gear or spline coupling. The symptoms are a pulsating orbit, with a large amount of vibration at exact whole fractions of running speed, such as $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, etc. In addition to the above, rotor systems may be strongly affected by more subtle phenomena. The most important of these phenomena are listed below.
- *Gyroscopics* describes the reluctance, due to conservation of angular momentum, of a rotating object of large radius (such as an impeller) from changing the direction of its axis.
- *Cross-coupling* is a force that develops perpendicularly to rotor motion when the rotor moves to an eccentric position within its bore clearances. It is caused by the higher pressure of the flow dammed up upstream from the minimum clearance pinch point, and it occurs in combination with the restoring bearing force that acts to support the rotor opposite in direction to the motion. The non-intuitive cross-coupling force acts in the opposite direction of the damping vector and can become an unexpected factor to cause decreased energy dissipation due to the vibration—in some circumstances to the point that the vibration becomes self-excited and unstable. After a certain threshold speed is passed, the self-excitement produces “rotor dynamic instability,” in which the rotor orbit takes up the entire clearance, whirls at about half rotor speed, and rapidly wears out the machine. Rotor dynamic instability is extremely destructive, but, fortunately, it is not common in the type of equipment used in pumping stations.
- *Added mass* refers to the effective inertia of the fluid surrounding the rotor. It is developed from three sources: (1) the fluid trapped in the impeller passages adds mass directly; (2) the fluid displaced by the presence of the impellers and shaft adds its

mass because, as the rotor vibrates within the fluid, it must displace this mass; and (3) the fluid in close clearances must accelerate off to the sides much faster than the rotor vibration acceleration to make room for the rotor motion. Item (3) can, potentially, add mass many times its displaced mass.

- *Lomakin effect* is an unexpected support force that occurs in pumps at annular seals such as wear rings due to the action of Bernoulli's effect during the normal leakage process. This effect, discussed in detail by Black [17] and Marscher [18], can change the rotor support stiffness dramatically and hence the rotor natural frequencies. The effect thereby either avoids or induces possible resonance between strong forcing frequencies at one and two times the running speed and at one of the lower natural frequencies. Although the effect is usually beneficial, it strongly depends on the annular seal diametral clearance, and loss of this clearance due to erosion and wear in service can lead to loss of Lomakin effect and the appearance of a resonance problem where there was none before.

Vibration Measuring Equipment

Modern vibration test equipment can be divided into three levels of detail and sophistication:

- *Hand-held sensor/meter packages.* The advantages of this equipment are that it is easily portable, easy to use, inexpensive, and the sensor is relatively lightweight and rugged. The disadvantages are (1) the frequency information it provides is not very distinct; (2) it is difficult to distinguish hydraulic forces and mechanical resonances from excitation harmonics of multiples of running speed caused by imbalance and misalignment; (3) only one probe at a time can be used, so the powerful modal testing discussed by Marscher [14] cannot be performed; and (4) the velocity probes used in these units rapidly lose accuracy outside of the range 10 to 17 Hz.
- *Single-channel FFT or real-time analyzers.* These devices are rather complicated and best used by experts. Velocity probes can be used with them, but accelerometers are better because they (1) have a broader accurate range of frequency (2 to 17,000 Hz), and (2) are robust. FFTs may also take input from eddy-current shaft proximity probes that sense the displacement of a rotating shaft relative to some stationary point on the pump

housing. The advantages of the single-channel FFT are (1) it is moderate in cost; (2) it gives accurate plots of running vibrations versus frequency (thereby allowing the broader frequency range taken up by resonances and hydraulic forces to be detected relative to the narrow frequency range vibration response of mechanical forcing frequencies); and (3) it provides enough frequency resolution to distinguish subtle but important frequency differences between resonances, hydraulic forces, and mechanical forces. An important disadvantage of single-channel FFTs is that experimental model analysis (EMA) testing and recording of shaft position versus time "orbits" cannot be obtained. Another disadvantage is that the timing relationship or frequency correlation between pressure pulsations or vibrations in a possible "problem source" area and vibrations in the problem "symptom area" cannot be obtained. These procedures often play an important role in vibration problem solving.

- *Multi-channel FFT or real-time analyzers.* As with single-channel FFTs, these are also based on the fast Fourier transform. They are heavier (7 to 36 kg or 15 to 80 lb) and cost more than single-channel units, but they allow measurements in two or more channels at once, so that the EMA, orbit, and correlation measurements mentioned above can be performed. As with single-channel FFTs, any type of voltage-output transducer can be used, including velocity probes. The more accurate accelerometers and displacement proximity probes are, however, preferred.

Data Types and Formats

Vibration test data are usually plotted in three different forms:

- A Cartesian plot of vibration amplitude versus frequency ("signature plot" or "spectrum"). Sometimes this plot may be combined with a plot of the phase angle (the "lag angle" from application of the exciting force to the responding vibratory motion) versus frequency. It is then called a "Bode plot."
- A Cartesian plot of vibration amplitude versus time, similar to a typical oscilloscope trace ("time" plot).
- A polar plot of vibration versus time in a plane perpendicular to the shaft axis (an "orbit").

The amplitude scales may be linear or dB (base 10 logarithm). A dB scale is often used to improve the resolution of natural frequency peaks. The amplitude

scale generally represents RMS (root mean square) values unless specifically scaled otherwise. To convert RMS values at a specific frequency to peak-to-peak, multiply by 2.83, and to convert from peak-to-peak to zero-to-peak, divide by 2.

22-4. Introduction to Vibration and Noise Calculations

The goals of the following 11 sections are to provide for a deeper understanding and appreciation of vi-

bration, to present means for evaluating a particular situation, and to offer suggestions for reducing vibration or noise to an acceptable level.

The calculation procedures presented here are generally simplified expressions that are meant only to give an idea of the severity of a given situation. If accurate predictions of noise and vibration levels are required, a qualified specialist should be consulted. It may not be easy to find qualified specialists, but a good place to start may be either the National Council of Acoustical Consultants, the Institute of Noise Control Engineers, or the Vibration Institute.

Nomenclature

Many of the following symbols are unique to this chapter and, hence, are omitted from Chapter 2.

Symbols

a	Acceleration amplitude of a vibrating body [m/s^2 (in./s^2)].
B	Bulk modulus of fluid [Pa (lb/in.^2)].
c	Speed of sound in air or speed of wave propagation in fluids or solids [m/s (ft/s)].
cm	Centimeters
d	Static deflection of vibration isolator [mm (in.)].
dB	Decibel.
dBA	A-weighted decibel.
D	Diameter of pipe or resonator [mm (in.)] (see ID and OD).
D_n	Noise dose (dimensionless).
e	Wall thickness of pipe [mm (in.)].
e_o	Radius of motion of eccentric mass [mm (in.)].
E	Modulus of elasticity [Pa (lb/in.^2)].
E_t	Total vibrational energy [$\text{N} \cdot \text{m}$ ($\text{ft} \cdot \text{lb}$)].
f	Frequency (Hz).
f_d	Pump driving frequency (Hz).
f_n	Resonance frequency of a multiple degree-of-freedom system (Hz).
f_o	Natural frequency of a single-degree-of-freedom system (Hz).
F	Dynamic force [N (lb)].
g	Acceleration of gravity [9.81 m/s^2 (386 in./s^2 or 32.17 ft/s^2)].
G	Modulus of elasticity in shear [Pa (lb/in.^2)].
Hz	Hertz (frequency in cycles per second).
I	Moment of inertia [m^4 (in.^4)].

ID	Inside diameter (of a pipe).
J	Mass moment of rotational inertia [$\text{kg} \cdot \text{m}^2$ ($\text{lb}_m \cdot \text{in.}^2$)].
k	Spring stiffness constant [N/m (lb/in.)].
K	Torsional rigidity [$\text{N} \cdot \text{m/rad}$ ($\text{lb} \cdot \text{in./rad}$)].
L	Length of pipe or shaft between support points [m (in.)].
lb	Pounds force.
lb_m	Pounds mass = $\text{lb/g} = \text{lb} \cdot \text{s}^2/32.2 \text{ ft}$.
L_p	Sound pressure level (dB or dBA).
$L_{p(\text{rev})}$	Reverberant field sound pressure level (dB or dBA).
L_{dn}	Day-night sound level (dB).
m	A small mass in kg (lb_m).
m	Mass per unit length [kg/m ($\text{lb}_m/\text{in.}$)].
M	Mass [kg (lb_m)].
NR	Noise reduction (dB).
NRC	Noise reduction coefficient (dimensionless).
NRC_a	NRC of added acoustical material (dimensionless).
OD	Outside diameter (of a pipe).
p	Acoustic pressure (μPa).
P	Static pressure of fluid in pipe [Pa (lb/in.^2)].
R	Radius of gyration [m (in.)].
rad	Radian ($1 \text{ rad} = 57.3 \text{ degrees}$).
RMS	Root mean square.
S	Surface area [m^2 (ft^2)].
S_o	Cross-sectional area [m^2 (in.^2)].
S_a	Surface area of added acoustical material [m^2 (ft^2)].
STC	Sound transmission class (dB).
STC_c	Composite (or effective) STC (dB).
T	Repetition time (period) of a vibrating object.
T	Temperature.
T	Transmissibility—the ratio of transmitted vibrational force with isolators to that without isolators.
v	Velocity amplitude of a vibrating body [m/s (in./s)].
w	Weight of pipe or shaft per unit length [N/m (lb/in.)].
W	Weight of a vibrating body [N (lb)].
x	Displacement amplitude of vibrating body [m (in.)].
γ_n	Coefficient (dimensionless) for bending resonances in piping (the subscript n is the resonance order).
δ	Damping coefficient [kg/s (lb_m/s)].

δ_c	Critical damping coefficient—the minimum damping required to allow a displaced system to return to the rest position without undergoing oscillations [kg/s (lb _m /s)].
δ_r	Damping ratio—the ratio of the damping coefficient to the critical damping coefficient (dimensionless).
ε	Efficiency of vibration isolator (percentage).
λ	Wavelength, m (ft).
μPa	Basic unit of acoustic pressure ($1 \mu\text{Pa} = 1.45 \times 10^{-10} \text{ lb/in.}^2$).
ρ	Fluid density [kg/m ³ (lb _m /ft ³)].
ω	Angular frequency [rad/s ($\omega = 2\pi f$)].

Definitions

A-weighted: A commonly used frequency weighting that closely approximates the frequency response of the human ear.

Absorption: Absorption refers to the conversion of acoustical energy into other forms of energy (usually heat). Sound-absorbing materials are usually rated by the noise reduction coefficient (NRC), a numerical value (usually between 0.1 and 1.0) that approximates the ratio of acoustical energy absorbed to the energy incident upon the material.

Acoustic pressure: The small pressure fluctuations that occur about the static atmospheric pressure (101.3 kPa or 14.7 lb/in.² at sea level) that is sound.

Amplitude: A quantitative descriptor of the level or strength of a noise or vibration signal or waveform.

Attenuation: The reduction in amplitude of a noise or vibration signal.

Blade passage frequency: The frequency associated with the motion of individual blades of a fan or pump. It is the same as the pump frequency for a centrifugal pump.

Critical damping: The minimum damping required to prevent oscillation in a displaced mechanical system.

Critical speed: Any rotational speed of a shaft that excites a resonance.

Damping: The mechanism by which energy is removed from a vibrating system.

Day-night sound level: The energy average of the A-weighted sound pressure level measured over a continuous 24-h period with the sound levels biased upward 10 dB between the hours of 10:00 P.M. and 7:00 A.M.

Decibel: A dimensionless unit used to quantify a relative amplitude of an acoustic or vibration

signal. Decibel is also used as the basic unit for the sound pressure level that is relative to a standard reference pressure of 20 μPa .

Direct field: The region near the source of a sound that is not influenced by the acoustical characteristics of other surroundings (e.g., room boundaries). In this region, the sound pressure level decreases approximately 6 dB with every doubling of distance (valid only for sources that are small in size compared with the distance).

Directivity: Directional radiation properties of a source that may make more noise in one direction than in another.

Displacement: The physical change in the position or angle of a body or particle as measured from its normal rest position.

Driving frequency: The frequency of forced vibration.

Dyne: The unit of force that causes 1 g to be accelerated at 1 cm/s² (= gram/9.81 m/s²).

Frequency: The inverse of the time required for a body or particle in a vibrating system to go through a full cycle of motion.

Fundamental: The mode of vibration with the lowest natural frequency.

Harmonic: A signal that is usually generated along with the fundamental. The frequency of a harmonic is always an integer multiple of the fundamental frequency.

Inertia base: A heavy mass (usually concrete) added to the frame of a piece of vibrationally isolated equipment.

Mode: A natural, repetitive pattern of harmonic motion in which every particle moves at the same frequency.

Natural frequency: The frequency of the natural or normal modes of vibration of a vibrating system. A system at rest will vibrate only at its natural frequencies after being displaced by an impulsive force.

Node: Any point in a vibrating system where the particle motion is zero for a given mode of vibration.

Period: The time required for vibrating particles to complete a full cycle of motion for a given mode.

Phase: The relative time or displacement between two points in a vibrating system. Any two points are said to be “in phase” (i.e., the phase shift is 0 degree) if their relative motion is always in the same direction at the same time.

Pump frequency: The primary frequency generated by the pump in hertz (cycles per second). It equals $n \times \text{rev/s}$ where n is the number of pressure pulses generated in the fluid with each full rotation of the shaft.

Resonance: A condition in which a vibrating system responds with maximum amplitude to a driving force. This condition occurs only at the natural frequencies of the vibrating system.

Reverberant field: The region in a room (usually far from the source) where the sound pressure level is reasonably constant with position and distance from the source.

Rotor frequency: The frequency (in hertz) of the shaft of a pump or other rotating machine.

Short circuit: Any mechanism that allows vibrational energy to bypass a vibration-isolating device, thus rendering it ineffective (e.g., a spring compressed until the coils touch).

Sound pressure: A synonym for acoustic pressure. The sound pressure level is defined as 10 times the common logarithm of the square of the ratio of the weighted or unweighted acoustic pressure to the standard reference pressure (20 μPa).

Sound transmission class: A single-number rating for a material that describes the ability of the material to resist sound transmission.

Spectrum: The frequency distribution of a sound or vibration signal.

Standing wave: A periodic wave that does not appear to propagate but remains stationary. The wave is actually formed by two or more progressive (moving) waves of equal frequency traveling in opposite directions.

Stiffness: The ratio of the change in force to the change in displacement or deflection of an elastic element.

Static deflection: The deflection of an elastic element caused by the static (dead) load of the mass it supports.

Transmission loss: A value (usually expressed in decibels and a function of frequency for a given material) proportional to the ratio of incident to transmitted energy in a sound- or vibration-isolation system. A wall or barrier provides a transmission loss of

sound. A spring or rubber mount is usually involved in the transmission loss of vibration.

Wavelength: The distance between “in phase” positions on an acoustic or vibration wave at any given instant. The wavelength is related to frequency, f , via the wave propagation speed, c , as $\lambda = c/f$.

Wave resonances: Resonances in vibration isolators that can permit the transmission of noise through the vibration isolator with little or no transmission loss. Wave resonances usually occur only at frequencies much greater than the fundamental resonance frequency of the vibration-isolation system.

22-5. Vibration and Noise Characteristics

To develop a good foundation for discussing vibration and noise issues in a pumping station, it is important to understand the characteristics and sources of vibration and noise as well as the techniques for measuring these quantities.

In reciprocating machinery such as engines, vibration is primarily due to the dynamic forces exerted on the machine by the reciprocating and rotating internal parts (pistons, crankshaft, etc.). The vibration levels of this equipment can be relatively high. The spectrum is typically dominated by the fundamental frequency (e.g., the cylinder firing rate in a diesel engine) and usually contains several (3 to 10 or more) prominent harmonics.

In centrifugal equipment, vibration is caused by unbalanced rotating parts, misaligned shafts, and fluctuations in the load. The vibration of centrifugal equipment is typically lower in level than comparable reciprocating machinery, and the spectrum usually contains only the fundamental shaft frequency, the blade passage frequency, and one or two harmonics.

Piping vibration is usually caused by the transmission of vibrations from the primary equipment to which it is connected. Piping can be excited by the pipe wall contact with the pump inlet or discharge (even through a flexible connection) or by the pressure fluctuations within the fluid in the pipe. In addition to the frequencies associated with the pump, pipes also vibrate at their own natural or resonance frequencies.

Building vibration results from dynamic forces within the primary vibration elements (i.e., equipment and piping). In addition to vibrating at the frequencies associated with the equipment and piping, the structural elements of the building also have their own natural frequencies. Excessive vibration of the building's structural elements (floors, walls, etc.) is usually a potential problem only when a structural resonance frequency is very nearly equal to one of the

driving frequencies associated with the primary equipment. It is also possible to transmit vibration into the earth and, eventually, into adjacent structures. Aspects of this problem are beyond the scope of this chapter.

Translational Vibration

Translational vibration is usually measured with a contact sensor (an accelerometer or velocimeter), which is a small device rigidly attached to the vibrating body. The accelerometer produces an electrical output (voltage) proportional to the instantaneous acceleration of the body, whereas a velocimeter produces a voltage proportional to the velocity. Translational vibration problems in pumping stations almost always occur at frequencies below 100 Hz, and usually the problem is associated with a resonance frequency.

The waveform describing the translational motion of a vibrating body at resonance is usually sinusoidal, as shown in Figure 22-1. The three curves describe

exactly the same motion (displacement, velocity, and acceleration) of a single point on the object that is vibrating—all as a function of time. All curves have the same shape and repetition period of T seconds. The only difference between these curves is the amplitude (vertical) scale and the relative phase (shift in time).

The acceleration, velocity, and displacement amplitudes of a sinusoidal signal are related to each other by

$$x = \frac{v}{2\pi f} = \frac{a}{4\pi^2 f^2} \quad (22-1)$$

where x is the displacement amplitude in meters (inches), v is the velocity amplitude in meters per second (inches per second), f is the frequency in hertz, and a is the acceleration amplitude in meters per second squared (inches per second squared). Vibration levels are sometimes reported as peak levels (the maximum deviation from the zero or “at rest” value), as peak-to-peak levels (the maximum total swing from positive to negative values), or the root mean square (RMS) value. The RMS value is 0.707 times the amplitude (or peak value) for sinusoidal waveforms (as shown in Figure 22-1).

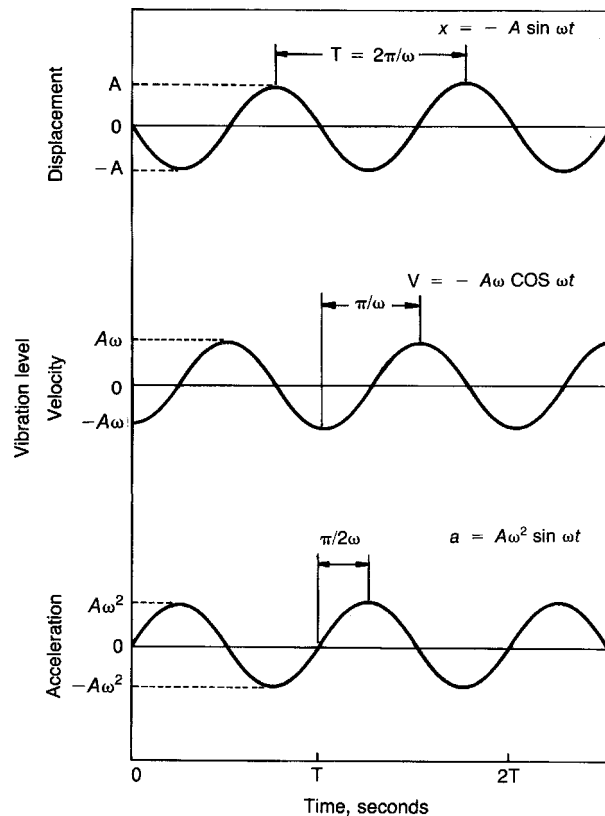


Figure 22-1. Displacement, velocity, and acceleration waveforms of a particle in simple sinusoidal motion.

For example, an object vibrating sinusoidally with a peak displacement of 1 mm (0.04 in.) at a frequency of 10 Hz (600 cycles/min) could be described accurately by any of the following (from Equation 22-1):

- 62.8 mm/s (2.47 in./s) peak velocity at 10 Hz
- 3.95 m/s^2 (12.8 ft/s) peak acceleration at 10 Hz
- 0.0444 m/s (0.571 ft/s) root mean square velocity at 10 Hz
- 7.90 m/s^2 (0.805 g) peak-to-peak acceleration at 10 Hz.

An important concept to keep in mind when analyzing vibrating systems is the buildup and dissipation of energy in the system. As a power source begins operation from rest, the system vibration rapidly builds up to a steady-state level. After the system reaches the steady state, energy is dissipated at a rate equal to the rate at which energy is put into the vibrating system. This rate is usually only a very small fraction of the total power consumption of the source. The energy dissipation is in the form of conversion to heat via various damping mechanisms as well as energy transmission to remote systems (such as building structure and the earth).

The total vibrational energy present in the system can be determined by computing the instantaneous potential energy at peak displacement (when the instantaneous velocity is zero) or by computing the instantaneous kinetic energy at peak velocity (when the instantaneous displacement is zero). Either way, for a point mass the result yields

$$E_t = 2\pi^2 M x^2 f^2 \quad (22-2)$$

where E_t is the total vibrational energy in the system in Newton-meters (inch-pound force), M is mass of the system in kilograms (pounds mass), x is the displacement amplitude in meters (inches), and f is frequency in hertz (see Section 22-4).

Solving Equation 22-2 for the dynamic displacement yields Equation 22-3, which demonstrates that, for a fixed amount of vibrational energy, the displacement is inversely proportional to the frequency and the square root of the mass. Thus, if the mass and energy remain constant, doubling the frequency results in a 50% reduction in the dynamic displacement.

$$x = \frac{1}{2\pi f} \sqrt{\frac{2E_t}{M}} \quad (22-3)$$

By combining Equations 22-1 and 22-2, it can be seen that the vibration velocity, v , is independent of

frequency at constant energy and the acceleration, a , is directly proportional to frequency at constant energy.

Torsional Vibration

Torsional vibration is difficult to measure because the shaft must be rotating during the measurement. The technique usually employed is to mount a lightweight, multitoothed gear near each end of the shaft at an accessible location and to install eddy-current sensors on a rigid foundation in close proximity to the gears. As the shaft rotates, the eddy-current sensor produces a pulsed output signal whose frequency is equal to the instantaneous shaft rotational speed in cycles per second (Hz) times the number of evenly spaced teeth on the gear. Torsional vibration of the shaft is detected as a periodic modulation of the output frequency, which indicates that the shaft speed is fluctuating at the sensor location. A frequency-to-voltage converter then transforms this pulsed signal into a low-frequency (usually below 100 Hz) waveform proportional to the instantaneous angular displacement in degrees or radians (or angular velocity in degrees per second or radians per second). This waveform can then be processed by conventional spectrum analysis techniques to obtain the torsional vibration levels at the frequencies of interest. Of course, for sinusoidal torsional waveforms, Equation 22-1 can also be used to convert angular displacement to angular velocity and angular acceleration if a consistent system of units is used.

Noise

Contrary to the characteristics of vibration, noise is usually only of concern at frequencies greater than 100 Hz. Noise problems are also almost never single-frequency problems and, as a result, some form of spectrum or frequency analysis is usually required to solve acoustical problems.

Noise is usually measured with a sound level meter, which is a hand-held instrument incorporating a microphone, an amplifying circuit, a frequency-weighting network, a display, and, perhaps, a set of filters (octave or one-third octave) for spectrum analysis. The microphone generates an electrical output proportional to the instantaneous acoustical pressure oscillations, and the sound level meter displays a frequency-weighted (or unweighted) average sound pressure level.

Noise levels are reported as a sound pressure level, L_p (in decibels), which is computed from the acoustic pressure, p , by

$$L_p = 20 \log_{10} (p/p_{\text{ref}}) \quad (22-4)$$

where p_{ref} is the standard reference pressure of 20 μPa (0.0002 dynes/cm²). The standard reference pressure is the approximate threshold of hearing for young adults in the ear's most sensitive frequency region (500–2000 Hz).

Subjectively, a noise level increase or decrease of 1 dBA is difficult to detect with the ear even though it does represent a 12% change in the acoustic pressure amplitude. A 10-dBA increase is generally perceived as a doubling of the loudness; conversely, a 10-dBA decrease is perceived as a 50% reduction in loudness. This nonlinear dependence is primarily due to the logarithmic response characteristic of the human ear.

Some typical A-weighted noise levels are given in Table 22-2. Noise levels can vary substantially with time, with location, and, in some instances, with changes in weather conditions. In reporting noise levels from an actual noise source or from an actual environment, it is extremely important to document the relevant conditions of the measurement, including the exact location of the microphone, the time and duration of the measurement, and the atmospheric conditions (primarily temperature, atmospheric pressure, wind speed, and direction). Without this back-up information, the validity of a noise measurement can be challenged.

Noise in and around pumping stations is usually broadband (meaning that it contains sounds at all frequencies simultaneously). A measure of an existing noise level is not complete unless the frequency weighting is also given. The most common frequency weighting is A-weighting, which closely approximates the frequency sensitivity of the human ear. Noise levels measured with this frequency weighting should be dimensioned as dBA for clarity.

To perform design calculations to predict noise levels with a reasonable degree of accuracy (± 5 dBA), it is necessary to measure or have information concerning the frequency content (spectrum) of the noise sources under consideration. An explanation of this level of analysis is beyond the scope of this chapter; if

this level of accuracy is required, consult a professional acoustical engineer.

22-6. Applicable Codes

Vibration

There are no established legal limits for vibration levels of equipment and piping associated with pumping stations, but there are guidelines for normally acceptable vibration levels of major pieces of equipment. Below 600 rev/min, the vibration criteria are based on peak-to-peak displacement; above 600 rev/min, the criteria are based on peak velocity (see Figure 22-1). The relationship between displacement and velocity is given by Equation 22-1. The Hydraulic Institute (ANSI/HI 9.6.4-2000) has published acceptable field vibration limits for clear liquid and nonclog horizontal and vertical centrifugal pumps. Vibration limits for reciprocating and rotary pumps are not available. Table 22-3 contains recommended vibration criteria (compiled from various sources) for major equipment used in pumping stations.

The vibration criteria listed in Table 22-3 refer to vibration levels at the shaft bearing housings. Vibration levels at other, less rigid portions of the machine may be higher. If the vibration level of a particular machine exceeds these values by more than 25% (particularly if the vibration level increases steadily with time), the unit and the system should be analyzed carefully to solve the problem. If the excessive vibration is present while the equipment on a rigid foundation is operating independently from the rest of the system, it should be returned to the manufacturer for balancing and/or alignment to correct the problem. If the vibration levels are only exceeded under load, the problem may be due to an interaction with another portion of the system.

Vibration criteria for piping systems are discussed in detail by Wachel and Bates [13] and are shown in Figure 22-2. The main danger of excessive piping vibration is failure due to repeated stress from distortion of the pipe. Stress in a piping system is primarily a function of the maximum displacement of the pipe as well as its vibratory mode shape. The allowable pipe displacement decreases with increasing frequency, as illustrated in the curves of Figure 22-2, because increasing frequency implies more vibrational energy input into the system and a greater number of stress cycles over the lifetime of the piping system. An alternative peak-to-peak pipe velocity criterion of 25.4 mm/s (1.0 in./s) has also been suggested

Table 22-2. Typical Noise Levels

Noise source or environment	Noise level (dBA)
Quiet suburb	40–55
Busy city	60–75
Gasoline-powered lawnmower ^a	80–90
Diesel truck (at 15 meters)	75–95
Chain saw ^a	95–120

^aAt operator position.

Table 22-3. Equipment Vibration Levels^a

Source	Peak-to-peak displacement below 600 rev/min		Peak velocity above 600 rev/min ^b	
	mm	in.	mm/s	in./s
Centrifugal pumps ^c				
Clear liquid	0.125	0.005	7.8	0.31
Nonclog	0.20	0.008	12.5	0.50
Electric motors ^d	0.08	0.003	5.0	0.20
Fans				
Centrifugal	0.15	0.006	9.5	0.38
Axial	0.10	0.004	6.3	0.25
Generator sets				
Diesel	—	—	20	0.80
Gasoline	—	—	15	0.60
Air compressors				
Reciprocating	0.4	0.016	25	1.00

^aLevels refer to filtered vibration amplitude at the vibration frequency and at the running speed (forcing frequency).

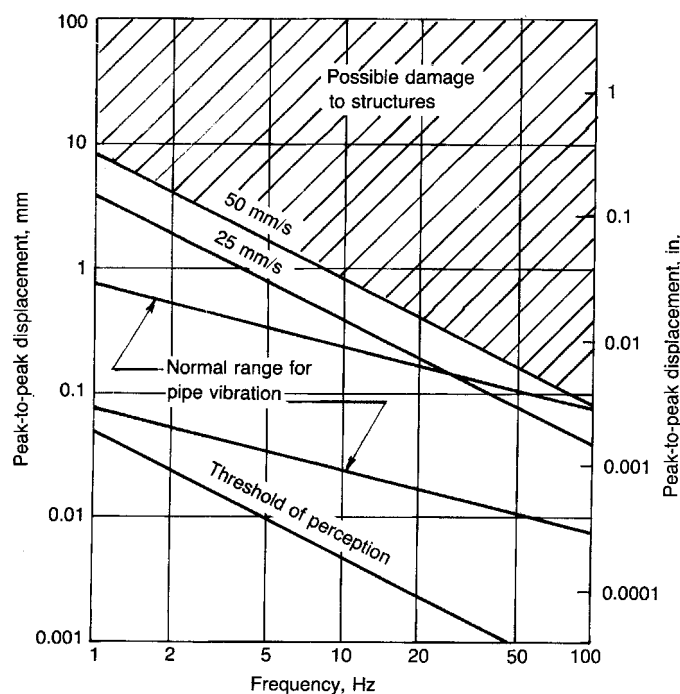
^bSee Equation 22-1 for the relationship to displacement.

^cSee ANSI/HI 9.6.4.4 for recommendations on vertical pumps.

^dLimits apply to base-mounted motors. Use pumps for pump-mounted motors.

by Wachel and Bates [13]. This velocity criterion, which includes allowances with a nominal safety factor for typical conditions found in real piping systems, is also shown in Figure 22-2 along with the approximate threshold of human perception.

Vibration criteria for structures are discussed briefly by Harris and Crede [19]. They state that a peak-to-peak level of 25.4 mm/s (1.0 in./s) is generally regarded as an upper limit for safe, continuous vibration levels of structural floors. This is not to say

**Figure 22-2.** Vibration displacement criteria.

that structural failure will result at higher levels. During earthquakes, peak-to-peak vibration levels sometimes exceed 50 mm/s (2 in./s) without damage. But the probability of eventual damage increases as the vibration level increases. Continuous vibration levels exceeding 50 mm/s (2 in./s) present a serious threat to the safety of most structures.

Noise

Legal limits on noise generally fall into two categories: (1) occupational exposure, and (2) community exposure. Occupational noise exposure limits have been established by the federal government and are administered by OSHA [20]. The federal noise exposure regulations apply to employers who engage in interstate commerce, but most states have passed similar legislation extending these limits (some states are even more restrictive) to virtually all occupational environments. Compliance requires substantial design and investigative effort. The intent of the law is to prevent hearing loss that can be induced by exposure to high noise levels over extended periods of time. Reducing noise levels in the work environment is also advantageous for improved speech intelligibility, which in turn improves worker safety. The relation-

ship between background noise level and the vocal effort required for adequate voice communication at various distances is shown in Figure 22-3.

In essence, the law restricts the length of time an employee can spend in a given noise environment during any 24-h period. Maximum allowable exposure times for various noise levels are listed in Table 22-4.

In most work environments, employees are exposed to noise that varies continuously throughout the day. The noise exposure under these conditions can be estimated by computing the noise dose, D_n ,

$$D_n = \sum_{i=1}^n \left(\frac{C_i}{T_i} \right) \quad (22-5)$$

where C_i and T_i are the cumulative exposure time and maximum allowable exposure times for the i th noise interval. If D_n exceeds 1.0, the employee is overexposed. Sometimes, it may be necessary to install portable dosimeters on the employees to monitor the actual noise level and compute the noise dose on a continuous basis.

If working conditions in any facility exceed the local or federal regulations, each employee should wear approved hearing protectors (earplugs or earmuffs) whenever the noise exceeds 90 dBA to avoid

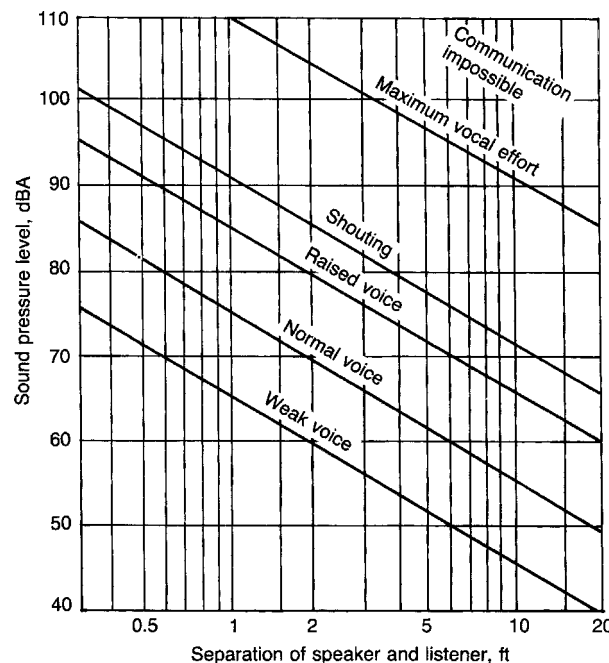


Figure 22-3. Relationship between voice and sound levels.

Table 22-4. Maximum Allowable Employee Noise Exposure

Exposure time (h/d) ^a	Noise level (dBA)
8.0	90
7.0	91
6.2	92
5.3	93
4.6	94
4.0	95
3.5	96
3.0	97
2.6	98
2.3	99
2.0	100
1.7	101
1.5	102
1.4	103
1.3	104
1.0	105
0.87	106
0.76	107
0.66	108
0.57	109
0.50	110

^aExposure times for all noise levels in the 80- to 130-dB range can be computed from the equation $T = 8/2^N$, where the exponent $N = (1/5)(L_p - 90)$.

permanent hearing loss. Good-quality hearing protectors provide an insertion loss (effective noise reduction to the inner ear) between 10 and 20 dB if they are installed properly. It is important to emphasize that extended exposure to high levels of noise can *permanently* damage hearing.

If the noise levels in a pumping station are expected to exceed federal or local regulations, steps should be taken during the design of the facility to mitigate the problem. These steps include the following:

- Design the facility in such a way that workers do not have to spend much time in noisy areas.
- Specify and/or select less noisy equipment.
- Install acoustical lining in noisy areas.
- Isolate noisy equipment with acoustical enclosures.
- Provide sound-absorbing barriers around noisy machinery.

Although there is no federal legislation affecting noise radiated into the environment, many local and state governments have such legislation to protect the welfare of the population living near permanent noise sources. Typically, these limits are much lower than the occupational noise exposure levels because the

intent is not to protect against hearing loss but to preserve a quality community environment.

A typical example of a community noise ordinance is illustrated in Table 22-5. The maximum allowable noise level is often a function of how the source and receiver property is zoned (not necessarily how the property is being used). Most ordinances also have allowances for somewhat higher noise levels for short periods of time as well as a requirement for lower noise levels (as much as 10 dB lower) on residential property during sleeping hours. The ordinance should also address the situation where existing ambient noise levels already exceed the ordinance requirements.

In critical situations, it may be necessary to monitor existing noise levels in the surrounding community prior to completing the design of the pumping station. This information on background noise can be used to assess the acoustical impact of the facility and make the appropriate design decisions to minimize the impact. The preferred descriptor of environmental noise levels is the day-night sound level, L_{dn} . The L_{dn} is a time-weighted energy average (over a 24-h period) of the continuously varying A-weighted sound pressure level at a point. The L_{dn} is usually determined by direct measurement with a special instrument called a “community noise monitor,” which is designed for computing noise statistics at permanent or semipermanent outdoor installations. In special circumstances where the noise level does not vary with time over the 24-h day, the L_{dn} is approximately equal to the continuous A-weighted sound pressure level plus 6 dBA.

The way in which noise is perceived by average residential communities is shown in Figure 22-4. For reference, typical L_{dn} values for various areas are given in Table 22-6.

In addition to meeting the legal requirements of any state or local noise ordinances, it may be necessary to provide additional noise control measures to prevent any adverse community noise impact. Such measures would particularly apply to communities with nonexistent or poorly written ordinances, or where the ordinance requirements were inadequate

Table 22-5. Commonly Used Maximum Permissible Community Noise Levels (dBA)

Source property	Receiving property		
	Residential	Commercial	Industrial
Residential	55	57	60
Commercial	57	60	65
Industrial	60	65	70

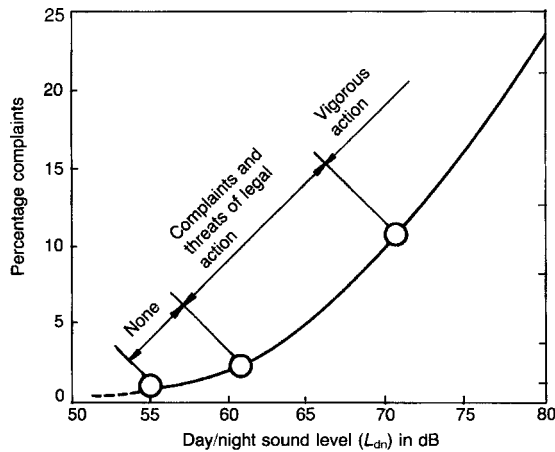


Figure 22-4. Average community reaction to noise.

to meet the needs of the community. To avoid an adverse community reaction, design pumping stations to meet the following four criteria:

- Do not increase the existing L_{dn} more than 3 dBA.
- Do not increase the existing L_{dn} more than 1 dBA if the existing L_{dn} is more than 60 dBA.
- Do not increase average hourly nighttime maximum noise levels more than 5 dBA in residential areas.
- Do not increase average hourly nighttime maximum noise levels more than 2 dBA in residential areas if the existing hourly nighttime maximum levels exceed 60 dBA.

The reason for restricting the hourly maximum noise level on residential property during the nighttime hours (usually defined as 10:00 P.M. to 7:00 A.M.) is to minimize sleep disturbance. Noise-induced sleep interference is more dependent on maximum noise levels than on average noise levels [21]. Maximum nighttime noise levels are quite volatile, that is, they usually vary by as much as 10 dB or more from hour to hour as well as on a daily basis. Consequently, it is best to base a design on an average noise level taken

Table 22-6. Typical Day-Night Sound Levels at Various Locations

Location	L_{dn} (dBA)
Near major international airport	75–85
Major city in downtown area	70–80
Noisy urban residential	60–70
Suburban residential	50–60
Quiet rural	45–55

over several nights. These four basic criteria can usually ensure little or no acoustical impact on community noise levels.

22-7. Equipment Vibration

The vibration limits recommended in Table 22-2 for major items of equipment in pumping stations apply to equipment mounted directly to a rigid foundation. Vibration levels are usually higher if the equipment is mounted on a flexible structure because both the support and the machine contribute to the total vibration level. All structures are flexible to some degree, but a rigid structure is defined in the ANSI/HI standards as one that has a fundamental natural frequency higher than 125% of the highest rotating speed of the machine.

When vibrating equipment is rigidly mounted to a massive structure (such as a building floor slab) the majority of the vibrational energy generated by the machine is transmitted into the structure. Much of this energy is absorbed by the building (which causes vibration of the building's structural elements), and some is dissipated into the earth via the foundation. If the building is structurally capable of handling this vibrational energy, and if the resulting noise radiated by the vibrating structure is within acceptable limits, the rigid mounting of equipment is preferable for most pumps because, unless the pump or motor excites a building resonance, it puts less stress on the machine.

If the rotational speed of the equipment happens to match one of the resonance frequencies of the supporting structure, the structure may vibrate at excessive levels. Obviously, if the structure supporting the equipment vibrates at an excessive level, this vibration also contributes to the vibration observed at the machine. Often, the equipment is thought to be out of balance when this occurs; in fact, it may be within acceptable tolerances if removed and tested on a rigid foundation. There are three potential solutions to the problem of excessive structure vibration caused by rigidly mounted equipment:

- Avoid machine speeds that excite structural resonances.
- Modify the structure to increase the resonance frequencies so that they are above the highest machine frequency.
- Isolate the equipment from the structure with vibration isolators.

Avoiding the resonance condition altogether is the preferred technique, but when this is not feasible consider vibration isolation of the equipment.

Vibration isolators reduce the amount of vibrational energy transmitted to the structure supporting the equipment and, hence, can reduce the vibration levels of the structure. Vibration isolators do not reduce the amount of vibrational energy generated by the machine, and, in fact, the vibration level of a machine mounted on them is usually greater than that of a rigidly mounted machine, particularly at low frequencies (i.e., in the region of the machine's shaft speed). The amount of additional vibration is primarily a function of the total mass of the machine. Equipment that is very massive (such as a large diesel generator set) usually has vibration levels that are almost independent of mounting, while lighter equipment (such as an air compressor) is subject to a significant increase in vibration level when moved from a rigid to an isolated mounting.

Because vibration isolation of equipment is an added cost, it should be considered only when there is a need for reduced noise or vibration. In general, vibration isolation of equipment does not reduce the airborne noise radiated by the equipment. Although vibration isolation can significantly reduce structure-borne noise, its effects are not noticeable unless the airborne noise is also controlled. In most centrifugal pumps, the vibrational energy is primarily low frequency, and structure-borne noise is not usually a problem. But structure-borne noise may be a problem for reciprocating pumps and rotary pumps if they have fundamental pump frequencies greater than 100 Hz. Therefore, most pumps do not require vibration isolation in well-designed pumping stations.

Equipment that should be vibration-isolated in most pumping stations includes generator sets, fans, and air compressors. This equipment usually contains sufficient low- and high-frequency energy to cause excessive structure-borne noise in other areas of the building. Structural vibrations from this equipment may also be a problem for computer equipment and other sensitive electronic instruments.

22-8. Vibration Isolation Theory

Equipment vibration in its simplest form can be modeled as a small mass, m , rotating about the center of gravity of a rigid body with mass M at a distance, e_o , from the center of gravity of mass M at a fixed rotational speed. The rotating mass produces an oscillating force, F_o , whose magnitude is

$$F_o = me_o\omega^2 \quad (22-6)$$

If the equipment is rigidly attached to the supporting structure, all of this force is transmitted into the structure, which causes it to vibrate at the frequency of rotation of the eccentric mass. The transmitted force (and resulting vibration) can be reduced by reducing (1) the eccentric mass, (2) the eccentricity, e_o , or (3) the rotational speed. Unfortunately, there are practical limits to equipment balancing (to reduce e_o), and equipment speed cannot always be reduced. The alternative method of reducing vibration transmission into the structure is by isolating the equipment with a resilient element such as a spring or a compliant pad. This isolating element, if properly designed and installed, can reduce the force transmitted into the structure and the resulting vibration dramatically.

The natural frequency of a simple, single-degree-of-freedom (one mass) isolation system mounted to a rigid structure (in the absence of damping) can be expressed as

$$f_o = \frac{1}{2\pi} \sqrt{k/M} = \frac{1}{2\pi} \sqrt{kg/W} = \frac{1}{2\pi} \sqrt{g/d} \quad (22-7)$$

where f_o is the natural frequency of vibration in hertz, k is the spring constant of the isolator in Newtons per meter (pounds force per foot), M is the total mass supported by the isolator in kilograms (pounds mass), W is the total weight supported by the isolator in Newtons (pounds force), g is the acceleration of gravity [9.81 m/s^2 (32.2 ft/s^2)], and d is the static deflection of the isolator due to the weight of the equipment in meters (feet). At frequencies much less than f_o the isolators provide no isolation and the system acts as though the isolators were not present. Within ± 50 of the resonance frequency, the isolation system reinforces and amplifies the vibration of the equipment as well as the vibration of the supporting structure. Only at frequencies much greater than f_o does the isolation system reduce the vibrational energy injected into the structure.

The efficiency of a simple, single-degree-of-freedom vibration isolation system is defined as

$$\varepsilon = 100 [1 - (F_2/F_1)] = 100 [1 - (x_2/x_1)] \quad (22-8)$$

where ε is the efficiency (a percentage), F is the vibrational force, x is the displacement, the subscript 2 means transmitted to the structure with isolators, and the subscript 1 means transmitted to the structure with rigid connections.

In general, the efficiency is a function of frequency, but usually the efficiency at the driving frequency or the resonance frequency of the supporting structure is

the main concern. The ratio F_2/F_1 is often called the transmissibility, T , of the system. At the lowest driving frequency, which is the rotational speed of the machine, the transmissibility is expressed as

$$T = \frac{F_2}{F_1} = \sqrt{\frac{1 + (2\delta_r f_d/f_o)^2}{(1 - f_d^2/f_o^2)^2 + (2\delta_r f_d/f_o)^2}} \quad (22-9)$$

where f_d is the driving frequency, f_o is the natural frequency of the isolation system, and δ_r is the damping ratio, δ/δ_c . Steel spring vibration isolators typically have damping ratios between 0.01 and 0.03 (1 to 3%). The transmissibility can be evaluated at any frequency, f , by substituting f for f_d in Equation 22-9. The frequency and damping ratio dependence of the transmissibility of the single-degree-of-freedom system is illustrated in Figure 22-5.

In addition to reducing the vibration levels of structures that support equipment, it is also important to reduce the vibration levels of the equipment itself. The general expression for the displacement amplitude of a rigid mass mounted on vibration isolators with a spring stiffness constant, k , is (for $f = f_d$)

$$x = \frac{F_o/k}{\sqrt{[1 - (f_d/f_o)^2]^2 + [2\delta_r(f_d/f_o)]^2}} \quad (22-10)$$

which is presented graphically in Figure 22-6 as a function of frequency and damping ratio. At low frequencies ($f \ll f_o$) the displacement amplitude asymptotically approaches the value F_o/k . At high frequencies ($f \gg f_o$) the displacement amplitude approaches the value $F_o/(M\omega^2)$. In the resonance region the displacement amplitude is controlled entirely by the damping factor.

Damping is the process by which vibrational energy is converted into heat. The damping ratio, δ_R , is the ratio of the damping factor, δ , and the critical damping factor, δ_c . The damping factor, δ , is a physical property of the isolator that is relatively constant (there is a slight temperature dependence) with frequency. Materials with high damping factors do not readily vibrate because the energy is rapidly converted into heat and it is not allowed to build up to high levels. Critical damping is the amount of damping required to prevent a displaced mechanical system from oscillating during its return to a rest position and is a function of the system parameters. For viscous damping, δ_c equals two times the mass times the natural frequency in radians per second. Consequently, there are no fixed values of δ_R for a given

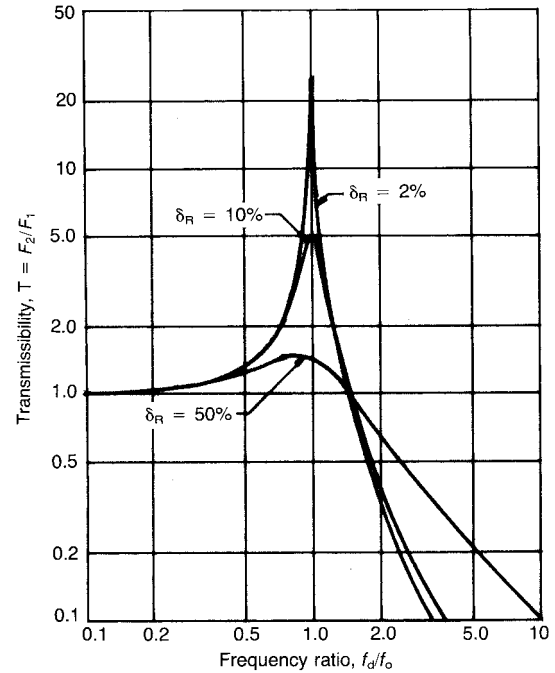


Figure 22-5. Transmissibility of a simple vibration isolation system.

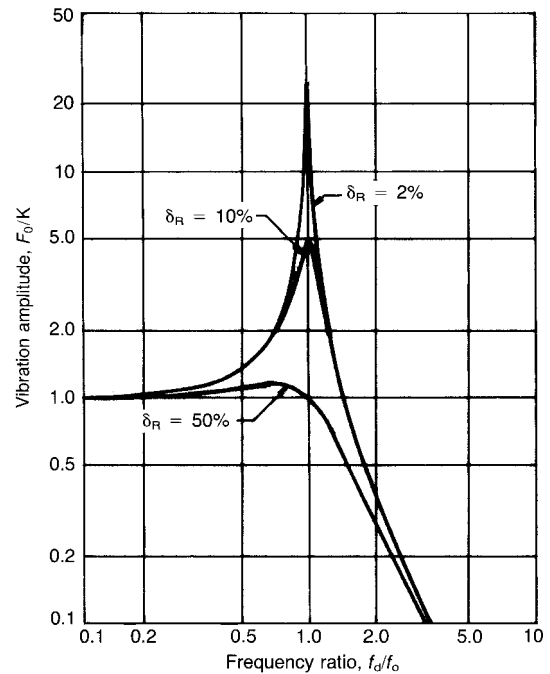


Figure 22-6. Amplitude response of a simple vibration isolation system.

material. Most spring-mounted vibration isolator systems, however, have damping ratios less than 5%. Systems with cork or rubber isolators may have δ_R values in the 10 to 20% range.

Although Figures 22-5 and 22-6 are similar in appearance, a completely separate but related phenomenon is described in each. Note that both curves show comparable response at low-frequency ratios. At high-frequency ratios (e.g., 2 or more), however, an increase in the damping ratio slightly reduces the vibration amplitude (Figure 22-6) of the equipment but increases the transmissibility (Figure 22-5) and, hence, increases the displacement amplitude of the building. Thus, depending on whether the equipment or the building is of greater concern, the addition of damping may be detrimental except at f_d/f_0 ratios between 0.4 and 1.5.

For high-efficiency vibration isolation, the lowest driving frequency should be 5 to 10 times the natural frequency of the isolation system, and the damping factor, δ_r , should be low. The relationships found in Equations 22-7 through 22-9 are summarized in Figure 22-7. The required isolator static deflection needed to achieve a particular isolation efficiency at a given rotational speed is shown for a damping ratio of 0.02. Be cautious because the assumption in the preceding analysis is a linear single-degree-of-freedom system (i.e., a rigid body mounted to a resilient element obeying Hooke's law with viscous damping) supported from an infinitely rigid structure. In reality, none of these assumptions is strictly true, but they

are approximately valid for frequencies less than $10f_0$ when the natural frequency of the supporting structure is much greater than $10f_0$. At higher frequencies, other factors (such as wave effects in which energy is transmitted directly into the structure via the isolation element) take over to make the isolation efficiency much less than would be predicted with the above equations. In practice, the vibration isolation efficiency very rarely exceeds 95% at any frequency.

Warning: Do not be confused with the isolation efficiency figures, which are a function of the linear ratio of the applied to the transmitted dynamic force (see Equation 22-8). Human perception of noise and vibration levels follows a logarithmic response, and the perceived reduction is always less than that derived directly from the isolation system efficiency.

22-9. Vibration Isolators

There are three major types of vibration isolators. The least expensive is the pad type, which is usually made of rubber, cork, fiberglass, or a combination thereof. These isolators generally provide static deflections of less than 5 mm (0.2 in.) and, hence, are effective (efficiencies greater than 90%) only for equipment with driving frequencies greater than 25 Hz (1500 rev/min).

The most common vibration isolator is the coil steel spring. Springs are manufactured in a variety of designs that provide static deflections from 10 to 100 mm (0.4 to 4.0 in.). Springs are suitable for isolating all types of equipment with driving frequencies greater than 300 rev/min. A typical open spring isolator (so called because the spring is not enclosed in a metal housing) is illustrated in Figure 22-8. Metal housings are often used to provide lateral restraint to resist forces due to earthquakes. The threaded rod in the top plate has height adjustment nuts for leveling the equipment after mounting. The neoprene rubber pad under the base plate reduces the transmission of high-frequency energy (via wave resonances in the steel coil) into the structure. Anchor bolts in the base plate prevent equipment creep, but they must be isolated from the base plate with a neoprene washer and grommet to prevent high-frequency energy transmission into the structure via the anchor bolt. For stability reasons, the coil diameter of open springs should be at least 0.8 times the operating height of the spring.

For extremely sensitive installations (which are not anticipated in pumping stations) or for very low-frequency equipment (less than 300 rev/min), air springs may be necessary. Air springs have an enclosed volume of air to support the equipment at

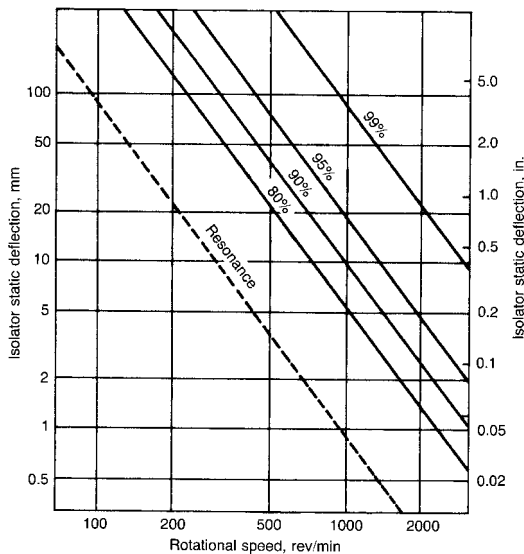


Figure 22-7. Vibration isolation efficiency.

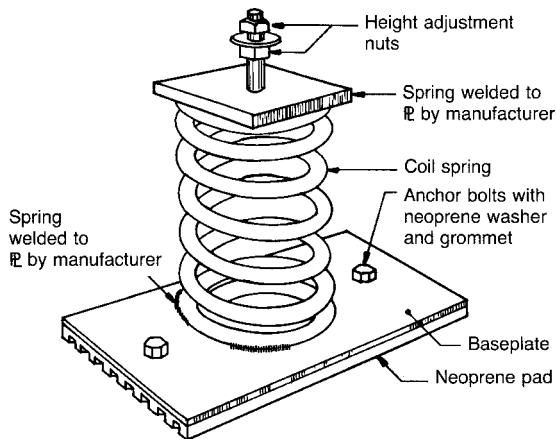


Figure 22-8. Open spring vibration isolator.

each support point. Air springs do not exhibit a linear load deflection curve (Hooke's law), and as a result, the simple theory presented above does not apply. Air springs do, in general, provide superior vibration isolation, partly because of the resulting low natural frequency and also because of the absence of high-frequency wave resonance effects. Air springs should be used only with the guidance of a qualified vibration consultant.

The recommended method of isolating a centrifugal pump from its supporting structure is illustrated in Figure 22-9. The machine is rigidly mounted to an inertia base, a concrete-filled frame designed to add mass to the equipment. The frame is then supported 50 mm (2 in.) above the housekeeping pad with steel springs. Seismic restraints should be provided at each support point with cushioned bumpers to limit mo-

tion to no more than 8 mm (0.3 in.) in any direction. Discharge piping and electrical power should be connected with flexible fittings.

The primary purpose of the inertia base in Figure 22-9 is to reduce the amplitude of the vibration of the pump, not to reduce the vibration level of the supporting structure. Adding mass to equipment that is already supported by vibration isolators may improve the theoretical efficiency slightly because it lowers f_0 . However, if enough mass is added to make a significant improvement in the isolation efficiency (by increasing the ratio f_d/f_0), the springs would probably overload, because springs typically can provide a safety factor of only 20 to 50% of load capacity before "short circuiting." The inertia base does not substantially decrease the vibratory force transmitted into the structure because it does not change the transmissibility. Furthermore, the inertia base does not reduce the driving force, because from Newton's first law ($F = ma$), the decrease in acceleration, velocity, or displacement is only due to a corresponding increase in mass. Thus, a 20% increase in mass results in a 20% decrease in equipment displacement (or velocity or acceleration) amplitude with no change in transmissibility or transmitted force into the structure. The efficiency of the isolation is controlled entirely by the selection of the springs (the static deflection), as shown in Figure 22-7.

In pumping stations, reciprocating air compressors and large, high-pressure centrifugal fans and pumps are usually the only machines that require an inertia base. The amount of additional mass required varies from 1.5 to 5 times the equipment weight and depends on the level of the vibration forces in the machine and the total weight of the machine. Heavy equipment (such as diesel generators) has sufficient

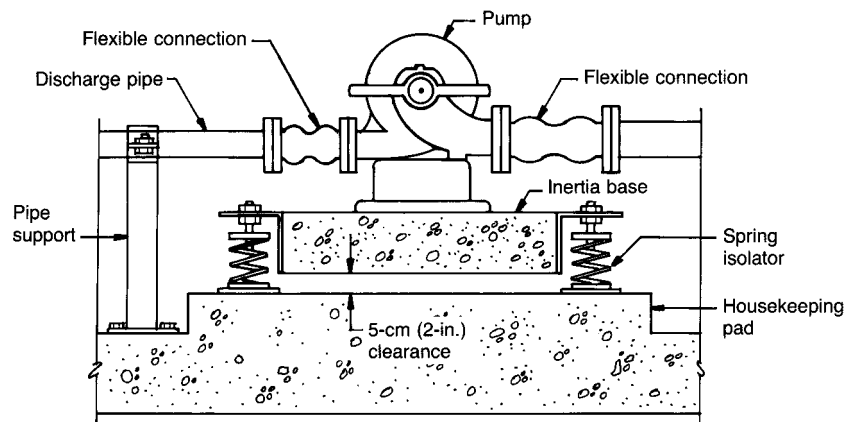


Figure 22-9. Vibration isolation of a centrifugal pump.

mass and rigidity to make inertia bases unnecessary. The only penalty for overdesigning the mass of the inertia base is the increased cost of the base and isolators.

If it is necessary to isolate a pump from the structure, it is absolutely mandatory that the pump, drive shaft, and motor (or engine) all be mounted on a common, rigid foundation (preferably a concrete inertia base) to avoid vibration-induced misalignment between these components. This isolation requirement automatically eliminates the possibility of

vibration isolation for some pumps (such as vertical turbines) with motors that must be supported by a separate structural element. So as not to degrade the overall isolation provided by the isolation mounts, piping and electrical connections must also be flexible. It is difficult to specify exactly how much relative motion must be allowed, but about 5 to 10 times the displacement amplitude of the pump is recommended. This displacement tolerance can usually be achieved by using flexible rubber hose (with nylon cord reinforcement) connectors.

Example 22-1 Vibration Isolation of a Centrifugal Pump

Problem: A 150-kW (200-hp) close-coupled centrifugal water pump is driven by an induction motor at 900 rev/min. The pump/motor assembly is rigidly mounted to a concrete pad 20 cm (8 in.) thick. The supporting floor vibrates at a peak-to-peak displacement of 1.0 mm (0.04 in.) at 15 Hz while the pump is running. During the test, the unit was operated without fluid in the connected piping, and it was found that the presence of the fluid in the system had little effect on the structural vibrations of the floor.

Design an isolation system for the pump-motor assembly that reduces the structural floor vibrations to meet the safe limit for the structure with a safety factor of 2 and provides reasonable equipment vibration levels. The total mass of the pump-motor assembly is 800 kg (1760 lb_m), and the manufacturer states that the out-of-balance force should not exceed that of an eccentric mass of 1 kg (2.204 lb_m) with an eccentricity of 25 mm (0.984 in.).

Solution: The recommended safe limit for structures is 25 mm/s (1 in./s) peak-to-peak. Compute the required efficiency of the isolation system. Begin with Equations 22-1 and 22-8.

SI Units

$$d = \frac{25}{2\pi 15} = 0.265 \text{ mm (peak to peak)}$$

$$d_{\text{desired}} = \frac{0.265}{2} = 0.132 \text{ mm}$$

$$\varepsilon = [1 - (0.32/1.0)] \times 100 = 87\%$$

U.S. Customary Units

$$d = \frac{1}{2\pi 15} = 0.0106 \text{ in. (peak to peak)}$$

$$d_{\text{desired}} = \frac{0.0106}{2} = 0.0053 \text{ in.}$$

$$\varepsilon = [1 - (0.0053/0.04)] \times 100 = 87\%$$

From Figure 22-7, the required isolator static deflection is about 12 mm (0.47 in.). This deflection cannot be achieved with most pad-type isolators but is easily obtained with conventional steel spring isolators. By definition, the spring constant equals force divided by deflection so the spring constant ($k = F/d$) for each of four springs must be about

$$k = \left(\frac{800}{4}\right) \text{ kg} \times 9.81 \frac{\text{m/s}^2}{0.012 \text{ m}} = 163,500 \text{ N/m} \quad k = \frac{(1760/4)}{0.47} = 936 \text{ lb/in.}$$

From Equation 22-6 ($F_o = m e_o \omega^2$), the maximum out-of-balance force for a mass of 1 kg at 25 mm (2.204 lb_m at 0.984 in.) is

$$F_o = (1)(0.025)(2\pi 15)^2 = 222.1 \text{ N} \quad F_o = \frac{2.204}{32.17} \left(\frac{0.984}{12}\right) (2\pi 15)^2 = 49.9 \text{ lb}$$

From Equation 22-7, the natural frequency of the isolation system is $f_o = (1/2\pi)\sqrt{k/M}$, where M is the pump mass supported by each spring and k is the stiffness of each spring:

SI Units

$$f_o = \frac{1}{2\pi} \sqrt{\frac{163,500}{(800/4)}} = 4.55 \text{ Hz}$$

U.S. Customary Units

$$M = W/g = \frac{(1760/4)}{32.17 \times 12} = 1.140 \text{ lb} \cdot \text{s}^2/\text{in.}$$

$$f_o = \frac{1}{2\pi} \sqrt{\frac{936}{1.140}} = 4.56 \text{ Hz}$$

The ratio of driving frequency to natural frequency is $f_d/f_o = 15/4.56 = 3.29$. From Equation 22-10, $x = F_o/k[(1 - f_d^2/f_o^2)^2 + (2\delta_v f_d^2/f_o^2)^2]^{-1/2}$ and assuming a 2% damping ratio, the maximum displacement amplitude of the pump is

$$\begin{aligned} x &= \frac{222.1}{163,500} [(1 - 3.29^2)^2 + (2 \times 0.02 \times 3.29)^2]^{-1/2} & x &= \frac{49.9}{936} [(1 - 3.29^2)^2 + (2 \times 0.02 \times 3.29^2)]^{-1/2} \\ &= 1.38 \times 10^{-4} \text{ m} & &= (0.053)/92.69 + 0.017 \\ &= 0.138 \text{ mm peak to zero} & &= 0.0054 \text{ in. peak to zero} \end{aligned}$$

From Equation 22-1 ($x = v/2\pi f$),

$$v = 2\pi f x = 2\pi(15)(0.138) = 13.0 \text{ mm/s (peak)} \quad v = 2\pi(15)(0.0054) = 0.51 \text{ in./s (peak)}$$

The computed value is almost double that allowable for clear liquid pumps given in Table 22-3. Hence, the vibration level of the machine must be reduced, so an inertia base must be added. If the inertia base has an added mass of 1000 kg (2204 lb_m) and if the spring constant is nearly doubled to 300,000 N/m (1713 lb/in.), the new static deflection of each of the four isolators will be

$$\begin{aligned} d &= \frac{(1000 + 800)}{4} \times \frac{9.81}{300,000} & d &= \frac{(2204 + 1760)}{4 \times 1713} = 0.579 \text{ in.} \\ &= 0.0147 \text{ m} = 14.7 \text{ mm} \end{aligned}$$

which still meets the 12-mm (0.47-in.) minimum. From Equation 22-7, $f_o = (1/2\pi)\sqrt{k/M}$, the new frequency would be

$$f_o = \frac{1}{2\pi} \sqrt{\frac{300,000}{1800/4}} = 4.1 \text{ Hz} \quad f_o = \frac{1}{2\pi} \sqrt{\frac{1713 \times 386}{(2204 + 1760)/4}} = 4.1 \text{ Hz}$$

The new dynamic displacement is calculated from Equation 22-10. Note that $f_d/f_o = (900/60)/4.1 = 3.659$.

$$\begin{aligned} x &= \frac{222.1}{300,000} [(1 - 3.659^2)^2 + (2 \times 0.02 \times 3.659)^2]^{-1/2} & x &= \frac{49.9}{1713} [(1 - 13.385)^2 + 0.04(3.659)^2]^{-1/2} \\ &= 5.98 \times 10^{-5} \text{ m} & &= 0.00233 \text{ in. peak to zero} \\ &= 0.060 \text{ mm peak to zero} \end{aligned}$$

From Equation 22-1 ($x = v/2\pi f$), the peak velocity is

$$v = 2\pi(15)(0.060) = 5.63 \text{ mm/s (peak)} \quad v = 2\pi(15)(0.00233) = 0.220 \text{ in./s (peak)}$$

which meets the criteria listed in Table 22-3.

22-10. Piping Vibration

The piping system attached to any pump vibrates as long as the pump is moving fluid inside the piping. The amount of vibration present in the pipe depends on many factors, including (1) the size, type, and

rotational speed of the pump; (2) the fluid characteristics; and (3) the size, construction, support, and layout of the piping system. The purpose of this section is to present and discuss the basic design information needed to avoid vibration problems in piping systems.

To understand the nature of pipe vibrations, it is important to realize that vibration can occur in many forms; the most common are shown schematically in Figure 22-10. In general, all three modes—compression, pure bending, and radial (as well as other, less important modes)—coexist in varying degrees in all piping systems.

In the compressional mode, the vibration displacement of the pipe wall is along the length of the pipe, and the instantaneous density of the pipe wall material oscillates slightly with time and distance along the pipe. This form of vibration does not tend to stress the pipe unduly, and, consequently, it is of less importance.

In the pure bending mode, the entire pipe displaces in a direction perpendicular to the length of the pipe, much like that of a violin string when plucked. This action can put significant stress on the pipe wall if the vibration amplitude is large enough.

In the radial mode of vibration, the pipe wall motion is perpendicular to the length of the pipe, but the pipe wall at a given location moves radially inward and outward. This mode of vibration is naturally induced by pumps that discharge the fluid into the piping system as a series of pulses that cause a sinusoidal variation of fluid pressure in the piping.

These three forms of vibration, called the “natural modes of free vibration,” occur only at particular discrete frequencies, called “resonance frequencies.”

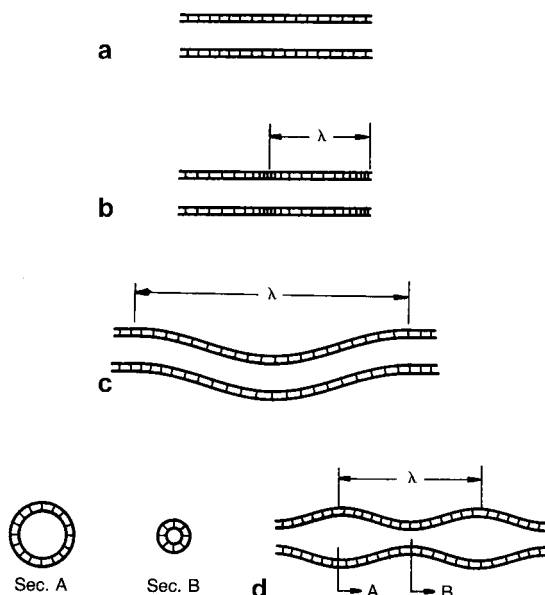


Figure 22-10. Modes of pipe vibration. (a) Pipe at rest; (b) compression; (c) bending; (d) radial.

Resonance frequencies are determined by the physical characteristics of the piping system.

Although each section of pipe has an infinite number of resonances in each of several forms of vibration, only a few of these are of concern to the designer because of the following:

- Vibrational energy from the pump is usually concentrated at the low-order pump frequencies, which are usually less than 100 Hz.
- Resonances occurring at low frequencies are the only ones that might cause large structural displacements and corresponding high stress.

In most situations, it is adequate to consider only the first two resonance orders for piping systems. The first-order resonance is called the “fundamental frequency.” The second-order resonance is defined in Figure 22-11.

Bending Modes in Piping Systems

At each bending mode resonance frequency of the piping system there is a characteristic shape that the pipe assumes. The pipe oscillates back and forth resuming the same shape at periodic time intervals equal to the inverse of the resonance frequency. The exact shape of the pipe for a given mode is slightly different from the theoretical shape because the pipe also simultaneously vibrates (to a lesser degree) at other modes.

The mode shape for a given resonance is determined by the boundary conditions (i.e., how the pipe section is supported) and the order of the mode. The lowest resonance frequency for a given system is called the “fundamental” or “first-order resonance.” This resonance is usually the most important because it has the greatest potential for large pipe displacements. The second-order resonance occurs typically at a frequency two to four times the fundamental and, therefore, has the potential for only about one-half to one-fourth of the displacement of the fundamental mode of vibration if equal energy input into each mode is assumed.

The boundary conditions describe how the pipe is supported (restrained from motion) at the ends of a pipe section. There are numerous ways of supporting pipe sections, but four of the more common are given in Table 22-7.

Mode shapes and corresponding natural frequencies for various combinations of the first three boundary conditions have been determined analytically [5, 22], and some are shown in Figure 22-11. The analysis of the vibration-isolated support does not seem to

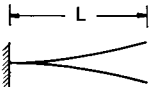
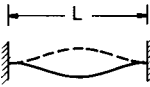
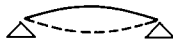
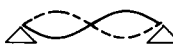
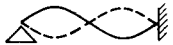
Support	Order, n	Shape	Coefficient γ_n	Support	Order, n	Shape	Coefficient γ_n
Cantilever	1		1.88	Anchored	1		4.73
	2		4.69		2		7.85
Simply supported	1		3.14	Combination	1		3.93
	2		6.28		2		7.07

Figure 22-11. Bending mode resonances.

have been published to date, but the mode shapes and natural frequencies are expected to fall somewhere between the free and simply supported conditions.

The general equation for computing bending mode resonance frequencies for a piping system of uniform diameter and physical properties can be expressed as

$$f_n = \frac{\gamma_n^2}{2\pi L^2} \sqrt{\frac{EIg}{w}} \quad (22-11)$$

where f_n is the resonance frequency in hertz, γ_n is the coefficient of bending resonance, L is the length of pipe between supports in meters (inches), E is the modulus of elasticity in Pascals (pounds force per square inch), I is the moment of inertia in meters to the fourth power (inches to the fourth power), g is the acceleration due to gravity [9.81 m/s^2 (386 in./s^2)], and w is the weight (including fluid) per unit length in Newtons per meter (pounds force per inch). In Equation 22-11, it is assumed that the pipe is straight and has a constant diameter. Values for γ_n are given in Figure 22-11 for the fundamental and second-

order resonance for various boundary conditions. The moment of inertia for pipe is

$$I = \frac{\pi}{64} [(OD)^4 - (ID)^4] \quad (22-12)$$

where OD and ID are the outside and inside diameters of the pipe. Weights of pipe are readily available from the pipe manufacturers (see also Tables B-1 to B-4).

Another condition of interest to designers is a straight section of uniform pipe with an added center mass, which may be a valve or some other device supported by the pipe. Blevins [5] has published an expression for the approximate natural frequency for simply supported boundaries:

$$f_1 = \frac{2}{\pi} \sqrt{\frac{3EI}{L^3(M_o + 0.49M_p)}} \quad (22-13)$$

where M_o is the central concentrated mass in kilograms (pounds mass) and M_p is the total mass of the pipe and fluid between the support points. The other terms are as defined previously.

An approximate solution for clamped boundaries and noncentered masses is also given by Blevins, but it is too lengthy to reproduce here. Techniques for computing the natural frequencies of piping systems containing elbows and other complexities are addressed by the Kellogg Company [22].

Forced Vibrations

The source of vibrational energy in a typical piping system is usually the pump (sometimes augmented by

Table 22-7. Boundary Conditions for Pipe Bending Modes

Support	Boundary conditions
Free	Pipe is unrestrained to any motion (moment and shear loads are zero)
Simply supported	Pipe is supported to prevent displacement but allow rotation
Clamped	Pipe is clamped to prevent both displacement and rotation
Vibration isolated	Pipe is resting on a compliant support (e.g., a spring) that allows both displacement and rotation

an engine drive). The energy can be transmitted to the pipe either through the coupling between the two or through the fluid in the pipe. The coupling between the pump and the pipe wall transmits vibrations from the pump housing, which is usually dominated by the rotational frequency of the pump shaft and the pump frequency (the shaft frequency times the number of blades, teeth, or pistons). The vibration of a pipe due to a continuous energy source (e.g., a pump) is called “forced vibration.” The term “forced vibration” is used because the system vibrates at the driving frequency of the source in addition to its own natural frequencies.

The amplitude of the forced vibration of a pipe depends on the amount of energy and frequencies introduced by the pump as well as the physical characteristics of the piping system. If the forced vibration occurs at one of the natural frequencies of the pipe, system vibration can be amplified to extremely high levels. This condition is called “resonance.” The amplification of vibration near resonance follows the same basic laws discussed in Section 22-7.

Most metallic pipes possess values of δ_r that are much less than 0.05 and even as low as 0.02. Consequently, during resonance ($f_d = f_n$) the vibration amplitudes, depending on the damping factor, can be as much as 10 to 25 times that of the nonresonant condition. Most vibration-induced piping failures are due to resonance. Resonance can best be avoided by designing the piping support system so that the driving frequency is either less than $f_1/2$ or greater than $3f_1/2$.

The problems introduced by variable-speed pumps are demonstrated in Example 22-2. The variable-

speed pump can also excite structural resonances of the building, a problem discussed in detail in Section 22-12.

The spacings between pipe supports should be approximately equal, but not necessarily exactly equal. Unequal support spacings simply add additional natural frequencies to the system, which reduces the frequency spacing between the driving frequency and the natural frequencies of the piping system. However, there is some merit to varying the pipe support spacing slightly (e.g., 10% or less) in an irregular manner, particularly if the system must operate near resonance. Adjacent sections of piping are more likely to reinforce vibration at resonance if the support spacings are identical. The damping factor is so low for most pipes that spacing variations as small as 2% should be adequate to minimize this problem.

If it is necessary to minimize bending mode vibrations in piping systems, observe the following design guidelines:

- Anchor the pipe to the structure at interval spacings that avoid natural frequencies in close proximity to forcing frequencies.
- Eliminate sharp elbows in the piping system—especially in the vicinity of pumps—since the fluid acceleration due to the change in direction at elbows helps transfer the energy of fluid pressure pulsations into bending mode vibrational energy in the pipe.
- Install flexible connections between the pump and piping at the inlet and discharge.

Example 22-2

Pipe Vibration with a Variable-Speed Centrifugal Pump

Problem: An end-suction centrifugal pump discharges clean water into a Class 53 ductile iron pipe 600 mm (24 in.) in diameter. The pump has a twin vane impeller that rotates at speeds between 560 and 800 rev/min. Design a piping support system free of bending mode resonance problems.

Solution: The rotating frequency of the pump shaft and impeller varies from $f_{\min} = 560/60 = 9.3$ Hz to $f_{\max} = 800/60 = 13.3$ Hz, and the fundamental blade passage frequency varies from $f_{\min} = 2(9.3) = 18.6$ Hz to $f_{\max} = 2(13.3) = 26.6$ Hz. The second and third harmonics (two and three times the fundamental, respectively) of the blade passage frequency vary as follows: $f_2 = 37 - 53$ Hz and $f_3 = 56 - 80$ Hz.

Support spacing requirements. Select the supporting spacing to avoid the resonance condition (within 50% as recommended previously). Because a variable-speed pump is used, virtually all frequencies above 18.6 Hz are possible driving frequencies or harmonics thereof. There is an available window between 13.3 and 18.6 Hz, but the window is not wide enough to meet the 50% clearance requirement. Because there is no practical method of completely

avoiding the pipe resonances, select a support spacing that puts the fundamental pipe resonance above 100 Hz. The resonance problem would thus be moved outside of the frequency region where vibration problems are most common, thus reducing the amplitude of vibration because displacement is inversely proportional to the frequency.

Resonance frequencies. Compute these by using Equation 22-11, which requires the determination of I (from Equation 22-12 and Tables B-1 and B-2), E (from Tables 4-1 or A-10), w (from Tables B-1 and B-2), γ_n (from Figure 22-11), and L .

SI Units

$$I = \frac{\pi}{64} [(0.655)^4 - (0.629)^4] = 0.00120 \text{ m}^4$$

$$E = 166 \times 10^6 \text{ kPa} = 1.655 \times 10^{11} \text{ N/m}^2$$

SI Units

$$w = w_{\text{water}} + w_{\text{pipe}}$$

From Tables B-1 and A-8, for 1 meter of length:

$$w_{\text{water}} = (0.308 \text{ m}^2)(998.2 \text{ kg/m}^3) \\ (9.81 \text{ m/s}^2) = 3020 \text{ N/m}$$

$$w_{\text{pipe}} = (170 \text{ kg/m})(9.81 \text{ m/s}^2) = 1670 \text{ N/m}$$

$$w_{\text{liner}} = (2.38 \times 10^{-3} \text{ m})\pi(0.627 \text{ m}) \times \\ (2400 \text{ kg/m}^3)(9.81 \text{ m/s}^2) = 110 \text{ N/m}$$

$$w \text{ per meter} = 4800 \text{ N}$$

$$\gamma_n = 3.14$$

$$\text{Try } L = 3 \text{ m}$$

$$f_1 = \frac{(3.14)^2}{2\pi(3)^2} \sqrt{\frac{(1.655 \times 10^{11})(0.00120)(9.81)}{4800}} \\ = 111 \text{ Hz}$$

$$f_2 = \left(\frac{\gamma_2}{\gamma_1}\right)^2 f_1 = \left(\frac{6.28}{3.14}\right)^2 \times 111 = 444 \text{ Hz}$$

U.S. Customary Units

$$I = \frac{\pi}{64} [(25.8)^4 - (24.9)^4] = 2880 \text{ in.}^4$$

$$E = 24 \times 10^6 \text{ lb/in.}^2$$

U.S. Customary Units

$$w = w_{\text{water}} + w_{\text{pipe}}$$

From Tables B-2 and A-9, for 1 in. of length:

$$w_{\text{water}} = (3.32 \text{ ft}^2)(62.3 \text{ lb/ft}^3) \\ (1 \text{ ft})/12 = 17.2 \text{ lb/in.}$$

$$w_{\text{pipe}} = (114 \text{ lb/ft})(1 \text{ ft})/12 = 9.5 \text{ lb/in.}$$

$$w_{\text{liner}} = (3/32 \text{ in.})\pi(24.7 \text{ in.})(150 \text{ lb/ft}^3) \\ (1/1728) = 0.71 \text{ lb/in.}$$

$$w \text{ per inch} = 27.4 \text{ lb}$$

$$\gamma_n = 3.14$$

$$\text{Try } L = 118.1 \text{ in.}$$

$$f_1 = \frac{(3.14)^2}{2\pi(118)^2} \sqrt{\frac{(24 \times 10^6)(2880)(386)}{27.4}} \\ = 111 \text{ Hz}$$

$$f_2 = \left(\frac{\gamma_2}{\gamma_1}\right)^2 f_1 = \left(\frac{6.28}{3.14}\right)^2 \times 111 = 444 \text{ Hz}$$

If these guidelines are followed, the worst situation that could result is a pipe system resonance excited by one of the higher harmonics of the pump. The displacement amplification may reach 20 or even 50, but the displacement of the pipe would also be reduced by the increase in vibration frequency and the decrease in energy available from the pump at higher harmonics. By increasing the natural frequency of the piping system from 15 Hz (the center of the available window between 13.3 and 18.6 Hz) to 112 Hz, the displacement amplitude is reduced by a factor of $15/112$ or 0.134 if equal energy in all harmonics is assumed. If harmonics above f_3 are assumed to have less than $1/10$ the energy of the fundamental

(a safe assumption for centrifugal pumps), then, even with a displacement amplification of 50 at 112 Hz (assuming the worst condition), the amplitude of the pipe vibration would be less than 0.134 times 50 divided by 10. This displacement is less than 70% of the value that would be expected at 15 Hz if a spacing of 10 m (30 ft) had been selected to fit into the 13.3- to 18.6-Hz window.

Another solution to Example 22-2 would be to switch to a constant-speed pump. If the specified pump were changed to a triple vane impeller centrifugal pump driven at a constant speed of 1200 rev/min, the driving frequencies would be 20 Hz (the shaft speed), 40 Hz (two times the shaft speed),

60 Hz (the pump frequency), 120 Hz, 180 Hz, and so on. A pipe spacing of 3 m (118 in.) would not be satisfactory because its resonance frequency (112 Hz) is close to the second harmonic of the blade passage frequency (120 Hz). With this pump, pipe resonance should ideally be about 90 Hz. Although the 90-Hz pipe resonance would provide only 30% clearance to the nearest driving frequencies, it would be preferred because the nearest driving frequencies are higher harmonics (which typically have less energy than the fundamental) and because the higher frequency would produce lower displacement amplitudes. Equation 22-11 can be rearranged to solve for L , which yields 3.33 m (10.9 ft) for 90 Hz.

Acoustic Resonances

Periodic pressure fluctuations caused by the pump can also excite acoustic resonances within the piping system. Just as there are an infinite number of bending mode resonances in a piping system, there are also an infinite number of acoustic resonances. An acoustic resonance is created when a discontinuity in the piping system (e.g., a termination, elbow, or branch) reflects a pressure wave. When a pressure wave propagating down the pipe reaches a discontinuity, a portion of the energy in the wave is reflected back toward the source while the rest of the energy in the wave continues down the pipe. If the length of the pipe and the speed of wave propagation in the fluid are such that the initial and reflected waves are in phase, the system operates at an acoustic resonance. In actuality, the pressure wave reflects back and forth at each discontinuity several times, creating what is called a “standing wave” in the pipe. The number of times a wave is reflected before it dissipates depends on the type of discontinuity (i.e., the strength of the reflection at each end) and the viscous losses in the system. Terminations or abrupt changes in area provide the strongest reflections. Reflections from small branches and 45-degree elbows are generally weak enough to be disregarded.

What occurs in the fluid when a system operates at an acoustic resonance is graphically illustrated in Figure 22-12. The instantaneous fluid pressure (with the movement in the pipe wall greatly exaggerated) is shown in a fixed section of pipe at four different times. At time $t = 0$, the pressure is unevenly distributed along the pipe in a sinusoidal manner with the peak pressure locations separated spatially by a distance, γ , the wavelength of the acoustic resonance in the fluid. At $t = f_n/4$, the pressure distribution and pipe shape are uniform. At $t = f_n/2$, the uneven dis-

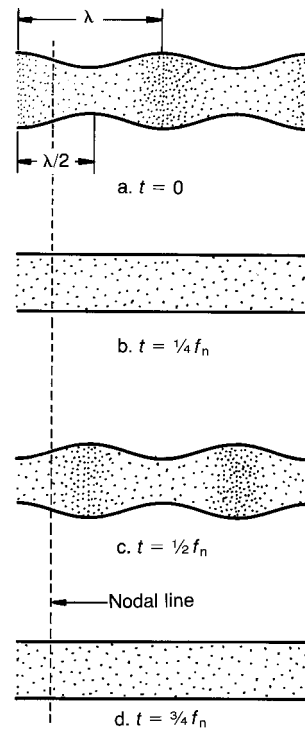


Figure 22-12. Internal pressure distribution at acoustic resonance.

tribution returns, except that the location that was at high pressure (at $t = 0$) is now a low-pressure zone, and vice versa. It is important to note that the average pressure in the pipe (averaged over a time interval of $1/f_n$ or its multiple) is the same at all points.

There are fixed locations along the pipe where the internal pressure does not oscillate about the average pressure. These points are called “nodes,” and they also are separated by a distance, $\lambda/2$, along the pipe. Note that the nodes and antinodes (which are regions of high-pressure fluctuations) of acoustic resonances are at fixed locations that are determined by the pipe system geometry. They do not move with the fluid in the pipe.

The natural acoustic frequencies of a straight, uniform length of pipe are determined by the pipe length, the boundary conditions at each end, and the compressional wave propagation speed in the fluid (speed of sound). Depending on the impedance presented by the discontinuity, the acoustic wave reflected by a discontinuity may be reflected with or without a 180-degree phase reversal. In general, a pipe opening into a large volume with an abrupt termination creates a large reflection with a 180-degree phase shift.

This termination is called an “open-end” condition. At the resonance frequency of the open-end condition, the system node is located at the termination. On the other hand, a pipe section that terminates with a blocked end cap, a tee connection, or a 90-degree elbow is said to have a “closed-end” condition, which provides no phase reversal for the reflected wave. Consequently, the dynamic pressure is greatest at the termination and the nodes occur at other positions along the pipe.

For straight runs of uniform pipe, the acoustic natural frequencies for pipes open at both ends or closed at both ends can be computed as follows:

$$f_n = \frac{nc}{2L} \quad (22-14)$$

where n is the resonance order ($n = 1, 2, 3, \dots$), c is the wave propagation speed of fluid, and L is the length of the column of fluid between discontinuities where a reflection is anticipated.

For pipes open at one end and closed at the other end,

$$f_n = (2n - 1) \frac{c}{4L} \quad (22-15)$$

Note that L has nothing to do with the spacing of pipe supports or the spacing of couplings.

In addition to individual sections of water columns, it is also possible to encounter acoustic resonances that incorporate one or more individual water columns, nonuniform sections of pipe, and other complexities. The equations for computing these frequencies are very complex—well beyond the scope of this text (see *Design of Piping Systems* [22]).

Speed of Sound

The speed of sound in liquids is

$$c_o = \sqrt{\frac{\gamma B}{\rho}} \quad (22-16)$$

where c_o is speed of sound in a free liquid in meters per second (feet per second), γ is the ratio of specific heats (1.0 for fresh water and 1.01 for sea water), B is the isothermal bulk modulus of the fluid in pascals (pounds force per square foot), and ρ is the density of the fluid in kilograms per cubic meter (slugs per cubic foot). In general, these quantities vary somewhat with temperature and pressure in a complex manner as shown in Tables A-8 and A-9. γ is further

defined as c_p/c_v , where c_p is specific heat capacity of the fluid at constant pressure and c_v is the specific heat capacity at constant volume. In SI units, an empirical expression for the speed of sound in fresh water is

$$c_o = 1403 + 5.02T - 0.055T^2 + 0.0003T^3 + 0.00173P \quad (22-17a)$$

where c_o is the speed of sound in meters per second, T is the temperature in degrees Celsius, and P is the gauge pressure in kilopascals.

In U.S. customary units,

$$c_o = 4248 + 13.23T - 0.072T^2 + 0.00016T^3 + 0.0392P \quad (22-17b)$$

where c_o is speed of sound in feet per second, T is the temperature in degrees Fahrenheit, and P is the gauge pressure in pounds force per square inch.

The speed of sound in salt water is slightly higher (by approximately 3%) than it is in fresh water due to the effects of salinity. The speed of sound can be lowered dramatically by air bubbles in the fluid. As long as the concentration of air is less than 0.01% by volume, the effect is insignificant. However, when the air concentration reaches 0.1%, the speed of sound is reduced by nearly 50% (see Streeter and Wylie [23] for more information on the effects of air bubbles).

In addition to its dependence on temperature and pressure, the speed of sound is also influenced by the pipe. The finite mass and compliance of the pipe wall decreases the speed of sound as the pipe wall thickness decreases and the pipe diameter increases. The expression for the speed of sound within a pipe is

$$c = \frac{c_o}{1 + B(ID/Ee)} \quad (22-18)$$

where c_o is given by Equation 22-17, B is the bulk modulus of the fluid, ID is internal pipe diameter, E is the modulus of elasticity of the pipe, and e is the thickness of the pipe wall. The speed of sound in fresh water within several Class 53 ductile iron pipes is shown in Figure 22-13 as a function of water temperature.

Acoustic resonances do not necessarily cause visible or perceptible vibration in the wall of the pipe, but they can cause very high dynamic pressures inside the pipe at the antinodes. It should be emphasized that the pressure rating for piping is based on internal static (steady) pressures. Safe limits for dynamic

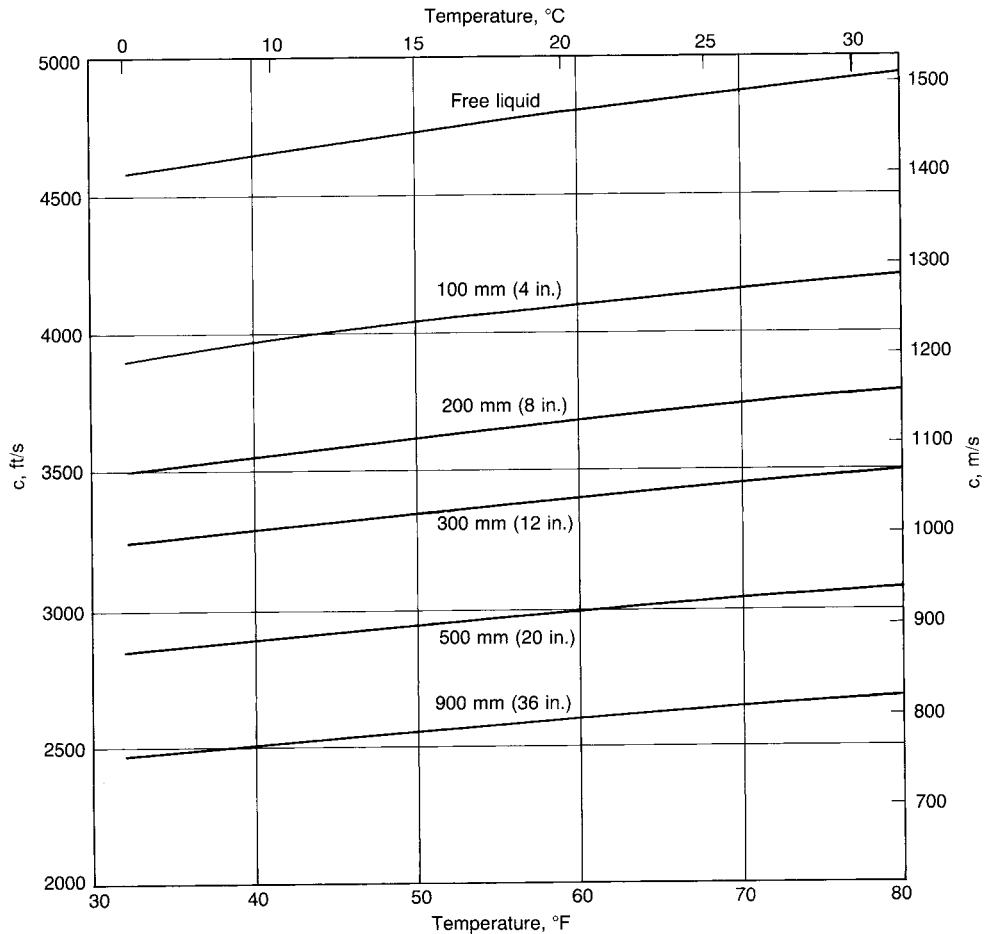


Figure 22-13. Speed of sound in fresh water at atmospheric pressure in Class 53 ductile iron pipe.

pressures are typically less than one-half of the static limits because of pipe fatigue. For most piping systems, this limits safe dynamic pressures to less than 690 kPa (100 lb/in.²) peak-to-peak.

Acoustic resonance may also increase the noise level radiated by the pipe if the frequency is high enough. Depending on the ambient noise level in the plant, and if the frequency is lower than 250 Hz (as it would be for most pumping station systems), radiated noise from an acoustic resonance should not be perceptible.

Suppression Techniques

Once a potential or existing piping vibration or acoustic resonance problem is discovered, one is usually interested in finding a remedy—or at least

a method of improving the situation with as little effort and expense as possible. The solution, of course, requires knowledge of the nature of the problem as well as some specifics concerning the frequencies and levels of vibration. For existing problems, this information is best obtained by direct measurement with an accelerometer and a spectrum analyzer. These measurements usually require the expertise of a qualified vibration consultant. Once the specifics of the problem are identified, a proper suppression technique can usually be applied.

Detuning

Because most vibration problems are associated with the mechanical resonance of one or more elements in the system, detuning is the simplest and most effective

method of solving a piping vibration problem. Detuning merely implies changing either the driving frequency of the pump or the natural frequency of the piping system. Prior to detuning any problem system, it is important to determine the exact driving frequency and natural frequency of the installed piping system, particularly if the anticipated solution is to make a modest change in pump speed. Without this information, it is possible to make the situation even worse if the new pump speed turns out to be closer to the piping's natural frequency than the original pump speed.

Flexible Connections

A flexible connection between the pump and the piping can be an effective means of reducing pipe vibration if the pump is vibration-isolated from the structure and if the flexible connection is truly flexible. When the pump and piping are not isolated from the structure, it is possible for this energy to get back into the piping via the common structure. As shown in Figure 22-14, the most effective flexible connectors are made from molded neoprene rubber. A series of "accordion" corrugations is even more effective. In addition to isolating the piping from the structural vibrations of the pump, this particular design also effectively reduces pulsations in the fluid by virtue of the expansion capabilities of the rubber walls. The use of neoprene rubber is limited to temperatures below 100°C (212°F) and pressures below 1000 kPa (150 lb/in.²).

The conduit for the motor should likewise be flexible to prevent transmitting vibration into the structure via the electrical conduit.

Resilient Supports

Resilient supports for piping systems are used in some industries to provide for expansion and contraction of the piping system as well as to reduce

structure-borne noise from the pipe. In these devices a resilient element (e.g., neoprene rubber or fiberglass) is placed between the pipe wall and the pipe clamp, which allows the pipe to flex slightly and independently of the structure. Theoretically, these clamps also add some additional damping to the bending modes of pipe vibration, but they cannot reduce pressure pulsations in the fluid. Because structure-borne noise is not usually a big problem in pumping stations, the use of these devices is probably not warranted.

Dynamic Absorbers

A dynamic absorber is a passive mass attached to a vibrating element or system with a resilient component (e.g., a spring). The auxiliary mass and spring are attached to the vibrating element and are designed (or "tuned") to vibrate out of phase with the primary system. This results in significant vibration, which is significantly reduced at the resonance frequency. The tuning of the auxiliary mass and spring is critical because these auxiliary elements create a two-degree-of-freedom system that may cause an increase of vibration level at other nearby frequencies. The use of a dynamic absorber is applicable only for a system operating at or near resonance and should be attempted only with the assistance of a qualified vibration consultant.

Expansion Chambers

Expansion chambers are simply enlarged sections of pipe with abrupt discontinuities at each end, as shown in Figure 22-15. The expansion chamber provides a loss of dynamic pressure (i.e., reduced pressure pulsations) primarily through the interaction of reflections within the chamber. The dynamic pressure losses occur at all frequencies except at 0 Hz and at the resonance frequencies of the expansion chamber (when the length, L , equals an integer multiple of one-half of the wavelength) where no losses occur. The expansion chamber cannot amplify the fluid pulsations in the downstream pipe.

The effectiveness of an expansion chamber can be described by the ratio of the outlet to inlet dynamic pressures:

$$P_{\text{out}}/P_{\text{in}} = \frac{1}{\{1 + [0.25(S_2/S_1 - S_1/S_2)^2] \sin^2(2\pi fL/c)\}} \quad (22-19)$$

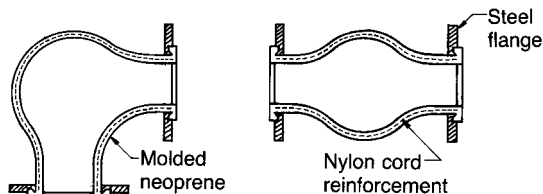


Figure 22-14. Vibration isolators for pipe. (a) Elbow coupling; (b) straight coupling.

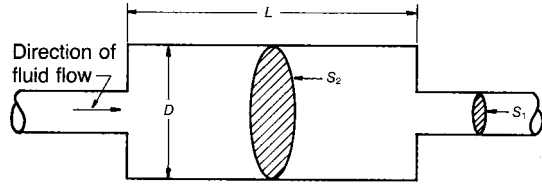


Figure 22-15. Expansion chamber.

where S_1 and S_2 represent the cross-sectional area of the inlet pipe and the expansion chamber, respectively. Note that from this expression there is no pulsation reduction whenever the argument of the $\sin^2$ function is an integer multiple of π . The maximum pulsation reduction occurs at the frequency where the argument of the $\sin^2$ function is equal to $\pi/2$, or

$$f = (2n - 1)c/4L \quad (22-20)$$

Thus, maximum reduction occurs when the length of the expansion chamber equals one-fourth of the wavelength, three-fourths of the wavelength, five-fourths of the wavelength, and so on.

In most pumping applications, the concern is with low-frequency performance where the expansion chamber length is only a fraction of the wavelength of the pump frequency. Under these circumstances, the expansion chamber is designed to ensure that the ratio S_2/S_1 and the pump frequency are high enough to produce the necessary pulsation reduction.

Tuned Resonators

Tuned resonators are sections of pipe that are added to the system in order to remove dynamic pressures in a very narrow frequency band. The simple example of a side branch resonator (a closed-end tube attached to the main pipe) is shown in Figure 22-16. If the effective length of the side branch resonator is exactly equal to one-fourth of the wavelength of the pump frequency, it can reduce the dynamic pressure in the downstream piping dramatically—by as much as 99%. It is important to realize that this device works only at the following particular frequencies:

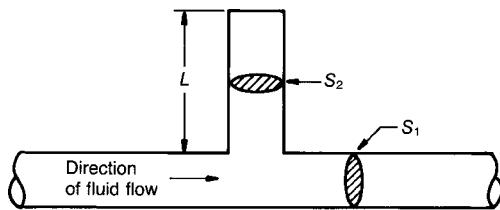


Figure 22-16. Side branch resonator.

$$f_n = (2n - 1)c/2L \quad (22-21)$$

where $n = 1, 2, 3, \dots$. The length of the resonator must equal one-half of the wavelength, three-halves of the wavelength, and so on. These lengths differ from those of the expansion chamber discussed previously, because the side branch resonator is closed at one end, whereas the expansion chamber is open at both ends. The sound wave in the expansion chamber can propagate the distance L in a single pass. But the sound wave in the side branch resonator travels the distance L in one direction, is reflected and travels back again, and gives rise to the different factors ($2L$ versus $4L$) in Equations 22-20 and 22-21.

At other frequencies the resonator is transparent to the system; that is, it acts as though it were absent or “invisible.” Therefore, consider this device only for single-speed pumps moving fluids with temperatures that remain constant. Side branch resonators are not economically feasible for low-speed pumps because the resonator length becomes very long.

Pulsation Dampers

A pulsation damper reduces the dynamic pressure fluctuations within a piping system, and it can also reduce the bending mode vibration levels of the piping system if the pipe vibration is induced by the pulsating fluid. Two types are commonly used. The most common pulsation damper is a fluid-filled spherical vessel mounted in line with the piping at the discharge of the pump. A sphere is commonly utilized to achieve maximum volume and maximum pressure rating with minimum weight and cost. The general design concept is the same as that of the expansion chamber discussed above. Typically, pulsation dampers reduce dynamic pressures by 75% and more.

The other type of pulsation damper is the air chamber (see the hydropneumatic tank in Chapter 7), which also functions as an accumulator. This device is essentially a pressurized expansion tank (partially filled with gas) connected to the piping near the pump discharge. The accumulator adds and accepts fluid from the piping through an orifice as the dynamic pressure fluctuates. The accumulator is not an in-line device; therefore, its effectiveness at reducing high-frequency (50 Hz and above) dynamic pressures in the downstream piping is limited. Neither the accumulator nor the pulsation damper has a sufficient volume of gas to control water hammer in most instances. Pulsation control devices should be considered for both the suction and discharge flow lines because these units protect the pump as well as the piping; usually, they also improve pump performance.

Example 22-3
Dynamic Pressure from a Plunger Pump

Problem: A reciprocating (single-acting plunger duplex) pump delivers fresh water at 3.15 L/s at 2413 kPa (50 gal/min at 350 lb/in.²) working pressure from a 5°C (50°F) reservoir at 400 rev/min into a Class 53 ductile iron pipe 300 mm (12 in.) in diameter that terminates in a large holding tank. Assume that the peak dynamic pressure generated by the pump is 25% of the static head (50% peak-to-peak). Compute the acoustic resonance frequencies and determine if the system is in danger of fatigue fracture if the pipe is relatively straight, has a total length of 64 m (210 ft), and has a yield strength of 290,000 kPa (42,000 lb/in.²).

Solution: The speed of sound in the pipe can be computed to be 1007 m/s (3303 ft/s) from Equations 22-17 and 22-18. If the effect of pressure is ignored (because it is so small), the speed of sound in the pipe can also be estimated from Figure 22-13. The driving frequency of the pump is $f_d = (400)(2)/60 = 13.3$ Hz, and the harmonics of the driving frequency are 26.6 Hz, 40 Hz, etc. The pipe is open at one end, and the pump itself closes the other end. From Equation 22-15 [$f_n = (2n - 1)c/4L$], the acoustic resonances are as follows:

SI Units**U.S. Customary Units**

$$f_1 = (2 \times 1 - 1)1007/(4 \times 64) = 3.93 \text{ Hz}$$

$$f_1 = (2 \times 1 - 1)3303/(4 \times 210) = 3.93 \text{ Hz}$$

The second- and third-order acoustic resonances are

$$f_2 = (2 \times 2 - 1)1007/(4 \times 64) = 11.8 \text{ Hz}$$

$$f_2 = (2 \times 2 - 1)3303/(4 \times 210) = 11.8 \text{ Hz}$$

$$f_3 = (2 \times 3 - 1)1007/(4 \times 64) = 19.7 \text{ Hz}$$

$$f_3 = (2 \times 3 - 1)3303/(4 \times 210) = 19.7 \text{ Hz}$$

The second-order acoustic resonance (11.8 Hz) is close (within about 10%) of the fundamental driving frequency of the pump ($f_o = 0.89f_d$), which could excite the first-order acoustic resonance within the pipe. A pulsation damper is required in this installation to reduce the dynamic pressure, but the pulsation damper will not change the resonance match between the driving frequency of the pump and the water column in the pipe.

Effect of damping. If the pulsation damper reduces the dynamic pressures in the discharge pipe by, for example, 75% and the pressure amplification for the acoustic resonance is estimated to be about 5 (from Figure 22-6, for 2 to 10% of critical damping), the maximum dynamic pressure can be computed as working pressure times 25% times (1 – pulsation damping reduction) times the acoustic resonance amplification factor, or

$$(2413)(0.25)(1 - 0.75)(5) = 754 \text{ kPa}$$

$$(350)(0.25)(1 - 0.75)(5) = 109 \text{ lb/in.}^2$$

These pressures meet the 50% of rated value criterion, but they are higher than the 690-kPa (100-lb/in.²) guideline. If the damping ratio were as low as 0.02, amplification factors could reach 20, which could bring the dynamic pressures inside the pipe to as high as 3020 kPa (440 lb/in.²), and the total (working plus dynamic) pressure would be 5440 kPa (790 lb/in.²). From Equation 4-2 ($s = PD/2e$), the hoop tensile stress in the pipe at peak pressure would be

$$s = \frac{5440 \times 0.335}{2 \times 0.0102} = 89,300 \text{ kPa}$$

$$s = \frac{790 \times 13.2}{2 \times 0.40} = 13,000 \text{ lb/in.}^2$$

This seems safe enough, but the maximum yield strength of 290,000 kPa (42,000 lb/in.²) is applicable only to static stress levels. Dynamic stress levels must be substantially lower to avoid fatigue failure.

Changing operating speed. In addition to the pulsation damper, the operating speed of the pump should be changed to avoid operation near the acoustic resonance frequencies and, thus, reduce the internal dynamic pressure. A suitable choice might be a smaller duplex pump running at 480 rev/min, which would yield driving frequencies of 16 Hz, 32 Hz, etc., and

would result in an estimated pressure amplification of less than about 2.0 for all values of the damping ratio (see Figure 22-6) for $f_d/f_o = 16/11.8 = 1.356$. The maximum dynamic pressure at the driving frequency of the pump would then be $(2420)(0.25)(1 - 0.75)(2) = 302 \text{ kPa}$ (44 lb/in.^2), which is safe by both criteria.

22-11. Vibration of Drive Shafts

Drive shaft vibration is a complicated phenomenon composed of two independent forms of vibration—translational and torsional. The approximation methods presented here are intended to provide only (1) a reasonable idea of safety margins, and (2) sufficient background for pumping station designers to judge properly the results presented by the shaft, engine, or (usually) pump manufacturer, who should be responsible for the design of the entire rotating system.

Translational Vibration

Translational vibration (which is sometimes called “lateral vibration”) involves the oscillatory motion of the drive shaft about its ideal “at rest” centerline. The shaft has natural frequencies and assumes related mode shapes just as piping systems do, and all of the equations and analyses on piping vibration in Section 22-10 also apply to drive shafts.

Drive shafts are much more sensitive to vibration problems than are piping systems, because the drive shaft rotates and any unbalance in the drive shaft or misalignment in the coupling causes the shaft to vibrate or deflect laterally, which results in “shaft whip.” This vibration is very difficult to measure directly, but it can be measured indirectly via sensors at the shaft support bearings. Avoiding rotational speeds that match or excite the natural or resonance frequencies of the translational vibration of the drive shaft is imperative. These speeds are called “critical speeds,” and they are determined entirely by the physical properties of the shaft.

If the drive shaft is assumed to be a uniform, straight, steel tube with a circular cross-section (or a solid cylinder if $ID = 0$), Equation 22-11 can be rearranged into a somewhat simpler form for a quick evaluation of the lowest natural frequency.

$$f_1 = \frac{\beta}{L^2} \sqrt{(OD)^2 + (ID)^2} \quad (22-22)$$

$$\beta = \frac{\gamma_1^2}{8\pi} \sqrt{\frac{E}{\rho}}$$

where f_1 is the first-order shaft resonance in hertz, OD is the outer diameter of the shaft in millimeters (inches), ID is the inner diameter of the shaft in millimeters (inches), L is the effective length of the shaft in millimeters (inches), and β is $2.03 \times 10^6 \text{ mm/s}$ in SI units for universal (pinned) joint couplings or $4.6 \times 10^6 \text{ mm/s}$ for direct-coupled (fixed) shafts. In U.S. customary units, β is $7.9 \times 10^4 \text{ in./s}$ for pinned joint couplings and $1.79 \times 10^6 \text{ in./s}$ for fixed shafts. A drive shaft also has other modes of lateral vibration at higher frequencies, but these are usually not problem frequencies because the rotational speed of shafts is usually below the first resonance.

When using Equation 22-22 to compute the resonance frequencies of the drive shaft, the length, L , of the shaft should be measured from the center of the universal joints on both the driving and the driven ends (see Figure 22-17). Steady bearings may be used as intermediate supports to reduce L and increase the resonance frequency out of the region of concern.

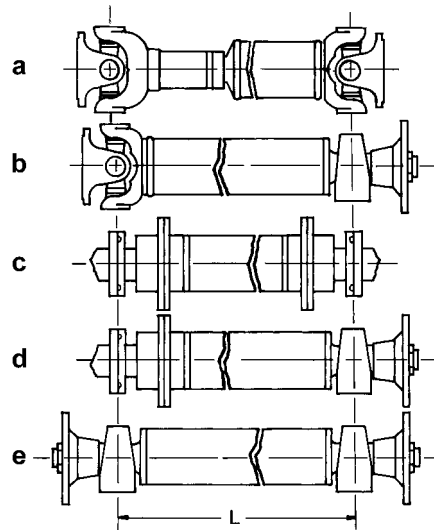


Figure 22-17. Typical drive shaft couplings. (a) Universal joint drive shaft. Vibration is accentuated if the two shafts are not exactly coaxial; (b) universal joint coupling shaft with right end flanged to connect with another shaft; (c) solid coupled shaft; (d) solid coupled coupling shaft; (e) middle shaft (with two support bearings) in a series of three shafts.

To avoid operating at a critical speed, the drive shaft should be designed to ensure that the highest rotational shaft speed is less than 75% of the lowest natural frequency. Furthermore, rotational speeds that are nearly equal to one-half of the lowest bending mode frequency of the shaft should be avoided because secondary moments from universal joints produce an excitation frequency at twice the shaft speed. Combining these two criteria gives two acceptable ranges for the lowest shaft frequency: (1) 1.4 to 1.6 times the shaft rotational speed, or (2) at least 2.5 times the shaft rotational speed.

Although the diameter of a drive shaft can be determined by the length of the shaft and the maximum operating speed, it is good design practice to limit the length to no more than 30 times the shaft diameter. Extremely long shafting must be divided into sections supported by steady bearings mounted on cross beams at the joints. If multiple sections are required, each section should, if possible, be of approximately equal length. The resonance frequencies of, say, four sections of nearly equal length are almost the same, but, if the lengths are substantially different, the additional resonance frequencies complicate the problem.

Support of Steady Bearings

When a drive shaft is divided into more than one section, the intermediate bearings must be supported adequately to resist not only the static load but also the dynamic load induced by the shaft rotation. Auxiliary steel members attached directly to the building structure usually support the intermediate bearings. A common rule of thumb is to make sure that the lowest natural frequency of the support structure is at least four times the shaft rotational speed. If the lateral support of the steady bearings does not meet this criterion, the boundary conditions will not approximate the “pinned” or “fixed” conditions and the first resonance frequency of the shafts will be lower than the calculated value.

The natural frequency of a single span support beam is given by Equation 22-11. The values of γ_n for clamped-clamped and pinned-pinned beams are given in Table 22-8 and, for modes 1 and 2, in Figure 22-11.

The length of the beam is often such that its lowest natural frequency is less than the recommended minimum of four times the shaft rotational speed. If so, it may be necessary to provide intermediate supports for the beam. If the main beam is supported at the center of the span, the same equation can be used (with modified values of γ_n) to compute the new resonance frequencies. Blevins [5] gives modified values of γ_n for

Table 22-8. Frequency Coefficients for Bending Modes of Straight Beams

Mode (n)	Clamped-clamped (γ_n)	Pinned-pinned (γ_n)
1	4.73	3.14
2	7.85	6.28
3	11.00	9.42
4	14.14	12.57

various boundary conditions and positions of the intermediate support. In the special case of a center support with pinned boundary conditions (no displacement, but unrestrained rotation), the modified values of γ_n are given in Table 22-9. Most real-world boundary conditions for “rigidly” attached beams are such that some displacement and rotation occur at the ends of the beam no matter how they are fastened to the structure. For this reason, a more suitable choice for the frequency coefficient, γ_n , is a number somewhere between the theoretical values for the clamped-clamped and pinned-pinned boundaries.

Table 22-9. Frequency Coefficients for Bending Modes of Straight Beams with Center Supports^a

Mode (n)	Clamped-clamped (γ_n)	Pinned-pinned (γ_n)
1	7.8	6.4
2	9.5	7.8
3	14.2	12.6
4	15.7	14.2

^aThe distance L is the full length of the beam—not the distance from the end to the center support.

Torsional Vibration

Torsional vibration problems occur less frequently than translational vibration problems, but they can be just as troublesome and sometimes catastrophic. Because they are not detected by our senses until failure occurs, it is important to select equipment components that preclude this possibility.

Torsional vibration can be magnified significantly if a torsional exciting frequency is close to the torsional natural frequency. The disturbing force in Figure 22-18 is the dynamic torque that oscillates about the average torque. At all frequencies less than 1.3 times the natural frequency, some magnification of the disturbance occurs, whereas at greater frequencies the disturbing force is damped. The maximum magnification occurs at resonance, and the value 3.3 shown is an arbitrary choice

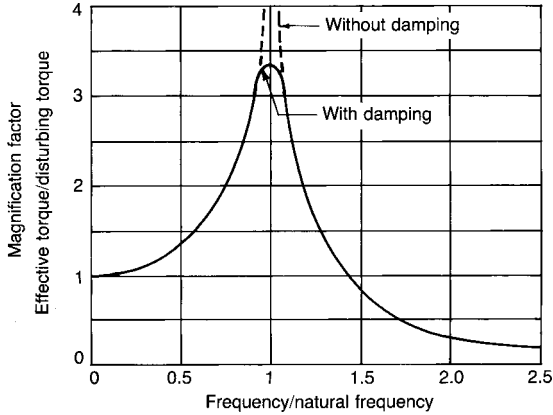


Figure 22-18. Magnification of a disturbing force in torsional vibration.

that is nevertheless typical. The maximum magnification can be calculated, but it depends on the damping factor, which is difficult to ascertain. The simplest and best procedure for designing shafts is to position the torsional natural frequency well outside any exciting frequencies. Typical exciting frequencies include: (1) the shaft rotational speed, (2) the pump vane passage frequency and its harmonics, and (3) the universal joints, which generate a frequency twice that of the shaft speed.

To avoid torsional resonances, keep the fundamental exciting frequencies less than 50% of torsional natural frequency. At times, it is necessary to run the system at speeds where harmonic exciting frequencies are above the natural frequency. This mode of operation is not a problem (1) if the exciting frequencies are more than 1.5 times the natural frequency, and (2) if the pass through the natural frequency range is accomplished quickly (which is normally true in electric motor pumping installations). If the exciting frequency is at least 1.5 times the torsional natural frequency, the effective disturbing system forces are less than the driver input disturbing forces. If the exciting frequency is less than 0.5 times the torsional natural frequency, the magnification factor will not exceed 1.3. Steel drive shafts typically have a minimum torsional elastic limit of 235 kPa (34,000 lb/in.²) or an endurance limit (infinite life) of 117 kPa (17,000 lb/in.²). The 1.3 stress magnification factor combined with the endurance limit means that a maximum allowable design torsional stress of $117/1.3 = 90$ kPa (13,000 lb/in.²) ensures safety if torsional exciting frequencies are less than 0.5 or more than 1.5 times the torsional natural (resonance) frequencies.

In the simplest of systems (a pump driven by an electric motor with intermediate shafting), the torsional natural frequency of the system can be computed in SI units as follows:

$$f_o = \frac{1}{2\pi} \sqrt{\frac{K(J_m + J_p)}{J_m J_p}} \quad (22-23)$$

where f_o is the torsional resonance frequency in hertz, K is the torsional rigidity of the shaft in Newton-meters per radian, J_m is the mass moment of inertia of the motor in meters squared times kilograms, and J_p is the mass moment of inertia of the pump in meters squared times kilograms.

In U.S. customary units, K has units of pounds force-inches per radian and J_m and J_p have units of inches squared times pounds mass. The mass moment of inertia of a motor rotor is approximated by that of a cylinder of equal weight and diameter in which

$$J = \frac{D^2 M}{8} = \frac{D^2 W}{8g} \quad (22-24)$$

where D is the diameter in meters (inches), M is the mass in kilograms (pounds mass), and W is the weight in Newtons (pounds force). Approximating the mass moment of inertia of the impeller and the water within the casing is neither easily done nor likely to be accurate, so obtain it from the equipment manufacturer. Accurate drive shaft torsional rigidity values can be obtained from computer programs (usually proprietary) or calculated directly by analyzing the shaft in sections and combining the rigidity constant, K , for each individual section to obtain the torsional rigidity of the entire shaft. The torsional rigidity of the stubs (the connecting link between two shafts), however, depends on the design—which is different for each manufacturer—so the gross torsional rigidity of shaft and stub must also be obtained from the maker.

The torsional rigidity of a uniform, hollow, circular shaft of length L is expressed as

$$K = \frac{\pi G}{32L} [(OD)^4 - (ID)^4] \quad (22-25)$$

where G is the modulus of elasticity of steel in shear, which is 82.7×10^9 Pa (12×10^6 lb/in.²), and linear dimensions are in meters (inches). For solid shafting, use $ID = 0$. If a shaft is made up of two or more sections of shafting with different physical properties, the torsional rigidity of the entire shaft can be computed from the torsional rigidity of the individual sections as

$$K = \frac{1}{\sum_{i=1}^n (1/K_i)} \quad (22-26)$$

Example 22-4
Torsional Vibration in a Shaft

Problem: A two-vane, end-suction centrifugal pump is driven by a 75-kW (100-hp), variable-speed, direct-drive electric motor at speeds varying from 840 to 1200 rev/min (14 to 20 Hz). The power shafting is a universal joint, 3.353-m (132-in.) drive shaft (see Figure 22-17a) with 2.67 m (105 in.) of 101.6-mm (4-in.) OD tubing and a wall thickness of 2.108 mm (0.083 in.). The manufacturer gives the mass moment of inertia as $J_p = 1.729 \text{ kg} \cdot \text{m}^2$ (15.3 lbm · in.²) and $J_m = 1.345 \text{ kg} \cdot \text{m}^2$ (11.9 lbm · in.²). The torsional rigidity of the drive shaft stubs (excluding the tubing between the stubs) is given as 31,400 N · m/rad (277,802 lb · in./rad).

Evaluate the translational and torsional resonances of the shafting and determine if the system is within acceptable guidelines. If intermediate support bearings are required, size the structural members within recommended guidelines.

Solution:

Lateral resonance. The rotating frequencies of the shaft are 14–20 Hz (fundamental) and 28–40 Hz (second harmonic and pump vane passage frequency). From Equation 22-22, the first-order lateral resonance of the shaft is

SI Units

$$f_1 = \frac{2.03 \times 10^6}{(3353)^2} \sqrt{(101.6)^2 + (97.384)^2}$$

$$= 25.4 \text{ Hz}$$

U.S. Customary Units

$$f_1 = \frac{7.99 \times 10^4}{(132)^2} \sqrt{(4)^2 + (3.834)^2}$$

$$= 25.4 \text{ Hz}$$

The maximum shaft speed is 20 Hz, which is 78% of the fundamental resonance of the shaft. Ideally, the shaft speed should not exceed 75% of the first resonance, so this shaft speed is a marginal condition.

Torsional resonance. To compute the torsional resonance frequency, first determine the shaft rigidity, K_s , from Equation 22-25.

$$K_{\text{tubing}} = \frac{\pi(82.7 \times 10^9)[(0.1016)^4 - (0.097384)^4]}{32(2.67)}$$

$$= 50,526 \text{ N} \cdot \text{m/rad}$$

$$K_{\text{tubing}} = \frac{\pi(12.0 \times 10^6)[(4)^4 - (3.834)^4]}{32(105)}$$

$$= 447,935 \text{ lb} \cdot \text{in./rad}$$

From Equation 22-26:

$$K_{\text{shafting}} = [(31,400)^{-1} + (50,526)^{-1}]^{-1}$$

$$= 19,368 \text{ N} \cdot \text{m/rad}$$

$$K_{\text{shafting}} = [(277,802)^{-1} + (447,935)^{-1}]^{-1}$$

$$= 171,463 \text{ lb} \cdot \text{in./rad}$$

Determine the torsional resonance frequency from Equation 22-23:

$$f_o = \frac{1}{2\pi} \sqrt{\frac{(19,368)(1.729 + 1.345)}{(1.729)(1.345)}}$$

$$= 25.4 \text{ Hz}$$

$$f_o = \frac{1}{2\pi} \sqrt{\frac{(171,463)(15.3 + 11.9)}{(15.3)(11.9)}}$$

$$= 25.4 \text{ Hz}$$

This is below the recommended minimum of 1.5 times 20 Hz (the highest shaft speed) and is, thus, unacceptable.

Increasing the shaft OD to 127 mm (5 in.) and the ID to 120 mm (4.72 in.) would increase the first-order lateral resonance to 31.6 Hz and the torsional resonance to 29.7 Hz. This modification would nearly meet the torsional criteria, but the lateral resonance may be excited by the second harmonic of the shaft speed at 948 rev/min (or 15.8 Hz). An alternative solution would be to retain the original shaft diameters and split the shaft into two equal sections with

an intermediate support bearing. The fundamental lateral shaft resonance then becomes 101.8 Hz, which is acceptable, because $f_1/f_d = 5.09$. The new torsional spring constant of each section is 101,052 N · m/rad, which yields a total spring constant of 23,956 N · m/rad and a torsional resonance of 58.2 Hz, which is also acceptable.

If the intermediate shaft bearing is supported by a section of steel angle 20.3 cm × 20.3 cm × 1.9 cm (8 in. × 8 in. × 3/4 in.) thick that weighs 568 N/m (38.9 lb/ft), has a moment of inertia about both axes of $2.9 \times 10^{-5} \text{ m}^4$ (69.7 in.⁴), and spans 6.10 m (20 ft) with fully restrained end points, the fundamental resonance frequency of this beam is, from Equation 22-11 and Figure 22-11,

SI Units

$$f_1 = \frac{(4.73)^2}{2\pi(610)^2} \sqrt{\frac{(2 \times 10^{11})(2.9 \times 10^{-5})(9.81)}{568}}$$

$$= 30.3 \text{ Hz}$$

U.S. Customary Units

$$f_1 = \frac{(4.73)^2}{2\pi(240)^2} \sqrt{\frac{(29 \times 10^6)(69.7)(386)}{(38.9/12)}}$$

$$= 30.3 \text{ Hz}$$

The computed resonance frequency of the beam is insufficient to meet the conservative criterion of four times the maximum operating speed, but if a center support (with pinned boundary conditions) were added to this beam, the lowest beam frequency would be increased by a factor of $(7.85/4.73)^2 = 2.75$. The fundamental beam frequency would be increased to $2.75 \times 30.4 = 83.6 \text{ Hz}$, which is 4.18 times the maximum operating speed of 20 Hz and does meet the recommended criterion. If the bearing were to be located at the center of the main beam (where the center support beam is attached), this problem would not arise because the vibrating force would be applied at a node of this first beam resonance, and it would not be excited.

22-12. Vibration of Structures

Pumping station equipment can cause structures to vibrate at levels exceeding normally acceptable criteria [19], particularly if the equipment is operating at a speed that matches one of the low-order resonance frequencies of the structure. In the design phase, it is too difficult to predict accurately the natural frequencies of supporting structures (building floor slabs, pedestals, etc.), but there are some simple calculations that can be made to determine whether a potential problem exists.

The lowest natural frequency of a supporting structure can be estimated from basic information concerning the static load-carrying capability of the structure. This information is often available directly from the structural engineer. Assuming that the structural member undergoes a static deflection directly proportional to the dead load of the equipment, the dynamic behavior of the structural member is similar to that of a steel spring (for small deflections). If the maximum static deflection of the structure (due to the

weight of the equipment), d , is expressed in millimeters, the natural frequency of the structure in hertz is approximately

$$f_o = 3.13\sqrt{1/d} \quad (22-27a)$$

If d is expressed in inches,

$$f_o = 15.8\sqrt{1/d} \quad (22-27b)$$

This expression is approximately valid ($\pm 25\%$) only when $d < L/500$, where L is the smallest cross-dimension (width or height) of the structure, but this condition should be satisfied by most pumping station buildings. Although the correlation between the vibration of a point mass on a spring and the distributed mass of a concrete floor may seem unreasonable, the two systems vibrate at nearly the same frequency, because the use of deflection in Equation 22-27 tends to compensate for the differences between the two systems.

Much more accurate determinations of structural resonance frequencies can be obtained by modal analysis and other more elaborate techniques, but these methods are usually beyond the scope of most pumping station projects. Determining the maximum structural deflection caused by the static weight of the equipment is sometimes complicated, but approximations that provide an upper limit for d are usually adequate. Practical solutions lie in the close cooperation between the project leader and the structural engineer, which illustrates why a project leader must be much more than an administrator who assigns tasks and responsibilities to others.

Ideally, vibrating equipment should not operate at a rotational speed near the fundamental frequency of the supporting structure. Although problems are unlikely if the rotating speed is 5% greater than or 5% less than the resonance frequency, a 25% safety margin on each side is recommended.

Natural frequencies of a variety of structural members are given by Blevins [5]. In general, the natural frequency of any structure is proportional to the square root of the composite moment of inertia, I , divided by the total mass of the system.

$$f_o \propto \sqrt{I/M} \quad (22-28)$$

Any efforts to change the natural frequency of an existing system, therefore, should be devoted to these two parameters. Notice that doubling the mass of the system (equipment plus structure) reduces the natural frequency by 29% ($1 - \sqrt{0.5}$) if it is assumed that mass can be doubled with no change in the moment of inertia. However, structural modifications usually affect both M and I , so the modifications should be evaluated carefully prior to making the change.

Example 22-5 Vibration of a Floor

Problem: A 300-kVA (400-hp) variable-speed induction motor (900 to 1200 rev/min) is to be mounted on a 100-mm- (4-in.-) thick housekeeping pad over a 200-mm- (8-in.-) thick concrete slab that is continuously supported by the perimeter wall as well as by a concrete beam that divides the slab in the center. The motor is mounted with the shaft vertical and is rigidly bolted to the housekeeping pad. The drive shaft penetrates the floor beside the beam and in the center of the pad. The structural engineer estimates the static deflection of the floor slab (due to the weight of the motor) to be 1 mm (0.04 in.). Evaluate the potential for a floor vibration problem and provide a solution if necessary.

Solution: The driving frequency range of the motor is 15 to 20 Hz for the fundamental and 30 to 40 Hz for the second harmonic. Note that because vibrating machinery produces excitation forces in all directions, the orientation of the motor is of little practical significance.

From Equation 22-27 the estimated natural frequency of the floor is as follows:

SI Units

$$f_o = 15.8 \sqrt{\frac{1}{1}} = 15.8 \text{ Hz}$$

which is in the operating range of the pump's fundamental frequency.

The natural frequency of the floor should be at least 25 Hz (1.25 times the maximum fundamental driving frequency of 20 Hz). Solving for d in Equation 22-27, the maximum allowable deflection is

$$25 = 15.8 \sqrt{\frac{1}{d}} \\ d = 0.4 \text{ mm}$$

U.S. Customary Units

$$f_o = 3.13 \sqrt{\frac{1}{0.04}} = 15.7 \text{ Hz}$$

$$25 = 3.13 \sqrt{\frac{1}{d}} \\ d = 0.016 \text{ in.}$$

This would require a reduction of more than 50% from the original conditions, and it might be very difficult for the structural engineer to design for so small a deflection. As an alternative, consider reducing the natural frequency of the floor so that the lowest driving frequency is about 25% above the natural frequency of the floor. This modification would require a floor

frequency of $15 \text{ Hz}/1.25 = 12 \text{ Hz}$, which (again from Equation 22-27) corresponds to a static deflection of

$$d = 1(15.8/12)^2 = 1.73 \text{ mm}$$

$$d = 0.04(15.8/12)^2 = 0.07 \text{ in.}$$

By directing the structural engineer to alter the supporting structure (decreasing the slab thickness, modifying the beam supports, etc.) to meet the desired static deflection, the floor resonance problem can be avoided. It should be noted that the floor would have other natural frequencies higher than the fundamental frequency approximated by Equation 22-27, and these natural frequencies may become resonant with the higher harmonics of the pump. Evaluating such a problem is so complex that it is beyond the scope of this chapter.

22-13. Noise

The intent of this section is to provide project leaders with some elementary tools to evaluate the severity of sound propagation through the air, including transmission through barriers. A discussion of structure-borne sound (i.e., structural vibrations resulting in subsequent noise radiation from solid surfaces) is of little importance in most pumping stations and, hence, is not included. Heed OSHA regulations and local ordinances, as explained in Section 22-6.

Outdoor Sound Propagation

Noise radiates from a source in all directions, and sound propagates in air at a nominal velocity of 344 m/s (1130 ft/s) at 20°C (70°F). The actual velocity varies somewhat with temperature and barometric pressure, but for practical purposes this dependence can usually be ignored. The intensity of sound decreases with the square of the distance from a source (much like the intensity of light from a light bulb) as the acoustical energy spreads out over a larger and larger surface area. In addition to this “spreading loss” there is also a slight loss due to atmospheric absorption (conversion of acoustical energy into heat), but this can be neglected in most applications.

The sound pressure level, L_p , at a point can be expressed as a function of the distance to the source, x , and the sound pressure level from the source at a reference distance, x_o , in the same direction. This spreading loss can be expressed as

$$L_p(x) = L_p(x_o) + 20 \log_{10} (x_o/x) \quad (22-29)$$

The effects of partial or full barriers, the reflections from surrounding surfaces, and the directivity

of the source are not considered in Equation 22-29, and in an enclosed room the equation may be valid only within a few feet of the source. For an outdoor source, the expression may be reasonably accurate for hundreds of meters provided that (1) there are no barriers or major reflecting surfaces (such as buildings), and (2) the direction of the reference measurement is the same as the direction of interest. In the special case of $x = 2x_o$ (where the location of interest is twice as far from the source as the reference position), the difference in the noise levels at the two positions is 6 dB. Thus, whenever the distance to the source is doubled, the sound intensity drops by 6 dB. Likewise, whenever the distance to the source is reduced by 50%, the sound intensity increases by 6 dB. These general rules are only strictly valid in a free field (i.e., an environment without reflective or absorptive surfaces), but they are approximately valid in many typical situations.

Indoor Sound Propagation

In an enclosed or partially enclosed room, noise radiating from equipment is reflected from the room boundaries and builds up to a higher noise level than if the equipment were located outside. This buildup of noise caused by acoustical energy being contained within the room creates a “reverberant area or field” in the room.

The reverberant field concept is illustrated in Figure 22-19, where noise level is plotted as a function of distance from a noise source. In the region near the source (the direct field), the noise level falls off at the 6 dB per double distance rate because the acoustical energy radiating directly from the source overpowers the reverberant field. As the listener moves farther from the source, the noise level reaches a sound pressure level that is relatively constant. It is this region that is called the reverberant field. In SI

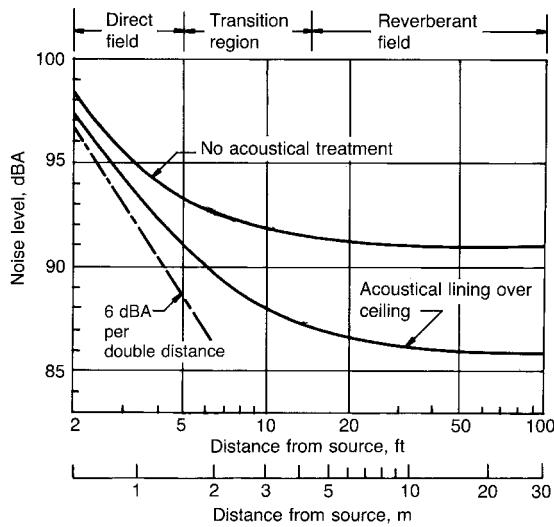


Figure 22-19. Noise level in a room.

units, the noise level in the reverberant field can be approximated by the following expression:

$$L_{p(\text{rev})} = L_p(1 \text{ m}) - 10 \log_{10}(NRC \times S) + 17 \quad (22-30a)$$

where $L_p(1 \text{ m})$ is the sound pressure level 1 m from the source, S is the total surface area of the interior surfaces of the room in square meters, and NRC is the average noise reduction coefficient of the room surfaces exposed to the noise.

In U.S. customary units, the expression is

$$L_{p(\text{rev})} = L_p(3 \text{ ft}) - 10 \log_{10}(NRC \times S) + 27 \quad (22-30b)$$

where S is in square feet.

The noise reduction coefficient is a single number (between 0 and 1.0) that approximates the ratio of the absorbed to the incident acoustical energy. For example, a material with an NRC rating of 0.60 would absorb 60% of the incident sound and reflect the other 40% of the sound back into the room. A list of common building materials and approximate NRC ratings is shown in Table 22-10. Note that most hard, smooth surfaces have low NRC ratings, while most soft surfaces have high NRC ratings.

The noise level in the reverberant field can be reduced by adding materials with high NRC values. The noise reduction achievable depends primarily on the product of the surface area and the NRC rating of the added acoustical material, but it also depends

on the average NRC rating and the total surface area of the existing room. For pumping stations with little or no existing acoustical materials, the noise reduction by adding acoustical materials can be approximated by

$$NR = 10 \log_{10} [1 + 10(S_a/S)NRC_a] \quad (22-31)$$

where S_a and NRC_a are the surface area and NRC rating of the added acoustical material. When a material is placed over an existing material (shielding it from the noise field), the NRC_a value used in Equation 22-31 should be the net increase in NRC . For example, if a 5-cm- (2-in.-) thick fiberglass insulation board is placed under a steel deck roof, the NRC_a should be $0.95 - 0.05 = 0.90$ (see Table 22-10). Hanging baffles (or unit absorbers) are sometimes rated in terms of metric or English sabins per unit rather than by NRC . These numbers can easily be converted to NRC values by dividing the sabins per unit by the total surface area of the absorber to yield NRC . Equation 22-31 is represented in graphical form by Figure 22-20.

Table 22-10. Values of NRC for Common Building Materials Based on Total Surface Area (All Sides of the Absorber)

Material	Approximate NRC
Concrete (poured or sealed)	0.05
Steel (and other hard metals)	0.05
Concrete block (porous)	0.05–0.10
Wood (all types)	0.05–0.10
Glass	0.05–0.10
Gypsum board	0.05–0.10
Floor tile (ceramic, vinyl, etc.)	0.05–0.10
Acoustical masonry blocks	0.35–0.75
Acoustical ceiling tile (mineral fiber)	0.45–0.75
Carpet	0.45–0.65
Acoustical roof decking	0.45–0.85
Fiberglass insulation board 25 mm (1 in.) thick	0.75–0.85
Fiberglass insulation board 50 mm (2 in.) thick	0.85–1.00
Acoustical baffles (hanging)	0.65–0.95
Acoustical panels (wall or ceiling mounted)	
12 mm (1/2 in.) thick	0.35–0.60
25 mm (1 in.) thick	0.65–0.80
50 mm (2 in.) thick	0.85–1.00
Spray-on acoustical system	
12 mm (1/2 in.) thick	0.35–0.60
25 mm (1 in.) thick	0.65–0.80
37 mm (1-1/2 in.) thick	0.85–0.95

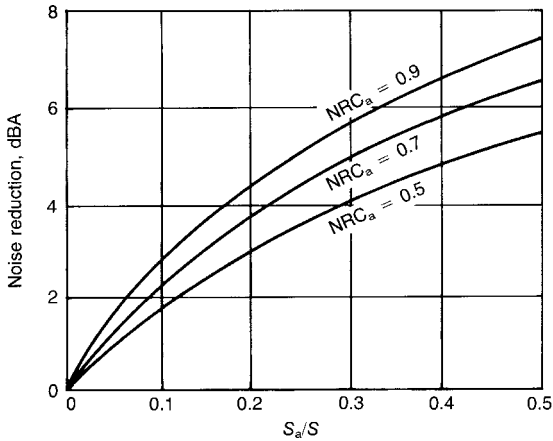


Figure 22-20. Reverberant field noise reduction with acoustical treatment on walls and ceiling.

Noise Sources

There are numerous noise-generating devices typically found in pumping stations, but the most common noise problems stem from engine generators, air compressors, fans, electric motors, and pumps.

Equipment manufacturers should be able to provide reasonably accurate noise data for a particular piece of equipment. The noise (sound pressure) levels in Tables 22-11 and 22-12 apply to the normal operation of equipment in good repair, but they are approximate and should be used only when manufacturer's data are unavailable or, for some reason, unreliable. Equipment with worn bearings or damaged internal parts may generate higher noise levels. Avoid comparing noise data in different formats. For

Table 22-11. Typical Noise Levels from Diesel Engines^a

		A-weighted noise level, [dBA at 1 m (3.28 ft)]	
Horsepower	(rev/min) ^b	Casing radiated	Exhaust radiated
under 50	1200	94	107
50–100	1200	96	109
100–200	1200	99	112
200–400	1200	102	115
400–800	1200	105	118
800–1600	1200	108	121

^aUse the same data for spark-ignition engines. They are quieter, so these data are more conservative.

^bFor other engine speeds add $30 \log_{10} [(\text{rev/min})/1200]$.

Table 22-12. A-weighted Sound Pressure Levels of Electric Motors [dBA at 1 m (3.28 ft)]

Power (kVA)	Horsepower	Speed (rev/min)	
		450–900	900–1800
7.5–37	10–50	88	92
37–75	50–100	92	96
75–150	100–200	95	99
150–300	200–400	98	102
300–600	400–800	101	105

example, sound power levels are not equivalent to sound pressure levels, even though they may appear to have the same units. (Actually, they do not have the same units.) Also be aware that sound pressure levels may be A-weighted, B-weighted, C-weighted, D-weighted, or even unweighted. They may also be octave band, one-third-octave band, or even narrow band. To be applicable to the expressions presented in this chapter, all noise levels must be A-weighted sound pressure levels at some identified direction and distance from the source.

Engine-Powered Generators

Most of the noise generated by engine-powered generators is caused by the engine firings within the individual cylinders. This acoustical energy escapes via direct transmission through the engine block and the exhaust stack into the surrounding environment. Although some noise exits the combustion air intake, it is usually much lower in intensity because it is blocked off during the ignition while the gases in the cylinder are exhausting. Mechanical noise from the internal rotating parts in the engine and the generator are usually insignificant. The radiator fan is the only other significant noise source on the machine (unless it is remotely cooled). Noise levels from these fans can be found in the subsection entitled “Fans,” below.

Approximate A-weighted sound pressure levels for diesel engines are given in Table 22-11. Casing-radiated and exhaust noise levels are shown separately because the exhaust stack outlet is usually remote from the engine. The conditions for the exhaust noise data are predicated on (1) no muffler in the exhaust pipe, (2) an exhaust directed upward, and (3) a measurement point 1 m (3.28 ft) to the side (90 degrees from the exhaust direction) at the elevation of the opening. Under these assumptions there is very little directionality to the source (i.e., the noise level is independent of azimuth).

The noise levels given in Table 22-11 are for steady operation at fixed speed. Start-up noise may be higher, particularly if compressed-air starters are used. Compressed-air starters should be avoided in residential areas because noise levels inside the plant can exceed 120 dBA.

Silencers are nearly always required to bring the exhaust stack noise levels down to meet environmental noise ordinances. Depending on the silencer design, the engine, and the operating speed, silencers usually reduce noise by 10 to 40 dBA. Most manufacturers provide exhaust noise data for their engines, with a variety of silencers, matched to the engine operating at fixed-governed speeds. In general, it is best to locate the silencer close to the engine exhaust with a flexible coupling between the two, as shown in Figure 22-21. This location minimizes the length of exhaust pipe with high internal noise, some of which radiates outward through the pipe wall. The preferred technique of silencing a diesel generator is illustrated in the figure.

Electric Motors and Pumps

Electric motors can be a significant source of noise in pumping stations. In most applications, the noise radiated by a motor-driven pump is generated by the electric motor, not the pump. There are some instances where the pump can be noisier than the motor, but these are usually limited to high-speed (pump frequencies greater than 200 Hz), high-pressure, low-volume pumps such as gear pumps, which are not common in pumping stations. Approximate A-weighted noise levels from induction motors are shown in Table 22-12.

Air Compressors

Noise levels produced by air compressors vary widely with design and application. Very large machines that are designed to deliver volumes of compressed air greater than 50 standard L/s (100 scfm) can generate noise levels in excess of 120 dBA near the source. Air compressors common in pumping stations require only a fraction of this capacity and, as a result, are much quieter (but still rather noisy). Typical noise levels from these machines at 1 m (3.28 ft) are in the 90- to 100-dBA range.

Fans

The noise levels produced by ventilating fans are a function of the fan blade and housing design, the number of blades, the tip speed of the blades, and the volume flow and static pressure created by the fan. The most popular types for use in pumping stations are propeller fans and centrifugal fans. Propeller fans are commonly used in an open condition (no housing or ductwork connected to the fan) to ventilate spaces through an opening in the wall or roof of the space. Under these conditions the static pressure requirements of the fan are minimal—usually less than 25 mm (1 in.) of water. Centrifugal fans, on the other hand, are commonly used in ducted applications where static pressure requirements are more substantial. Noise levels radiated from centrifugal fans are computed as shown in the ASHRAE handbook [24].

The approximate noise levels from propeller fans of various diameters and speeds are given in Figure 22-22. The blade design (airfoil, flat, etc.) also has some effect on the resulting noise level, but the noise

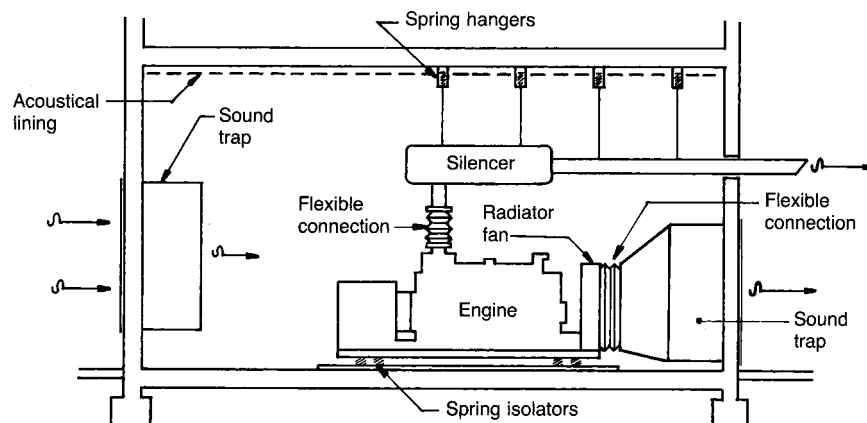


Figure 22-21. Noise control for a diesel generator.

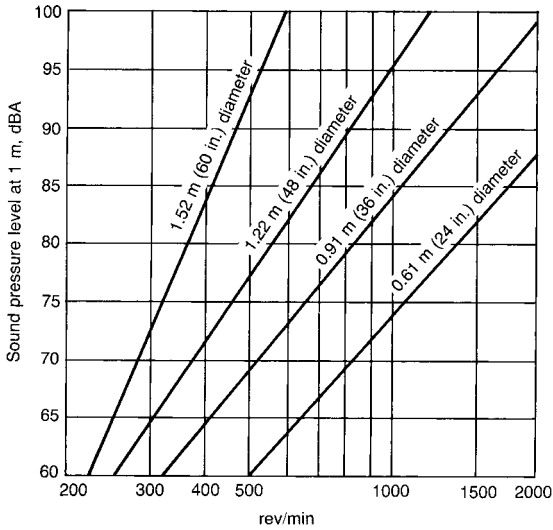


Figure 22-22. Noise levels of propeller fans.

produced by most propeller fans should fall within 5 dBA of these curves.

Multiple Sources

When two or more sources of noise operate simultaneously, the total noise level at a point from all sources can be computed from the following expression:

$$L_p = 10 \log_{10} \sum_{i=1}^m 10^{N_i/10} \quad (22-32)$$

where L_p is the total noise level in dBA, N is $L_i/10$, L_i is the noise level of the i th source, and m is the total number of sources. This expression is valid for multiple sources in a free field (i.e., outdoors) or within a room as long as the noise level from the direct field and the reverberant field are each evaluated independently as separate sources. If all sources have the same noise level at a point, Equation 22-32 reduces to

$$L_p = L_i + 10 \log_{10} m \quad (22-33)$$

Noise Reduction by Enclosures

Noise levels from mechanical equipment can be reduced by (1) modifying the source, (2) adding acous-

tical materials to the room, or (3) isolating the source within a complete or partial enclosure. Techniques for modifying the source (e.g., reducing the speed or changing the equipment design) can often be very effective, but they may be costly and are usually undesirable for one reason or another. Adding acoustical materials to the room is somewhat expensive and only partially effective because it does not reduce the noise near the source. Usually a 5- to 8-dB noise reduction is the most that can be achieved (see Figure 22-20). Consequently, the most common method of reducing noise from machinery is to enclose it within partitions and/or barriers.

The noise reduction, NR , provided by a partition or wall separating two spaces is defined as the difference between the noise level on the source side, L_{p1} , and the receiver side, L_{p2} , of the partition. In general, the noise reduction is primarily a function of the transmission loss of the partition, the total area of the common partition through which the noise radiates, the physical size and acoustical properties of the rooms in question, and the frequency distribution of the noise on the source side. A complete description of the equations required to compute the noise reduction accurately is well beyond the scope of this text. However, the following simple equation is usually accurate to within ± 8 dBA:

$$NR = L_{p1} - L_{p2} = \text{STC} - 5 \quad (22-34)$$

The STC term in Equation 22-34 refers to the sound transmission class, STC, of the partition. The STC rating of a partition is a single number descriptor determined from laboratory measurements of the sound transmission loss as a function of frequency. The STC ratings of some common building materials are listed in Table 22-13.

For materials not listed in the table, the STC ratings can be obtained from data in Harris [25] or approximated from the surface weight of solid materials using the following formula:

$$\text{STC} = 8 + 17 \log_{10} w \quad (22-35a)$$

where w is the surface mass of the material in kilograms per square meter.

In U.S. customary units, the expression is

$$\text{STC} = 20 + 17 \log_{10} w \quad (22-35b)$$

where w is the surface weight in pounds mass per square foot.

Table 22-13. Approximate STC Ratings for Common Building Materials

Material	STC ^a
Poured concrete, 150–300 mm (6–12 in.) thick	50
Hollow concrete block, 200 mm (8 in.) thick, unpainted	40
Steel acoustical panel, 100 mm (4 in.) thick	40
Drywall partition, ^b 150 mm (6 in.) thick	35
Drywall partition, ^c 150 mm (6 in.) thick	40
Safety glass, 6.35 mm (1/4 in.) thick	30
Hollow metal door, without seals	20
Metal louver, more than 50% open area	2

^aNote that the STC values are *not* additive for composites: two partitions each with an STC of 35 *do not* yield an STC of 70. Each composite must be separately tested.

^bMetal studs with one layer of gypsum board on each side.

^cBatt insulation between metal studs and gypsum board on each side.

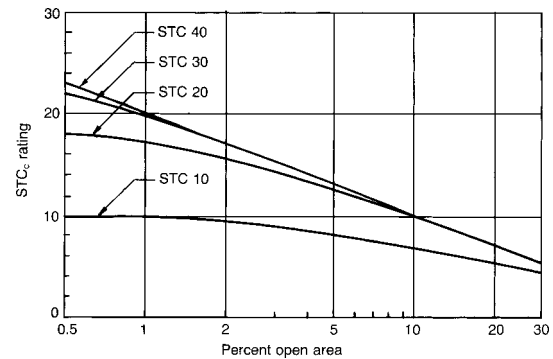
In most applications, an enclosure is made from more than one material. For example, the door and roof are usually constructed differently than the walls, and there may be openings for ventilation. To evaluate this complex situation, calculate the composite sound transmission class, STC_c. The composite STC can be determined from substituting the exposed areas and STC of each component of the barrier system into the expression

$$\text{STC}_c = -10 \log_{10} \left[\frac{1}{s} \sum_{i=1}^n A_i 10^{-N_i} \right] \quad (22-36)$$

where N is STC_i/10, A_i and STC_i are the exposed surface area and STC rating of the i th material, and s is the total exposed surface area of all components in the direction of the listener. In most situations, the isolation provided by a composite system is only as good as its weakest component, unless that component has a surface area less than one-tenth of the other components. A unique but common example is the enclosure with an opening because openings have an STC of 0. The dependence of the composite STC on opening size is illustrated in Figure 22-23. Note that even small leaks can cause serious degradation of performance, particularly for partitions with high STC ratings.

Sound Traps

In most practical applications, enclosure openings are required for ventilation. Louvered openings are typ-

**Figure 22-23.** Dependence of composite STC rating on openings in partitions or enclosures.

ically more than 50% open area, so these openings present a major acoustical problem—a problem usually solved with sound traps. A sound trap or duct silencer is usually a rectangular steel box (but sometimes round for circular ducts) open at both ends with a packing of acoustical materials that create parallel baffles, as in Figure 22-24. The air passes straight through most units without changing direction. These devices are prefabricated in a variety of sizes and are furnished with laboratory-tested acoustic and aerodynamic performance data. The STC of an opening can, therefore, be significantly increased by inserting one or more sound traps into the opening in the barrier and sealing the perimeter airtight with acoustical (nonhardening or resilient) caulking.

The acoustical performance of a sound trap depends on its size and internal design. The length (the dimension in the direction of air flow) of the sound trap is the most important factor affecting performance. Sound traps are available in lengths ranging from 0.3 to 3 m (1 to 10 ft). Within each length, most manufacturers produce several designs that essentially vary in the size of the air passage (i.e., the percentage of open area). Units with larger air passages have lower acoustical ratings and lower aerodynamic pressure drops. Sound traps are rated by the dynamic insertion loss (in decibels) at a variety of frequencies (and not by STC values). The dynamic insertion loss is the difference in noise level with and without the sound trap in the system (with air flowing through the silencer). Aerodynamic and acoustic performance data for a few typical sound traps are given in Table 22-14. The STC of a sound trap can be approximated roughly by the manufacturer's dynamic insertion loss rating in the frequency band nearest the primary frequency of the noise source. The primary frequency would be the blade passage frequency for a fan or the firing rate frequency for a

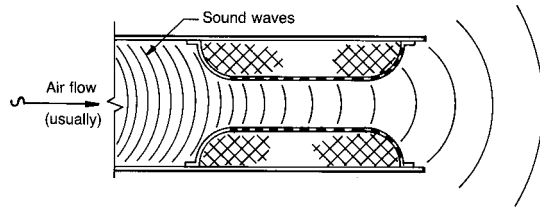


Figure 22-24. A typical duct silencer.

diesel engine. If the primary frequency of the noise is unknown, the insertion loss at 250 Hz may be used to approximate the STC of the sound trap.

Noise Reduction of Barriers

It is often neither practical nor economical to enclose a noise source completely. An often-used compromise is to erect a barrier between the noise source and the receiver to block the direct transmission of noise to the listener. For the barrier to be effective, it must (1) be solid (nonporous), and (2) completely block the straight-line path between the source and the listener.

Table 22-14. Performance Characteristics of Sound Traps. Face Velocity of 5.08 m/s (1000 ft/min).

Length m	Pressure drop ft	Approximate mm Hg	in. H ₂ O	STC rating
0.3	1	1.40	0.75	6
1.0	3	0.09	0.05	10
1.0	3	0.19	0.10	15
1.0	3	0.37	0.20	25
1.5	5	0.15	0.08	18
1.5	5	0.28	0.15	25
1.5	5	0.37	0.20	30
2.1	7	0.19	0.10	25
2.1	7	0.28	0.15	30
2.1	7	0.48	0.25	35

Chain-link fences, shrubs, or rows of trees are virtually useless as noise barriers, but earth berms and concrete walls have been used successfully. Barriers do not provide as much noise reduction as enclosures because the noise tends to go around the barrier.

The noise reduction provided by a barrier is primarily dependent on (1) the material of construction, (2) the height of the barrier, and (3) the presence of other adjacent reflecting surfaces. For an infinitely long solid wall blocking the line-of-sight transmission between two points (as shown in Figure 22-25) with no nearby reflecting surfaces, the noise reduction is approximated by the following dimensionless equation from Beranek [26]:

$$NR = 20 \log_{10} \left(\frac{\sqrt{2\pi N}}{\tan h \sqrt{2\pi N}} \right) + 5 \quad (22-37)$$

where N is the dimensionless Fresnel number, which is defined by

$$N = \frac{2f}{c} (A + B - r) \quad (22-38)$$

in which f is the primary frequency of the noise source in Hz and c is the speed of sound in air in meters per second (feet per second). See Figure 22-25 for the definitions of A , B , and r , which are in meters (feet). Although very high barriers might reduce noise by more than 20 dBA, they usually do not because of sound transmission either through the barrier material or by another flanking path. In general, the barrier material should have an STC rating about 15 dB greater than the noise reduction predicted by Equation 22-37. Note that the noise reduction given by Equation 22-37 represents the change in sound level at the receiving position with and without the barrier.

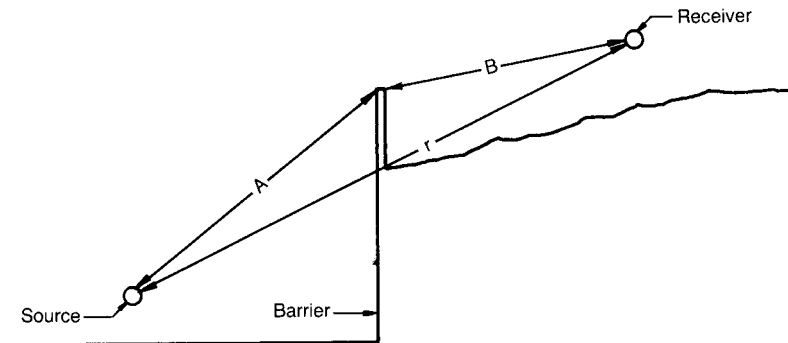


Figure 22-25. Noise reduction of barriers.

22-14. Reducing Exterior Noise

As shown in Example 22-6, the exterior noise radiating from a pumping station may stem from several sources, such as motors, pumps, engine generators, and cooling fans.

Noise is attenuated by (1) walls, (2) partitions or other enclosures (Table 22-13), (3) distance, and (4) silencers and sound traps. Reducing reverberant noise by acoustical treatment [acoustical tile, masonry blocks (if specially manufactured to do so), or thick fiberglass] also reduces radiating noise unless the source is near an opening in a wall. Placing motors below grade or using submersible pumps and motors is very effective. (But engines should not be below grade.) Fan noise can be reduced by using (1) larger diameter blades at lower rotational speeds, (2) more effective sound traps, and/or (3) exhausting through the roof. Reducing fan noise by using larger blades and lower air velocity carries the penalty of a larger louver, which decreases the STC of the wall (or roof), but the penalty can be overcome by permitting

a higher pressure drop in the sound trap, thus making it more effective for suppressing both fan noise and generator-casing noise. In some situations, it may be desirable to provide cooling by refrigeration—an expensive solution.

When designing for noise suppression, contributing sources with noise levels that are 10 dBA or more below the total noise level allowed can usually be ignored because (from Equation 22-32) the effect on total noise level is less than 1 dB.

An additional margin of safety is possible, but it may become quite expensive. Generally speaking, noise control measures tend to follow the law of diminishing returns; that is, the cost of the first 10 dBA of noise reduction is generally much less expensive than the second 10 dBA, and so on. In situations where noise is important, it is usually cost-effective to hire a consultant who can take the responsibility of performing a detailed frequency analysis and specifying exactly what is required to meet code requirements without excessive safety margins.

Example 22-6
Reducing the Sound Level at a Pumping Station

Problem: A pumping station with concrete exterior walls 200 mm (8 in.) thick houses two 500-kVA (670-hp), 1800-rev/min standby diesel generators that supply power to six 200-kVA (270-hp), 600-rev/min direct-drive centrifugal pumps, two of which are back-ups. The layout of the pumping station is such that the nearest residential property is 60 m (197 ft) from the edge of the building, as shown in Figure 22-26. Compute the expected noise levels inside and

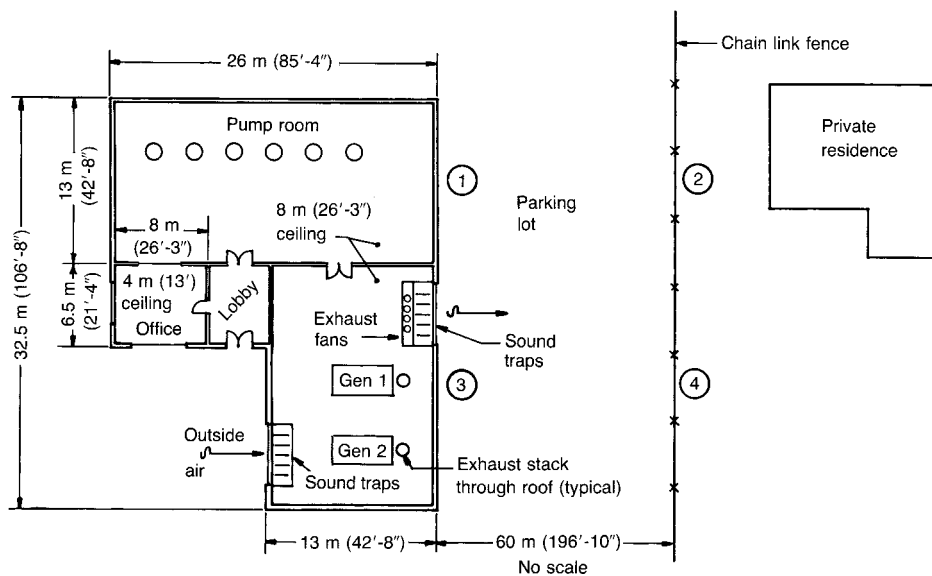


Figure 22-26. Pumping station for Example 22-6.

outside of the building with no special acoustical considerations, then determine the appropriate measures needed to reduce the acoustical impact to an acceptable level. Assume that the NRC for the untreated interior spaces is 0.10. An NRC of 0.05 is not realistic for real-life conditions because of absorption from miscellaneous fixtures, etc.

Solution:

Interior noise levels in pump room and office. The main noise sources in the pump room are the six electric motors that power the pumps. From Table 22-12, the expected noise level from each pump is 98 dBA at 1 m (98.8 dBA at 3 ft). The noise level in the reverberant field from each pump is evaluated from Equation 22-30. Assume the room dimensions to be $13 \times 26 \times 8$ m ($43 \times 85 \times 26$ ft). The total surface area is $S = 1300 \text{ m}^2$ ($14,000 \text{ ft}^2$). Use Equation 22-30 and Table 22-10.

SI Units

$$\begin{aligned} L_{p(\text{rev})} &= 98 - 10 \log_{10} [(0.10)(1300)] + 17 \\ &= 93.8 \text{ dBA} \simeq 94 \text{ dBA} \end{aligned}$$

U.S. Customary Units

$$\begin{aligned} L_{p(\text{rev})} &= 98.8 - 10 \log_{10} [(0.1)(14,000)] + 26.5 \\ &= 93.8 \text{ dBA} \simeq 94 \text{ dBA} \end{aligned}$$

The total noise level in the pump room is evaluated by using four noise sources for m in Equation 22-33. (Two of the six pumps are always off.)

$$L_p = 94 + 10 \log_{10} 4 = 100 \text{ dBA}$$

$$L_p = 94 + 10 \log_{10} 4 = 100 \text{ dBA}$$

The noise level in the office can be computed by estimating the sound transmission through the 32 m^2 (345 ft^2) gypsum board partition with a 3 m^2 (32.4 ft^2) safety glass window separating the office from the pump room. Assuming the partition has an STC of 40 and the window has an STC of 30 (from Table 22-13), the composite STC is determined from Equation 22-36

$$\text{STC}_c = -10 \log_{10} \frac{1}{32} [(29)(10)^{-4} + (3)(10)^{-3}] = 37.3 \quad \text{STC}_c = -10 \log_{10} \frac{1}{325} [(313)(10)^{-4} + (32)(910)^{-3}] = 37.3$$

From Equation 22-34, the noise reduction is

$$NR = 37.3 - 5 = 32.3 \simeq 32$$

$$NR = 37.3 - 5 = 32.3 \simeq 32$$

and the noise level in the office is approximately

$$L_p = 100 - 32 = 68 \text{ dBA}$$

$$L_p = 100 - 32 = 68 \text{ dBA}$$

Interior noise in generator room. Noise data from the manufacturer indicate that the casing-radiated noise level from each unit (including the radiator fan) is 98 dBA at a distance of 3 m (9.84 ft), which can be converted to 1 m (3 ft) by Equation 22-29.

$$98 = L_p(1 \text{ m}) + 20 \log(1/3)$$

$$98 = L_p(3 \text{ ft}) + 20 \log(3/9.84)$$

$$L_p(1 \text{ m}) = 98 + 9.5 = 107.5 \text{ dBA}$$

$$L_p(3 \text{ ft}) = 98 + 10.3 = 108.3 \text{ dBA}$$

The reverberant field level is computed using $S = 13 \times 19.5 \times 2 + (13 + 19.5) 2 \times 8 = 1027 \text{ m}^2$ ($11,055 \text{ ft}^2$), which yields (from Equation 22-30)

$$\begin{aligned} L_{p(\text{rev})} &= 107.5 - 10 \log_{10} (0.10 \times 1027) + 17 \\ &= 104.4 \text{ dBA for each unit.} \end{aligned}$$

$$\begin{aligned} L_{p(\text{rev})} &= 108.3 - 10 \log_{10} [(0.1)(11,055)] + 26.5 \\ &= 108.3 - 30.4 + 26.5 \end{aligned}$$

$$= 104.4 \text{ dBA for each unit.}$$

When both units are operating, the noise level (from Equation 22-33) increases to

$$L_p = 104.4 + 10 \log_{10} (2) = 107.4 \text{ dBA}$$

$$L_p = 104.9 + 10 \log_{10} (2) = 107.4 \text{ dBA}$$

The other major noise sources in the generator room are the six propeller fans 1 m (36 in.) in diameter located in the louvered opening. These fans rotate at 600 rev/min, and each fan generates a noise level of 75 dBA at 1 m, as shown by Figure 22-22. (For U.S. customary units, first use Equation 22-29 to obtain the noise level at 3 ft.) From Equation 22-30 the reverberant field noise level from each fan is

$$L_{p(\text{rev})} = 75 - 10 \log_{10} (0.10 \times 1027) + 17$$

$$L_p(3 \text{ ft}) = 75 + 20 \log_{10} (3.28/3) = 75.8$$

$$\begin{aligned} L_{p(\text{rev})} &= 75 - 20.1 + 17 = 71.9 \text{ dBA} \\ &\simeq 72 \text{ dBA} \end{aligned}$$

$$\begin{aligned} L_{p(\text{rev})} &= 75.8 - 10 \log_{10} [(0.1)(11,055)] + 27 \\ L_{p(\text{rev})} &= 75.8 - 30.4 + 26.5 = 71.9 \\ &\simeq 72 \text{ dBA} \end{aligned}$$

and from all six fans, Equation 22-33 gives

$$L_p = 72 + 10 \log_{10} 6 = 79.8 \text{ dBA}$$

$$L_p = 72 + 10 \log_{10} 6 = 79.8 \text{ dBA}$$

From Equation 22-32, the total noise from the six fans and the two generators is

$$\begin{aligned} L_p &= 10 \log_{10} [(10)^{7.98} + (10)^{10.74}] \\ &= 107.4 \text{ dBA} \end{aligned}$$

$$\begin{aligned} L_p &= 10 \log_{10} [(10)^{7.98} + (10)^{10.74}] \\ &= 107.4 \text{ dBA} \end{aligned}$$

Exterior noise levels. Independently compute the noise levels from each source type and add them together by using Equation 22-32. First, consider the pump room. The interior noise level is 100 dBA (the reverberant field level) because the wall is not within the direct field of any of the pumps (see Figure 22-19). The STC rating of the partition is 50 (see Table 22-13), so if there are no penetrations or openings in the wall, the exterior noise level at position 1 (Figure 22-26) is, from Equation 22-34,

SI Units

U.S. Customary Units

$$L_p = 100 - NR = 100 - (50 - 5) = 55 \text{ dBA}$$

$$L_p = 100 - NR = 100 - (50 - 5) = 55 \text{ dBA}$$

Estimating the average distance from position 1 to the four motors as 12 m (40 ft) and from position 3 to the four motors as 72 m (235 ft), the pump noise level at position 3 is, from Equation 22-29,

$$L_p(1) = 55 + 20 \log_{10} (12/72) = 39 \text{ dBA}$$

$$L_p(2) = 55 + 20 \log_{10} (40/235) = 39 \text{ dBA}$$

The generator room noise is evaluated in a similar manner. The main difference here is the fact that the generator room has a louvered opening containing six propeller fans in the wall facing the receiving property. The total area of the opening is 14 m² (151 ft²), or approximately 9% of the 156 m² (1680 ft²) exterior wall area. From Equation 22-36, the STC-50 partition is reduced to a composite (effective) STC_c rating of

$$\begin{aligned} \text{STC}_c &= -10 \log_{10} \left\{ \frac{1}{156} [14(10)^{-0.2} + 142(10)^{-5.0}] \right\} \\ &= 12.5 \text{ dBA} \end{aligned}$$

$$\begin{aligned} \text{STC}_c &= -10 \log_{10} \left\{ \frac{1}{1680} [151(10)^{-0.2} + 1529(10)^{-5.0}] \right\} \\ &= 12.5 \text{ dBA} \end{aligned}$$

The STC_c rating from Figure 22-23 is 11, a satisfactory check. From Equation 22-34 again, the generator casing noise level at position 3 with both generators running is

$$\begin{aligned} L_p(3) &= 107.4 - NR = 107.4 - (12.5 - 5) \\ &= 99.9 \\ &\simeq 100 \text{ dBA} \end{aligned}$$

$$\begin{aligned} L_p(3) &= 107.4 - NR = 107.4 - (12.5 - 5) \\ &= 99.9 \\ &\simeq 100 \text{ dBA} \end{aligned}$$

Assuming the distance from position 3 to the generators to be 5 m (16.4 ft), the generator casing noise level at position 4 is, from Equation 22-29,

$$\begin{aligned} L_p(4) &= 99.9 + 20 \log_{10} (5/65) \\ &= 77.6 \text{ dBA} \end{aligned}$$

$$\begin{aligned} L_p(4) &= 99.9 + 20 \log_{10} (16.4/213) \\ &= 77.6 \text{ dBA} \end{aligned}$$

The propeller fan noise is evaluated from Figure 22-22. As shown, each of the fans is expected to generate a noise level of 75 dBA at a distance of 1 m (3 ft). Thus, at position 4, each fan yields a noise level of

$$\begin{aligned} L_p(4) &= 75 + 20 \log_{10} (1/60) \\ &= 39.4 \text{ dBA} \end{aligned}$$

$$\begin{aligned} L_p(4) &= 75 + 20 \log_{10} (3.28/197) \\ &= 39.4 \text{ dBA} \end{aligned}$$

The noise level from all six fans (from Equation 22-33) would be

$$L_p = 39.4 + 10 \log_{10} (6) = 47 \text{ dBA}$$

$$L_p = 39.4 + 10 \log_{10} (6) = 47 \text{ dBA}$$

The manufacturer lists the exhaust stack noise from each unit to be 62 dBA at a distance of 15 m (49 ft) when equipped with the standard residential-grade silencer. Using Equation 22-29, the noise level from one of the silenced exhausts at position 4, which is 63 m (207 ft) distant, would be

$$L_p(4) = 62 + 20 \log_{10} (15/63) = 49.5 \text{ dBA}$$

$$L_p(4) = 62 + 20 \log_{10} (49/207) = 49.5 \text{ dBA}$$

From Equation 22-33, the total noise level from both stacks would be

$$L_p(4) = 49.5 + 10 \log_{10} (2) = 52.5 \text{ dBA}$$

$$L_p(4) = 49.5 + 10 \log_{10} (2) = 52.5 \text{ dBA}$$

The total exterior noise level at the residential property from all sources originating from the pump station property is greatest at position 4; from Equation 22-32, it is

$$\begin{aligned} L_p &= 10 \log_{10} [10^{7.76} + 10^{4.7} + 10^{5.25}] \\ &= 77.6 \text{ dBA} \end{aligned}$$

$$\begin{aligned} L_p &= 10 \log_{10} [10^{7.76} + 10^{4.7} + 10^{5.25}] \\ &= 77.6 \text{ dBA} \end{aligned}$$

which is well above the maximum allowable noise standards for residential communities (see Table 22-5).

Potential interior noise reduction measures. Noise levels in both the pump and generator rooms may exceed OSHA standards [20] for 8 h/d employee noise exposure. Assume that on an average day an employee is expected to spend time as follows:

Time, (h)	Place	Noise level, dBA	Allowable time per Table 22-4	D_n from Eq. 22-5
1.0	Pump room	100	2.0	0.5
0.25	Generator room	108	0.66	0.38
6.75	Office	68	No limit	0.00
Total				0.88

The computed D_n value is below the allowable limit of 1.0, and there should be no problem complying with OSHA noise laws provided that all employees spend a considerable amount of

time in the quiet areas. But the anticipated noise levels would make vocal communication difficult between co-workers in the pump room and the generator room.

The generator room has a total surface (floor, ceiling, and walls) of 1027 m^2 ($11,055 \text{ ft}^2$), and if the ceiling, which has an area of 253.5 m^2 (2728 ft^2), were entirely covered with 5-cm- (2-in.-) thick fiberglass insulation board, the reverberant field noise level would be reduced by (from Equation 22-31 and Table 22-10)

$$NR = 10 \log_{10} [1 + 10(253/1027)(0.95)] \\ = 5.2 \text{ dBA}$$

$$NR = 10 \log_{10} [1 + 10(2728/11,055)(0.95)] \\ = 5.2 \text{ dBA}$$

This 5-dBA noise reduction would (from Figure 22-3) nearly double the distance for which voice communication is possible. If the pump room ceiling were also covered with fiberglass insulation board, the D_n noise dose would be reduced as follows:

$$NR = 10 \log_{10} [1 + 10(13 \times 26/1300)(0.95)] \\ = 5.4 \text{ dBA}$$

$$NR = 10 \log_{10} [1 + 10(42.7 \times 85.3/14,000)(0.95)] \\ = 5.4 \text{ dBA}$$

Time, (h)	Place	Noise level, dBA	Allowable time per Table 22-4	D_n from Eq. 22-5
1	Pump room	95	4.0	0.25
0.25	Generator room	102	1.5	0.17
Total				0.42

Example 22-7 Reducing Exterior Noise from a Pumping Station

Problem: The pumping station of Example 22-6 must meet a community noise ordinance that limits the maximum noise level at adjacent residential property to 55 dBA. Find and evaluate the modifications necessary to comply with the local ordinance.

Solution: All three major noise sources (generator casing, engine exhaust, and fan noise) must be reduced. From Example 22-6, the noise level at position 3 (Figure 22-26) emanating from the pump room is only 39 dBA, so further reduction of the noise from this source is not necessary.

The following noise levels at position 4 are obtained from Example 22-6, for which there are no special noise control features:

Generator casing:	77.6 dBA
Propeller fans:	47.0 dBA
Generator exhaust:	52.5 dBA
Total noise level:	77.6 dBA

To meet the maximum noise level of 55 dBA requires a reduction of 23 dB. Try STC-30 sound traps in the louver. The pressure drop for these 1.5-m- (5-ft-) long traps is about 0.37 mm Hg (0.20 in. WC) at 5.08 m/s (1000 ft/min). To reduce this pressure drop to the recommended 0.09 mm Hg (0.05 in. WC), the velocity must be reduced. Because the pressure drop is proportional to the square of the velocity,

$$\frac{\text{new pressure drop}}{\text{original pressure drop}} = \left(\frac{\text{new velocity}}{\text{original velocity}} \right)^2$$

SI Units

$$\frac{0.09 \text{ mm Hg}}{0.37 \text{ mm Hg}} = \left(\frac{v}{5.08 \text{ m/s}} \right)^2$$

from which $v = 2.51 \text{ m/s}$

The original 14-m^2 (151-ft^2) louver sized for a face velocity of 3.81 m/s (750 ft/min) must be enlarged to

$$A = 14(3.81/2.51) = 21 \text{ m}^2$$

The net wall area is

$$A = (19.5 \text{ m})(8 \text{ m}) - 21 \text{ m}^2 = 135 \text{ m}^2$$

From Equation 22-36,

$$\begin{aligned} \text{STC}_c = & -10 \log_{10} [(1/156)(21 \times 10^{-30/10} \\ & + 135 \times 10^{-50/10})] = 38.4 \text{ dB} \end{aligned}$$

Recompute the noise level at position 4. Noise from the generator casing is reduced by an STC_c 38.4-dB barrier instead of the original STC_c 12.5-dB wall (without sound traps). The net noise reduction is $38.4 - 12.5 = 25.9 \text{ dB}$.

The propeller fan noise is reduced by the STC rating of the sound traps (30 dB) minus the STC rating of the louver (2 dB) for a net noise reduction of 28 dB. The components at position 4 are, therefore,

$$\begin{aligned} \text{Generator casing } (77.6 - 25.9) &= 51.7 \text{ dBA} \\ \text{Propeller fans } (47 - 28) &= 19 \text{ dBA} \\ \text{Generator exhaust} &= 52.5 \text{ dBA} \end{aligned}$$

and the total noise from Equation 22-32 is

$$\begin{aligned} L_p &= 10 \log_{10} [10^{51.7/10} + 10^{19/10} + 10^{52.5/10}] \\ &= 55.1 \text{ dBA} \end{aligned}$$

U.S. Customary Units

$$\frac{0.05 \text{ in.}}{0.20 \text{ in.}} = \left(\frac{v}{1000 \text{ ft/min}} \right)^2$$

from which $v = 500 \text{ ft/min}$

$$A = 151(750/500) = 226 \text{ ft}^2$$

$$A = (64 \text{ ft})(26.25 \text{ ft}) - 226 \text{ ft}^2 = 1454 \text{ ft}^2$$

$$\begin{aligned} \text{STC}_c = & -10 \log_{10} [(1/1680)(226^{-30/10} \\ & + 1454^{-50/10})] = 38.4 \text{ dB} \end{aligned}$$

$$\begin{aligned} L_p &= 10 \log_{10} [10^{51.7/10} + 10^{19/10} + 10^{52.5/10}] \\ &= 55.1 \text{ dBA} \end{aligned}$$

This sound level is just slightly above the 55-dBA limit. To make further reductions, choose critical-grade exhaust silencers for an additional 5-dB noise reduction on the exhaust noise. Note that both engines must be silenced with critical-grade silencers to achieve this result.

With the critical-grade exhaust silencers and the STC-30 sound traps, the noise level at position 4 is

$$\begin{aligned} \text{Generator casing:} & 51.7 \text{ dBA} \\ \text{Propeller fans:} & 19.0 \text{ dBA} \\ \text{Generator exhaust:} & 47.5 \text{ dBA} \\ \text{Total noise level:} & 53.1 \text{ dBA (from Equation 22-32)} \end{aligned}$$

This meets the noise ordinance with a small margin of safety. For an increased margin of safety, STC-35 sound traps could be used for the louver opening. This application would require sound traps that are 2.1 m (7 ft) long, which would require considerable floor space—one of the several reasons for avoiding retrofit in favor of planning for noise reduction during the first stages of layout.

22-15. References

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